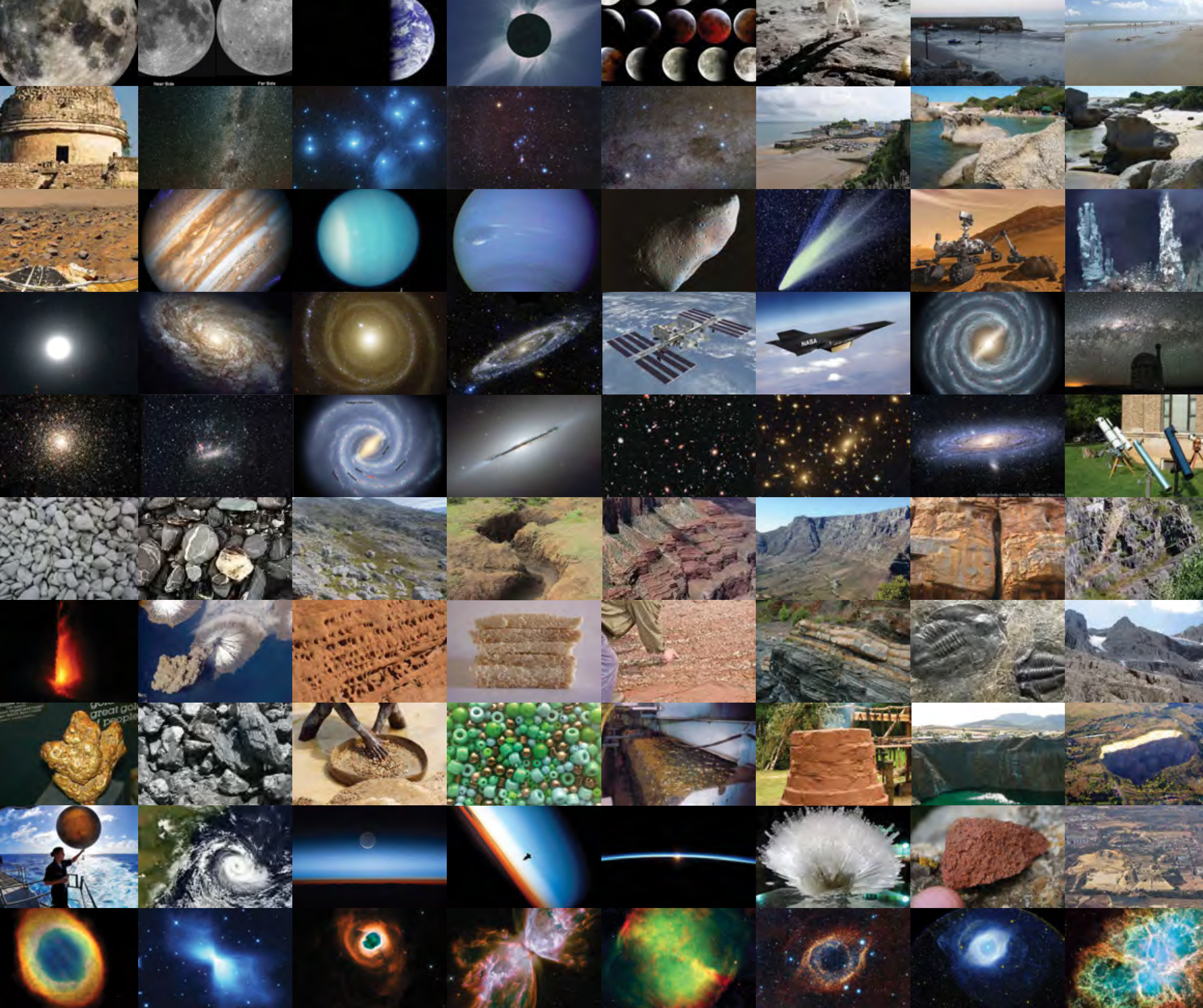
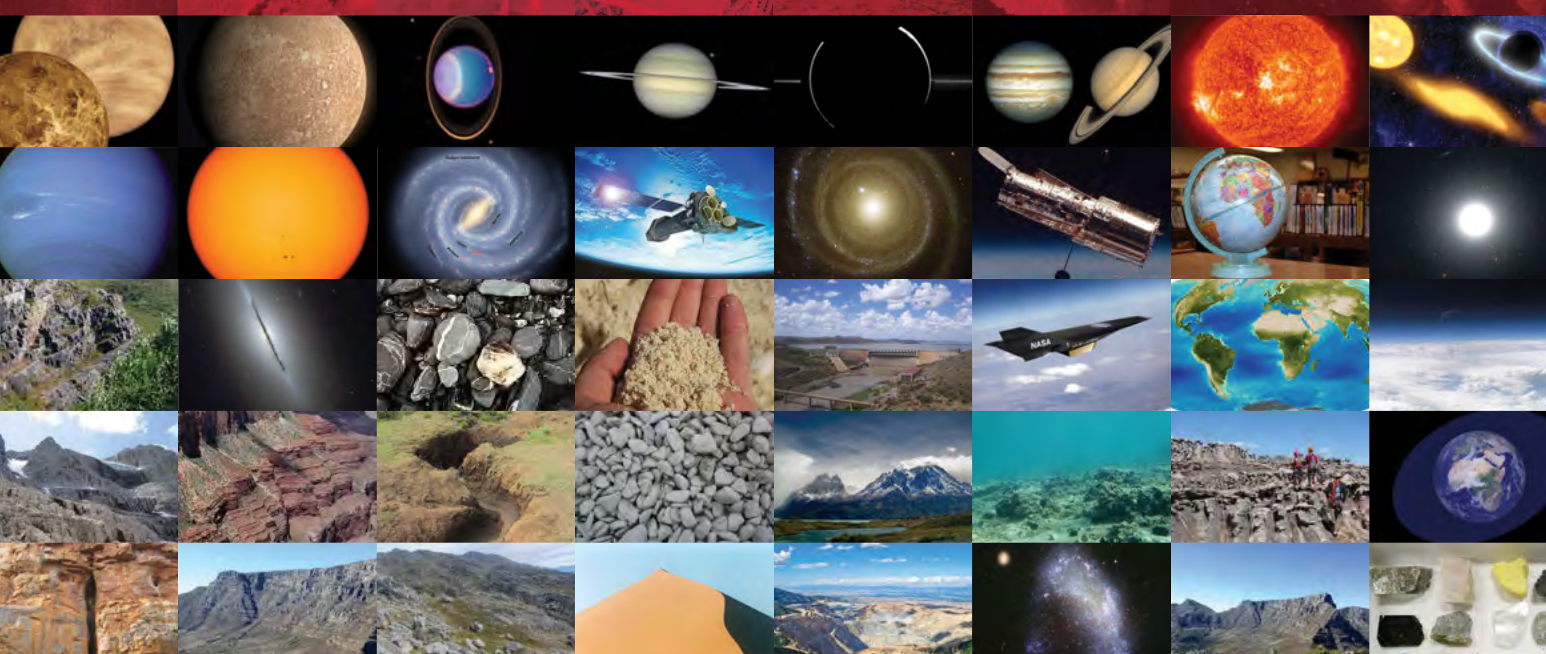


PLANET EARTH AND BEYOND



1 The solar system

KEY QUESTIONS:

- How does the Sun produce its energy?
- How can we observe the Sun without damaging our eyes?
- What objects are in orbit around the Sun in our solar system?
- Why are there two types of planets?
- How do the planets in our solar system differ?
- What are asteroids and comets?
- What is the difference between a planet and a dwarf planet?
- Why is life possible on Earth?

NEW WORDS

- solar system
- star
- nuclear fusion
- convection
- sunspot
- solar wind

Our solar system includes the Sun and all the objects that orbit around the Sun. As you will find out, a variety of objects are in orbit around the Sun: eight planets, many dwarf planets, asteroids, Kuiper Belt objects and comets.

1.1 The Sun

Before we look at the Sun close up, let's summarise what you learned about the Sun in Grades 6 and 7:

1. The Sun is our closest star and is very important for life on Earth as it provides us with light and heat.
2. The Sun is located at the very centre of our solar system.
3. The Earth and other planets all orbit around the Sun, held in orbit by the force of gravity.

What do you think the Sun would look like if it was further away, like the other stars we see at night?

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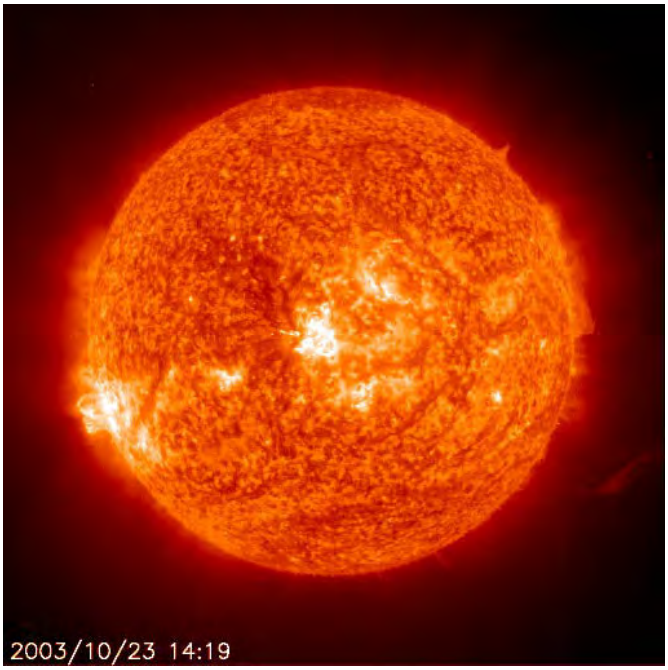
VISIT

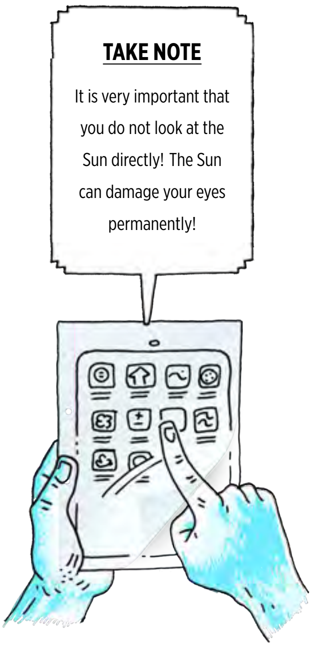
Secrets of a dynamic Sun (video)

bit.ly/1h0io4b

Let's look at the Sun in more detail.



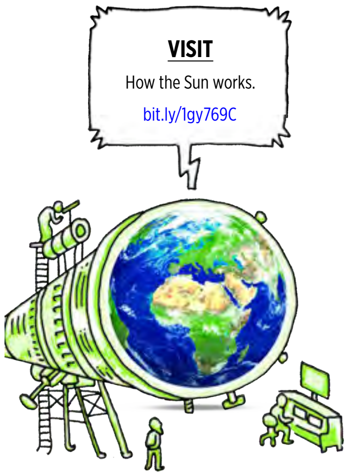
An image of the Sun taken with the SOHO space satellite.



Do you know what the Sun is made of? The Sun is mostly made up of hydrogen gas (about 71%), and also helium gas (about 27%) with a tiny amount of other gases. The temperature at the Sun's surface is very high, around 5500 °C. However, that is nothing compared to deep inside the Sun. At the Sun's centre, or core, it is about 15 million °C. It is so hot at the Sun's centre that **nuclear reactions** can occur, which change atoms from one element to another. In the Sun's case, four hydrogen nuclei are squeezed or fused together to form a new helium nucleus. This process is called **nuclear fusion**.

This nuclear fusion reaction releases energy because the new helium nuclei produced have very slightly less mass than the four hydrogen nuclei used to make them. How can this be? Well, according to the famous scientist Albert Einstein, energy and mass are equivalent. Some of the mass in the hydrogen nuclei is converted and released as energy when the nuclei fuse to make helium. A very large amount of energy is released. This energy travels outwards from the Sun's core towards its surface. The energy eventually reaches the Sun's surface somewhere between 17,000 and 100,000 years later! The Sun's energy then spreads out into the solar system in the form of heat and light.

You are now going to observe the Sun to look at its surface features. **Remember, you should never look directly at the Sun as it can permanently damage your eyes.** You can use either a telescope with a filter on it or a pinhole to project an image of the Sun onto a screen to safely view the Sun's image.



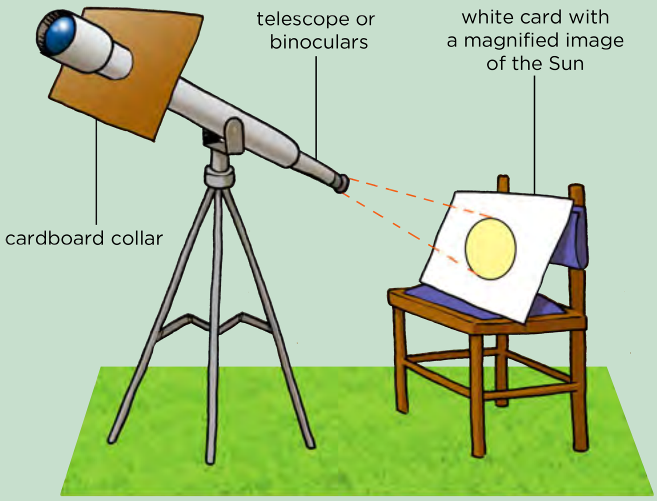
ACTIVITY: Observing the Sun using a telescope

MATERIALS:

- telescope
- white card
- chair to rest the card on
- cardboard to make a shade collar
- pair of scissors
- pencil

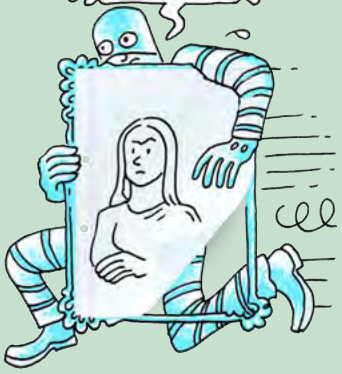
VISIT

Interact with this simulation to visualize the effects of gravity on orbital paths of the Sun, Earth and Moon.
bit.ly/1a2mJCL



TAKE NOTE

NEVER look directly at the Sun, even with sunglasses on as you can permanently damage your eyes.



INSTRUCTIONS:

1. Take a piece of cardboard and place it up against the narrowest end of the telescope.
2. Draw an outline around the edge of the telescope on the card to use as a guide for cutting to make the collar.
3. Cut out inside the circle you just drew so that the cardboard can fit over the telescope as shown in the figure above. You can cut a single slit into the circle from the edge of the card as shown in the diagram
4. Place the collar on the telescope. Adjust the size of the cut out circle if necessary (for example if your telescope is slightly wider in the middle than at the end, you may want to make your circle slightly larger). This collar shades the area, where the image will fall, from stray light.
5. Select the lowest magnification eyepiece lens you have and insert it into the telescope's eyepiece.
6. Focus the telescope by looking at a distant object (NOT the Sun).
7. Point the telescope at the Sun (do NOT look through the telescope to do this).
8. Place a chair behind the telescope and rest a white piece of card on it. The card should be tilted towards the telescope.
9. Adjust the direction in which the telescope is pointing until the image of the Sun appears on the white paper card. This may take some time.
10. Keeping the telescope still, move the white card toward or away from the eyepiece until the image of the Sun fits neatly in the middle of the card.

Adjust the chair's position as needed.

11. Adjust the tilt of the white card until the Sun's image is circular.

QUESTIONS:

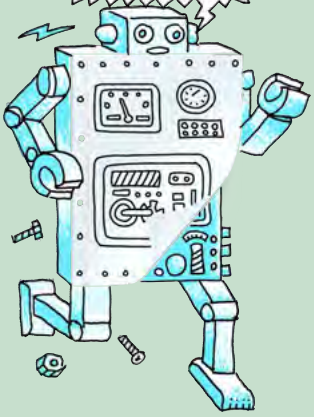
1. Looking carefully you should see that the Sun's image moves slowly across the white card. What causes this motion?

---

2. Draw a picture of what the surface of the Sun looks like on the white card in the circle below.

TAKE NOTE

Revise the model of the atom that you learned about in Matter and Materials if you are unsure of some of the terms used here, such as nucleus, which is at the centre of an atom, and consists of protons and neutrons.



Alternatively, if you do not have access to a telescope or binoculars, you can perform the following activity to view the Sun.

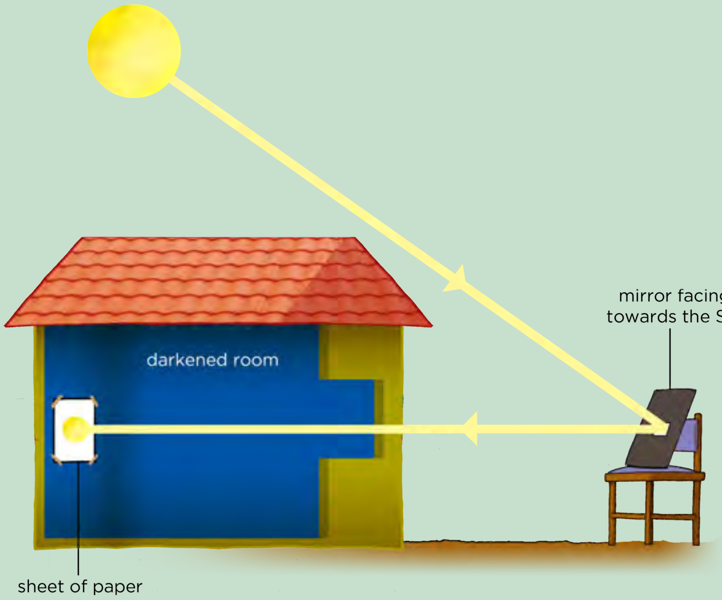
ACTIVITY: Observing the Sun with a pinhole camera

In this activity you will reflect an image of the Sun onto a white card or screen for your learners to observe. This method has the advantage of not needing a telescope or binoculars, however, the solar image produced will be a bit fuzzy. However, it should be good enough to show large sunspots. This activity is designed as a teacher-led demonstration. If you have a sunlit window or door to your class you can do this activity in the classroom. If you do not have a classroom with a sunlit window, or if your class is very small, you can do the activity outdoors, reflecting the Sun's image onto a shaded wall or back into a darkened classroom.

VISIT
Three years of the Sun in three minutes.
bit.ly/19nCfGu

As a rough guide, begin with a distance of around 8 m between the white card and the mirror. The further away you place the mirror from the white screen the fainter and larger the image will appear. At closer distances the image will be brighter but it may not be in very good focus.

As mentioned in the previous activity, sunspots are sometimes (not always) visible on the Sun's surface. Therefore, you could repeat this activity over the course of several days to see if any sunspots or sunspot groups change shape, size, or position over time.



VISIT
Where does the Sun get its energy?
bit.ly/1azFmsM

MATERIALS:

- small pocket mirror or hand mirror
- piece of plain cardboard (or paper) to fit over the mirror (or alternatively tape)
- white cardboard screen
- bin bags or curtains for darkening the classroom

METHOD:

1. Cut the plain cardboard or paper so it fits over the mirror.
2. Cut or punch a very small hole, about 5 mm, in the middle of the plain cardboard.
3. If you do not have cardboard, you can use tape to cover all but a small portion of the surface of the mirror.
4. Place the mirror on a window sill in the Sun and tilt it so that it catches the sunlight and reflects it into the classroom. If your classroom is very small, placing the mirror outside on a chair may be a better option in order to get a larger image.
5. Darken the classroom using curtains or bin bags, excluding where the mirror is.
6. Reflect the sunlight from the mirror onto a wall of the darkened room.
7. Put the white cardboard or paper on the wall where the reflected light showing the Sun's image falls.
8. Observe the image of the Sun.

VISIT
E = mc^2 explained (video).
bit.ly/16mVFNI

9. Remove the white cardboard from the wall and take three steps towards the mirror with the cardboard still facing the mirror. Note what happens to the image of the Sun on the cardboard.

QUESTIONS:

1. As you moved the white cardboard screen closer towards the mirror, what did you notice happened to the image of the Sun?

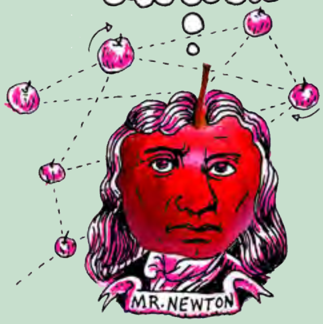
---

2. Draw a picture of what the surface of the Sun looks like on the white card in the circle below.

3. When the Sun reflects off the surface of the mirror, what can you say about the angle of incidence and the angle of reflection of the ray?

DID YOU KNOW?

Albert Einstein explained the mass-energy equivalence with the famous equation $E = mc^2$.

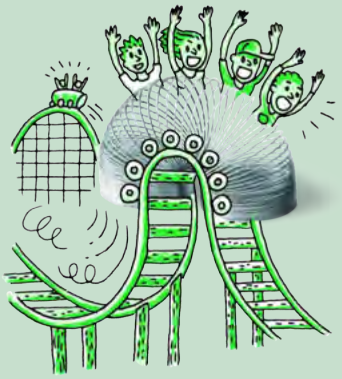


VISIT

Fiery looping rain on the Sun (video)

bit.ly/16qmriQ

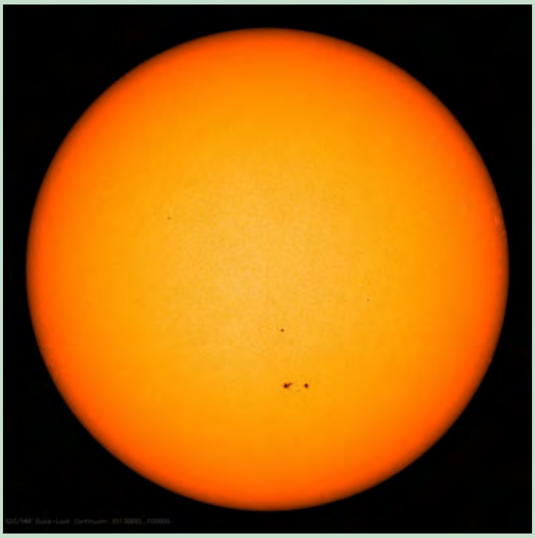
Did you notice any features on the Sun's surface when you viewed it in class? Let's find out what some of these surface features could have been in the next activity.



ACTIVITY: Observing sunspots on the Sun's surface

INSTRUCTIONS:

1. Look at the images of the Sun which were taken in June 2013.
2. Answer the questions that follow.

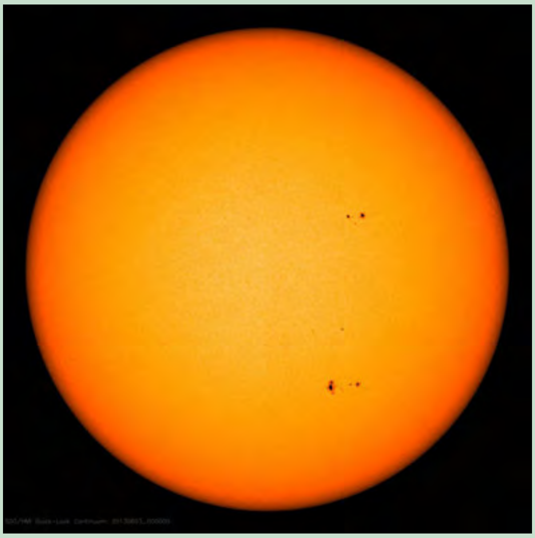


A: DATE: 02.06.2013

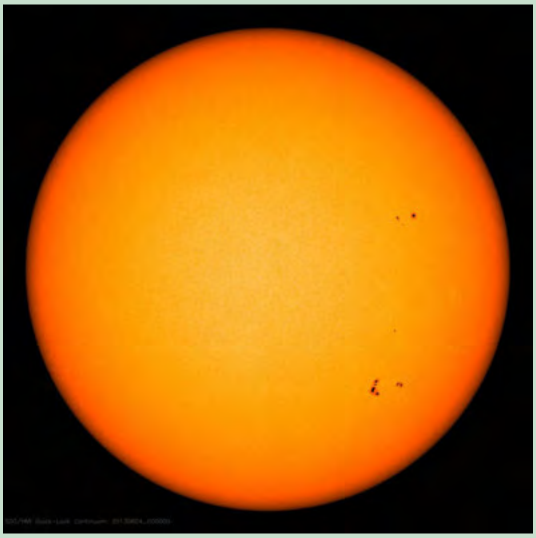
VISIT

Learn more about the research that NASA is doing about our Sun with the Solar and Heliospheric Observatory (SOHO).

bit.ly/1fQhd8u



B: DATE: 03.06.2013



C: DATE: 04.06.2013

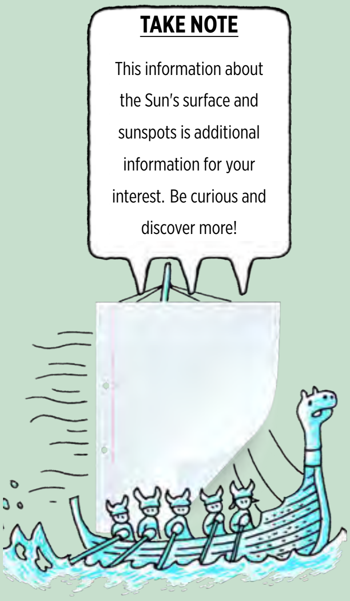
QUESTIONS:

1. How many groups of dark spots do you see in each image?

2. What do you notice about the positions of the spots in each image?

3. Why do you think the spots have moved?

4. What do you think these spots are?



Sunspots and the Sun's surface

The Sun's surface often has little blemishes on it. These dark spots on the Sun are called sunspots. They are areas that are slightly cooler than the rest of the Sun's surface. The Sun's surface is typically about 5500 °C and a typical sunspot has a temperature about 3900 °C.

VISIT

View real time images of the Sun and track sunspots.

bit.ly/19ZoU6c

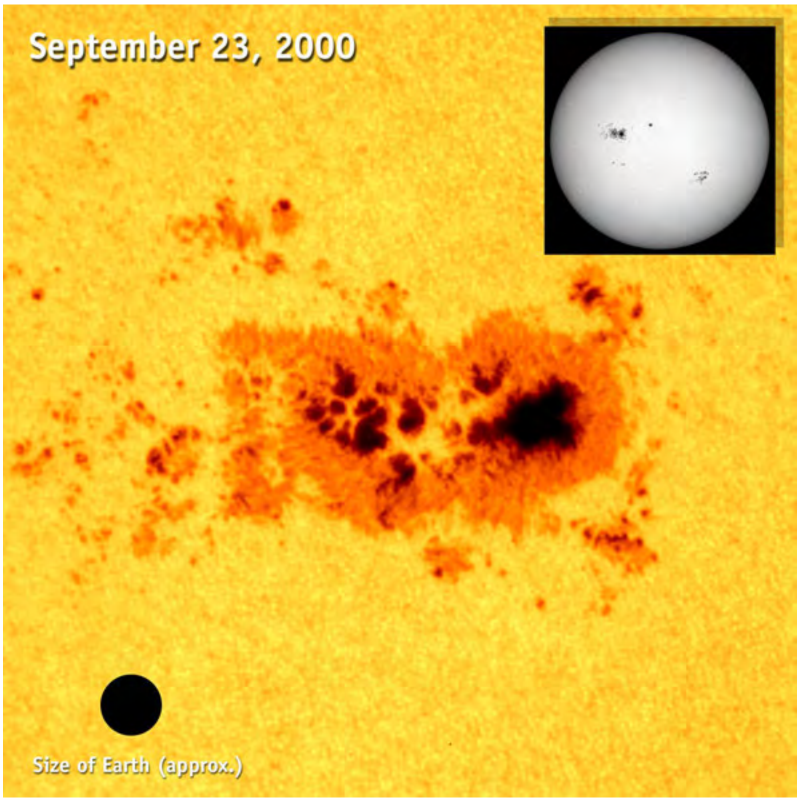


Image of a sunspot. For perspective, take note of the size of the Earth in the lower left.

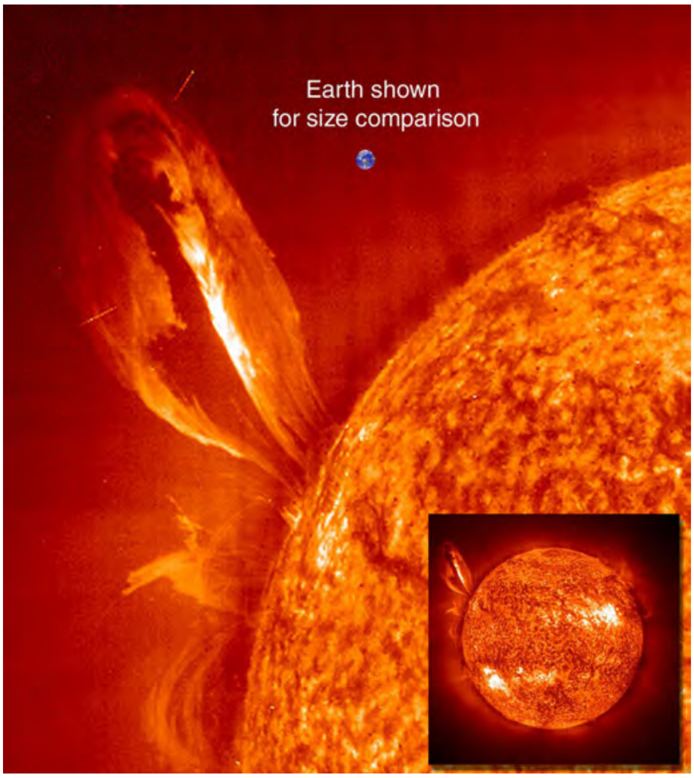
DID YOU KNOW?

The number of sunspots on the Sun increases and decreases in a regular pattern which repeats every 11 years. When there are more sunspots the Sun is more active and there are more solar storms and more of the Sun's energy reaches the Earth.

As the Sun is made up of gas, there is no solid surface like on Earth. So when one says that you are looking at the Sun's surface what are you actually looking at? Imagine that you are standing in thick fog (mist) with a friend. You can see things close to you, like your hand in front of you and your friend standing next to you. However, because the fog is so thick you cannot see far into the distance. Similarly, when we look at the Sun, we cannot see right into the centre of the Sun. As you go deeper and deeper in towards the centre of the Sun the gas begins to get thicker and thicker so that we cannot see through it. The deepest depth that we can see into the Sun's gas is what we call the Sun's surface.

Sunspots are areas that are slightly cooler, and therefore darker, than the rest of the Sun's surface. A typical sunspot only lasts a few days. When a sunspot lasts for several days you can observe it move across the Sun's disc. The sunspot appears to move across the Sun because the Sun is spinning slowly on its own axis.

The outer atmosphere of the Sun is called the **corona**. Gas particles from the corona are constantly escaping into space, forming the **solar wind**. When the Sun is very active, violent eruptions called solar flares occur on its surface.



A large loop of gas extending over 35 Earth diameters out from the Sun's surface.

VISIT

Explore the solar system from your computer with this 3D environment bit.ly/1c9rpbM and view any objects in the solar system with this interactive simulator bit.ly/1gyasJR

1.2 Objects around the Sun

NEW WORDS

- terrestrial planet
- gas giant
- dwarf planet

The Sun is by far the largest and most massive object in our solar system making up 98% of the total mass of the solar system. Due to the Sun's massive size, its large gravitational pull causes the planets and other objects in the solar system to orbit around it.

In orbit around the Sun are the eight planets along with their moons, dwarf planets and many much smaller objects like asteroids, Kuiper belt objects and comets. You will learn all about these objects later on in this chapter.

The four planets closest to the Sun are Mercury, Venus, Earth and Mars. These are called **terrestrial planets** because they have solid rocky surfaces. Further out, lie the **gas giants** Jupiter, Saturn, Uranus, and Neptune. These are much larger than the terrestrial planets and are mainly made of gas with small cores of rocky materials. In between the terrestrial planets and the gas giants lies the asteroid belt and out beyond the orbit of Neptune lies the Kuiper belt.

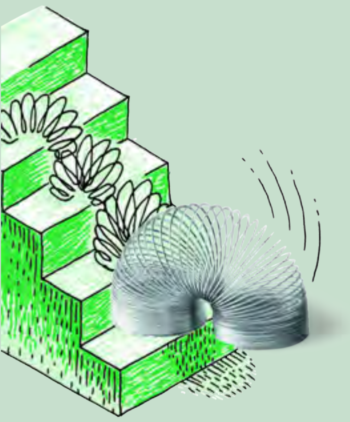
TAKE NOTE

'terra' is the Latin word for land or earth.

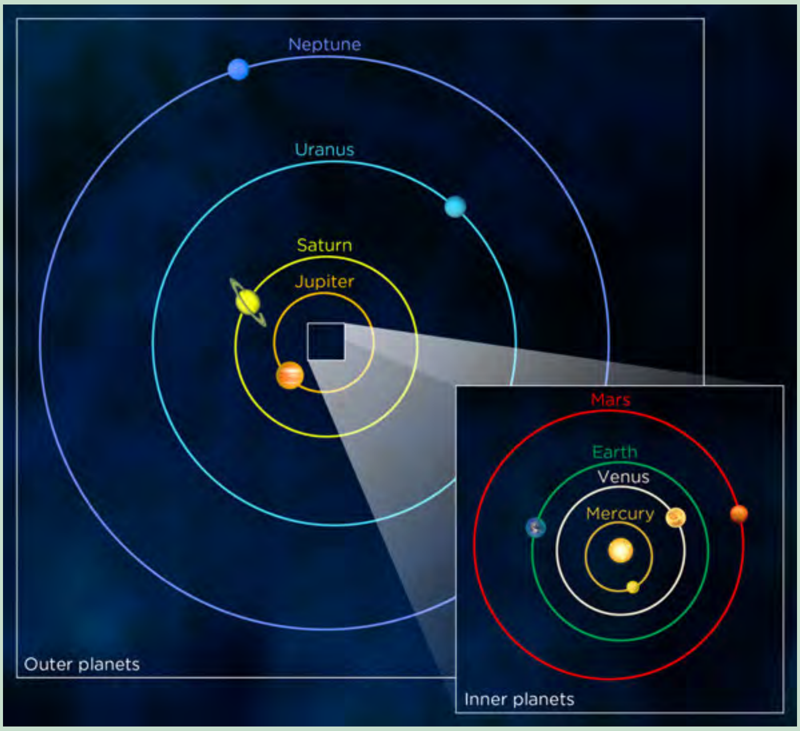
As you can see, there are lots of different types of objects orbiting the Sun, and not all of them are planets! To be classed as a planet, an object must:

1. orbit around the Sun
2. be large enough that its own gravity pulls it into a spherical shape
3. clear out smaller objects in its orbit, by either flinging them into another orbit or by attracting and then sticking them to itself (this means that there are no other similar sized objects orbiting in their vicinity)

You will learn about planets and the other objects orbiting the Sun in more detail later on in this chapter. Let's begin by learning more about the size and scale of the solar system.



ACTIVITY: The scale of the solar system



The orbits and planets in the solar system which we are going to model.

MATERIALS:

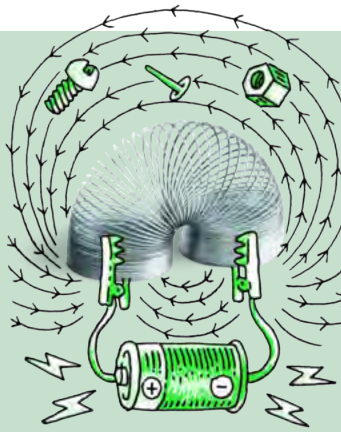
- grapefruit
- peppercorns
- salt grains
- poppy seeds
- pea
- grape
- measuring tape

INSTRUCTIONS:

1. Go outside to a large field for this activity. Start at one end of the field.
2. Put the grapefruit on the ground, this represents the Sun.
3. Measure 4.2 m away from the grapefruit and put a grain of salt on the ground. This represents Mercury. If you do not have a measuring tape then count four big strides away from the Sun instead.
4. Repeat this for each of the planets in the solar system. Your teacher will tell you the distance each planet lies from the Sun and will give you the appropriate object to represent your planet.
5. Guess how far away you think the next closest star after the Sun is.

Let's now make a smaller model of the solar system.

ACTIVITY: Make a hanging solar system



MATERIALS:

- cardboard about 30 cm across
- paper
- string or thread
- pair of scissors
- tape
- string
- pencil, crayons, or markers
- compass (for drawing circles)
- nail (for making a hole in the cardboard)

VISIT

Compare the planets using this tool from NASA.

bit.ly/16qoflJ

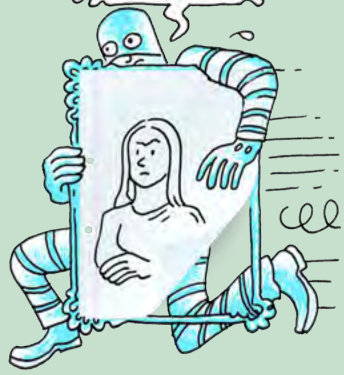
INFORMATION TABLE:

Object	Orbit radius (cm)	Object radius (cm)
Sun	-	5.0* - this is NOT to scale
Mercury	0.4	0.2
Venus	0.7	0.8
Earth	1.0	0.8
Mars	1.5	0.4
Jupiter	5.0	5.1
Saturn	9.2	4.1
Uranus	18.6	1.6
Neptune	29.1	1.6

* Note that if the Sun were drawn at the same scale as the rest of the planets, its radius should be 50 cm rather than 5 cm!

TAKE NOTE

The scale of the orbits differs from the scale of the object sizes in the table here. If they were on the same scale then the Sun and planets would be much much smaller.



INSTRUCTIONS:

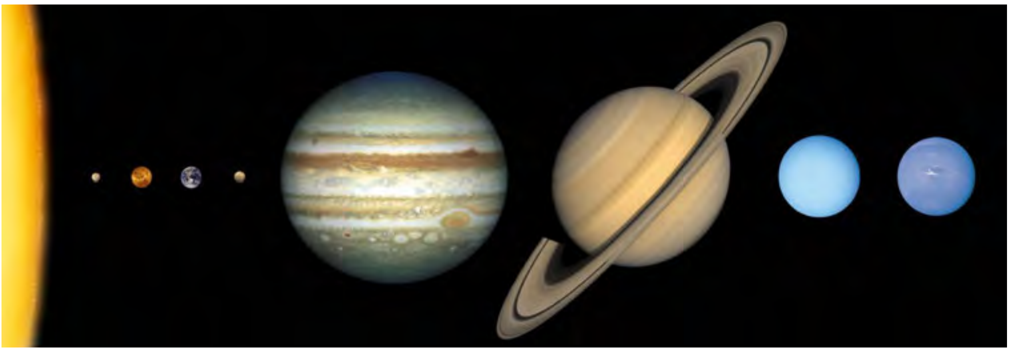
1. Cut out the cardboard into a circle of radius 15 cm. Use a compass and pencil to mark out the circle for cutting.
2. Mark the centre of the circle. This will be the position of the Sun.
3. Using a compass, draw the orbits of the 8 planets on the card. The first four planets orbit relatively close to the Sun, then there is a gap (the asteroid belt), then the last five planets orbit very far from the Sun. The radius of each circle, representing each planet's orbit, is shown in the table above.
4. Using the sharp point of the scissors' blade, or a large nail, punch a hole in the centre of the card (this is where the Sun will hang).
5. Punch one hole on each circle (orbit); a planet will hang from each hole.
6. Cut out one circle from the paper to represent the Sun.

7. Repeat this for each of the planets. The range in size of the Sun and the planets is far too large to represent accurately, so as a rough representation use the radii listed in the table to make your circles. The sizes of Mercury and Mars are very small in relation to the other planets. If you are battling to cut circles this size, then make them slightly bigger.
8. Colour in each planet and the Sun according to the pictures later in this chapter.
9. Tape a length of string or thread to the Sun and each planet.
10. Lace the other end of each string or thread through the correct hole in the large cardboard circle.
11. Tape the end of the string to the top side of the cardboard.
12. After all the planets and the Sun are attached, adjust the length of the strings so that the planets and Sun all fall to the same depth when the circle is held up in the air.
13. To hang your model, tie three pieces of string to the top of the cardboard around the edge. Then tie these three together and tie them to a longer string (from which you'll hang your model).

QUESTION:

Why did you adjust the string lengths so that the Sun and all the planets hang at the same height?

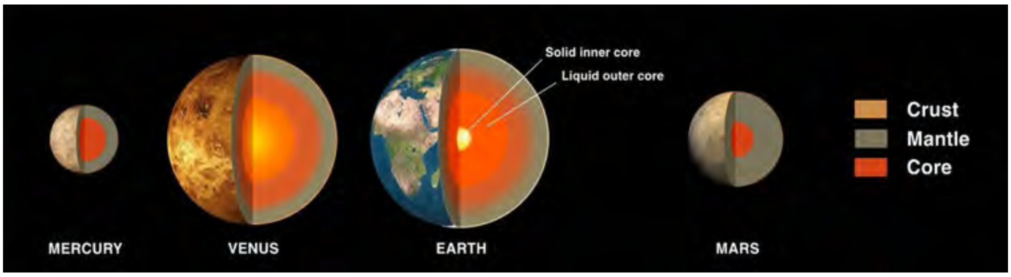
Now that you have an idea of the size and scale of the planets in our solar system, let's compare the two groups of planets, the inner worlds, Mercury, Venus, Earth and Mars with outer worlds, Jupiter, Saturn, Uranus and Neptune, in more detail. Look at the following pictures which compare the features of the two groups of planets.



The relative sizes of the terrestrial planets and gas giants, from left to right: Mercury, Venus, Earth, Mars, Jupiter, Saturn, Uranus and Neptune. Note that the planets are not spaced at equal separations from each other, but are shown in this way to fit on the page.

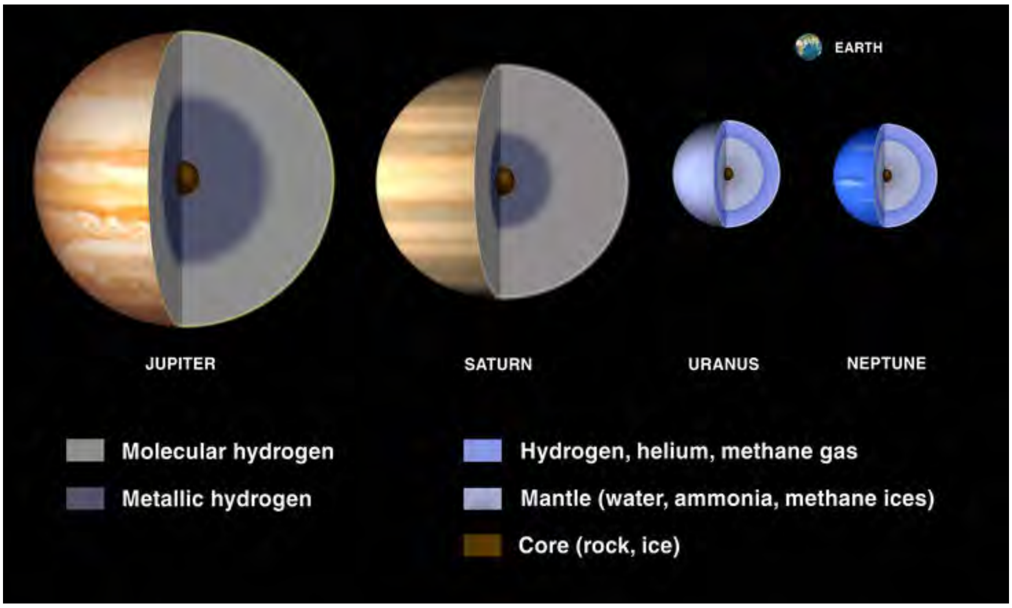
How do the sizes of the terrestrial planets and gas giants compare with each other?

Let's now look at the compositions of the two types of planets.



The above image shows the internal structure of the terrestrial planets. They all have a metal core, a rocky mantle and a thin outer crust. They also have a thin atmosphere (Mercury has an extremely thin atmosphere). The Earth's atmosphere is unique in the solar system in that it contains abundant oxygen, which is necessary to sustain life on Earth.

The image below shows the structure of the gas giants. They are mostly made of hydrogen and helium gases and are much less dense than the rocky terrestrial planets.



As you go deeper into the atmospheres of Saturn and Jupiter their atmospheres get denser and denser until they gradually become a liquid. This liquid hydrogen is called metallic hydrogen. Deeper down they have a solid core made of rocky materials.

Uranus and Neptune have thick atmospheres which have methane in addition to hydrogen and helium. The methane gives them their blue colour. Scientists think that below their atmospheres they have a slushy mantle made of water, ammonia and methane ices. At their centres they have a rocky-icy core.

Look at the pictures below. They show images of the gas giants. What features do you see that the gas giants all have in common?

DID YOU KNOW?

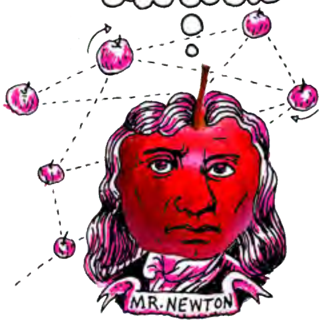
When it is winter on Mars you can see polar ice caps forming on the planet, like on Earth. However, unlike the Earth's polar ice caps which are made of frozen water, the ice caps on Mars are made of frozen carbon dioxide. This frozen carbon dioxide comes from Mars' atmosphere.

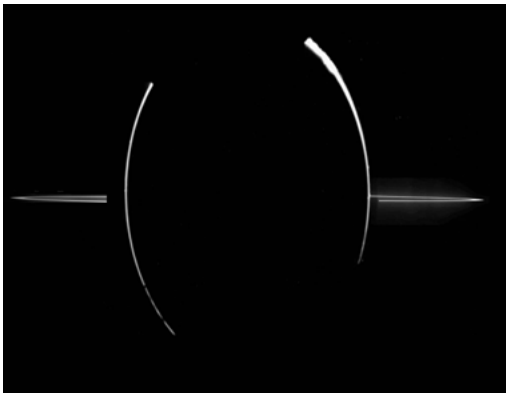
DID YOU KNOW?

Pluto was reclassified from planet to dwarf planet in 2006. Although Pluto orbits the Sun and is almost round, it has not cleared out other objects in its orbit, and so it cannot be classified as a planet. There are many more dwarf planets at similar distances from the Sun as Pluto.

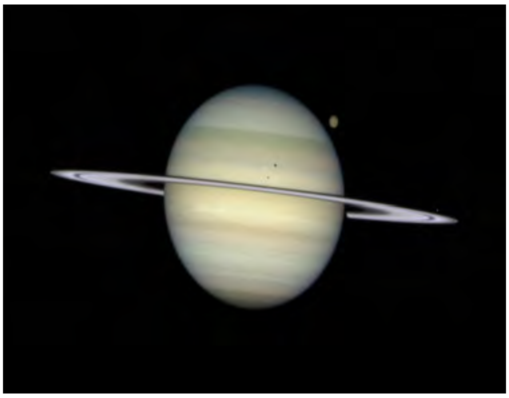
DID YOU KNOW?

Hydrogen is a liquid deep inside Jupiter and Saturn because the hydrogen molecules have been squeezed together due to the enormous pressure at those depths caused by the weight of the planet's atmosphere above.

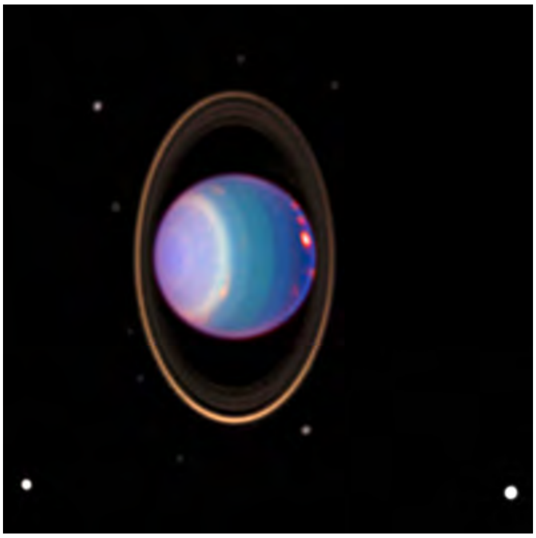




This image of Jupiter in shadow was taken by the space probe Galileo as it studied Jupiter in 1998.



This image of Saturn was taken with the Hubble Space Telescope. Can you see some of its moons?



Uranus, taken with the Hubble Space Telescope. What do you notice that is strange about Uranus?

VISIT

Discover more online at NASA's Solar System Exploration site.

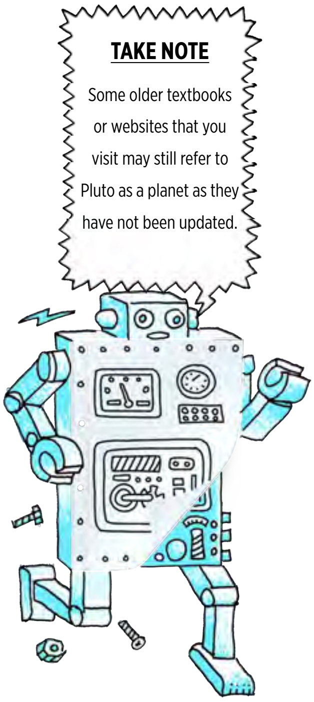
bit.ly/1azHL6M

Neptune is to the bottom right of this picture, just out of view. This image was taken by the space probe Voyager 2 as it flew past Neptune in 1989.

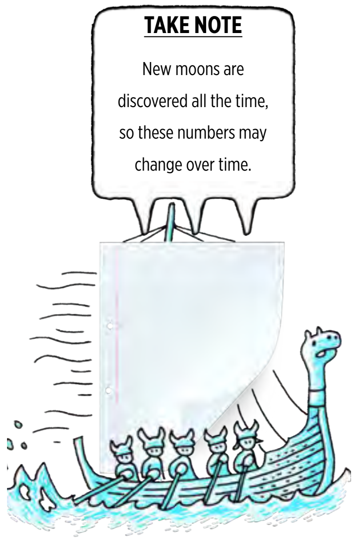
You can see that *all* the gas giants have rings. None of the terrestrial planets have rings.

Another difference between the inner rocky and outer gas giant planets, are the number of moons orbiting each planet. Look at the table below which shows the number of moons each planet in our solar system has.

Planet	Number of Moons
Mercury	0
Venus	0
Earth	1
Mars	2
Jupiter	67
Saturn	62
Uranus	27
Neptune	13



What can you say in general about the number of moons that the two types of planet have?



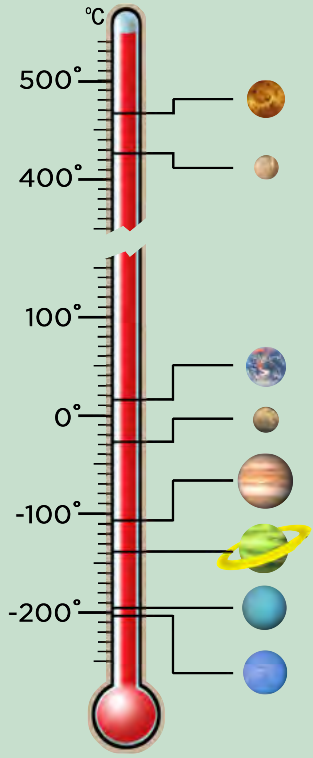
The terrestrial planets are much closer to the Sun than the gas giants. Because of this, the terrestrial planets orbit the Sun in less time than the gas giants, because they have a shorter distance to cover.

Lets see how the distance from the Sun affects the planets' temperatures.

ACTIVITY: Planetary Temperatures

INSTRUCTIONS:

1. Look at the table, it shows the surface temperatures of each of the planets.
2. Correctly label each of the planets on the thermometer using the temperature information provided in the table.



TAKE NOTE

Ice does not just refer to water ice, but other frozen elements and compounds too. Also, the rocky-ice materials do not resemble any rock or ice you would see on Earth, since the temperatures and pressures on these planets and gas giants are much, much higher.

Planet	Temperature (°C)
Mercury	167
Venus	464
Earth	15
Mars	-65
Jupiter	-110
Saturn	-140
Uranus	-195
Neptune	-200

QUESTIONS:

1. Which planet has the lowest average temperature?

___

2. Why do you think this is?

___

3. What do you notice about the average temperatures of the terrestrial planets compared with the gas giants?

___

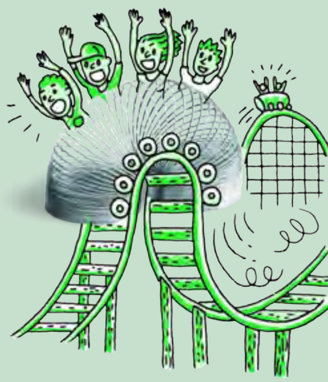
4. If you exclude Venus, how does the ordering of the planets from the Sun compare with their average temperature?

___

___

Clearly the terrestrial planets and gas giants have very different properties. Let's compare them.

ACTIVITY: Comparing terrestrial planets and gas giants



INSTRUCTIONS:

1. The table below compares the two types of planet. Fill in the missing gaps.

Terrestrial Planets	Gas Giants
close to the Sun	___ from the Sun
closely spaced orbits	widely spaced orbits
small masses	large masses
small radii	___ radii

Terrestrial Planets	Gas Giants
mainly rocky	mainly ___________________
solid surface	___________________ surface
high density	___________________ density
slower rotation	faster rotation
___________________ moons	many moons
___________________ rings	many rings
thin atmosphere	___________________ atmosphere
warm	___________________

Why do you think the two types of planets are so different?

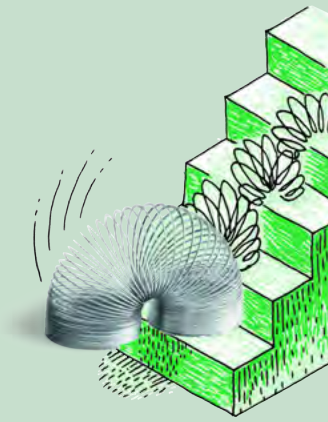
____

____

When the solar system was forming, the difference in temperature across the early solar system caused the inner planets to be rocky and the outer ones to be gaseous. Close to the Sun it was hot and only materials with very high melting points, such as metals, could remain solid and form planets. Further away from the Sun, where it was cold, compounds like water and methane were frozen. Astronomers call these frozen compounds *ices*. Therefore the cores of the gas giants contain rocky and icy compounds. As the abundance of metals in the universe is very small, the inner planets are much smaller than the gas giants. The gas giants could also attract large amounts of hydrogen and helium to their atmospheres due to their size.

Let's continue to compare the rocky planets and the gas giants.

ACTIVITY: Comparing the inner and outer planets



INSTRUCTIONS:

Use the information in the table below to answer the questions that follow.

Planet	Density (kg/m^3)	Diameter (km)	Distance from the Sun (million km)	Day length (hours)	Year length (Earth days)
Mercury	5427	4879	57.9	4222.6	88
Venus	5243	12104	108.2	2802.0	224.7
Earth	5514	12756	149.6	24.0	365.25
Mars	3933	6792	206.6	24.7	687.0
Jupiter	1326	142984	740.5	9.9	4331
Saturn	687	120536	1352.6	10.7	10747
Uranus	1271	51118	2741.3	17.2	30589
Neptune	1638	49528	4444.5	16.1	59800

QUESTIONS:

1. Given that the density of water is 1000 kg/m^3, which of the planets would float on water? Explain your answer.

____

____

2. Compare the densities of the rocky planets and the gas giants. Which type of planet tends to be more dense? Explain why.

____

____

3. Which planet has the shortest day?

____

4. Compare the day length for the rocky planets and the gas giants. Which type of planet tends to have the shortest day? What does this tell you about how fast the two types of planet rotate on their axis?

____

____

5. Which planet orbits around the Sun the fastest? Why is this?

6. Which planet's year is shorter than its day?

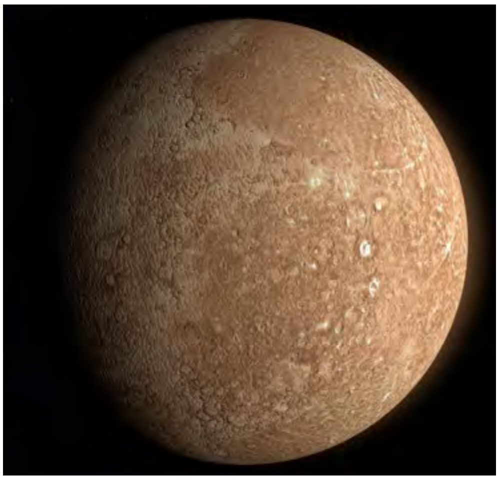
7. Plot a bar graph to show the distance each planet is from the Sun. Use the following space for your graph.

VISIT

Solar System 101: NASA's 'Homework helper' can show you where to look to find out more.

bit.ly/H6nbDD

Mercury



Mercury, imaged by the Messenger spacecraft, is covered with craters like our Moon.

- Mercury's atmosphere is very thin and constantly being lost into space because the planet's gravity is too small to hold onto it.
- Mercury has the most extreme temperatures in the solar system, reaching 426 °C during the day and -173 °C during the night.

TAKE NOTE

The following pages provide some interesting, extra information about the planets in our solar system.

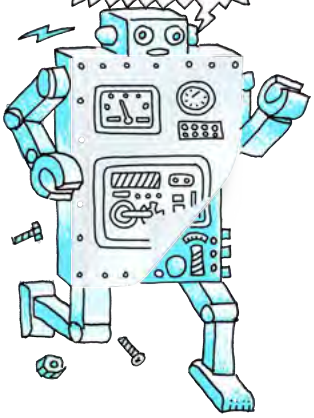
Venus

The surface of Venus in false colour (bottom left) and the top of the atmosphere (top left) as seen with the Magellan spacecraft.

- Venus is the hottest planet in the solar system, the temperature is hot enough to melt lead!
- Venus has clouds of sulphuric acid.
- Venus rotates in the opposite direction to all the other planets.

TAKE NOTE

Venus has a thick dense atmosphere mostly made up of carbon dioxide which is an effective greenhouse gas. This is why Venus has the highest surface temperature, as you saw in the activity of Planetary Temperatures.



DID YOU KNOW?

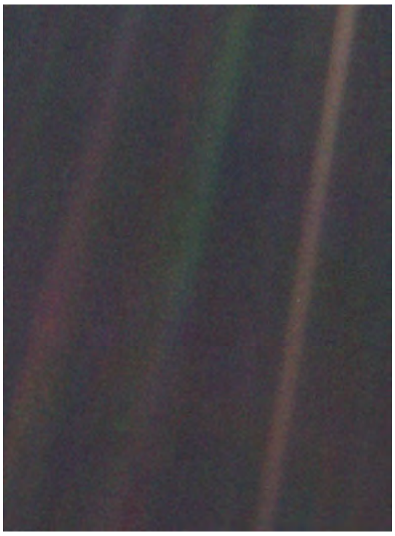
As Voyager 1 was leaving the solar system, Carl Sagan, a famous astronomer, requested that they turn the camera around to take a photograph of Earth across a great expanse of space.

VISIT

Carl Sagan - Pale Blue Dot (video)

bit.ly/1h0msBx

Earth



This famous image is a photograph taken of Earth in 1990 by Voyager 1 from 6 billion kilometers away. Earth appears as a tiny dot (the blueish-white speck approximately halfway down the brown band to the right). The coloured bands are scattered light rays from the Sun.

- To date, Earth is the only planet in the universe known to harbour life.
- The average distance between the Sun and Earth is called an *astronomical unit* (AU) and is equivalent to 150 million kilometres.

Mars

Mars is nicknamed the Red Planet because of its red surface, as the rocks on are rich in iron. The white smudges in the middle are water-ice clouds.

- Mars' surface is like a dry red desert. Mars has mountains, volcanoes and valleys just like Earth.
- Mars is home to the deepest and longest valley in the solar system, *Valles Marineris*, which is almost as wide as Australia!

DID YOU KNOW?

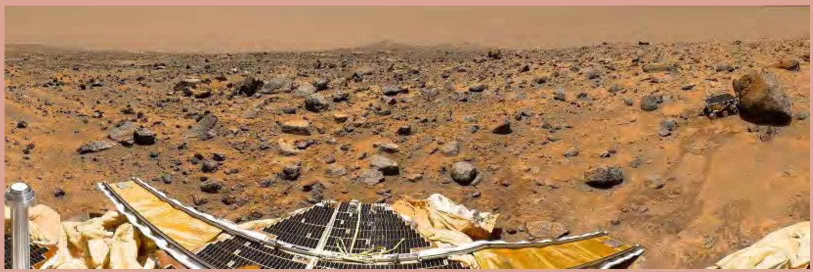
The largest volcano in the solar system, Olympus Mons, is on Mars and is three times taller than Mount Everest.

Mars and the Search for Life

Scientists are interested in Mars because they think that Mars might have once had liquid water on its surface, and perhaps life. Channels, valleys, and gullies are found all over Mars, suggesting that liquid water might have once flowed through them. Although there is no liquid water on the planet's surface now, scientists think that there may still be some water in cracks and tiny holes in underground rock. Mars has been visited many times by robotic landers.

The first lander, NASA's Viking 1, landed on Mars in 1976, a long time before you were born! It took the first close-up pictures of the Martian surface but found no evidence of life. Water ice has been discovered below the planet's surface, and minerals indicating that liquid water was once present have also been found by Mars landers. The latest lander currently exploring Mars is NASA's Mars Science Laboratory mission, with its rover named Curiosity. Curiosity landed on Mars in August 2012 and is busy investigating the planet's rocks near a giant crater called the Gale crater. One of the main aims of the Mars Science Laboratory is to determine whether Mars ever had an environment capable of supporting life.

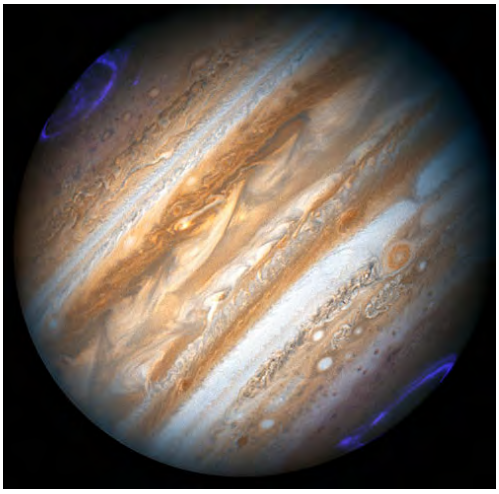
The Curiosity rover.



One of the first colour images of Mars' surface taken by the Curiosity rover. You can see part of the rover at the bottom of the photograph.

VISIT
Saturn close-up (video).
bit.ly/15XvvyG

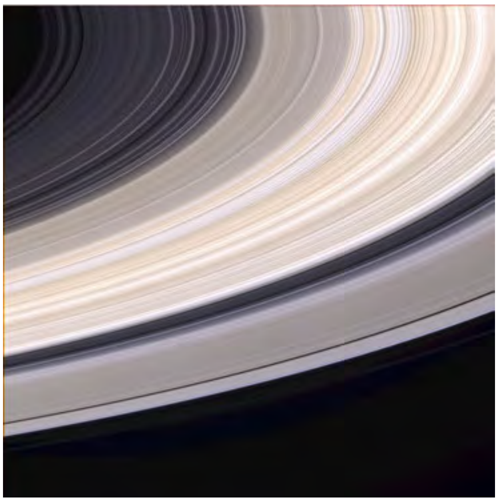
Jupiter



Magnetic storms cause the aurorae seen on Jupiter near its poles.

- Jupiter's diameter is over ten times the Earth's diameter.
- Jupiter rotates slightly faster at the equator (remember it is not a solid object, but a large ball of gas).
- Jupiter's famous great red spot, is a giant hurricane that has been raging for at least 300 years. This storm's area is larger than the Earth.

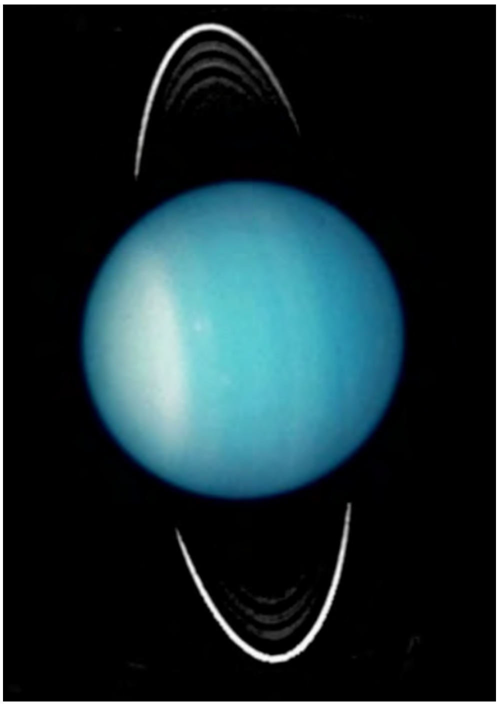
Saturn



Saturn's beautiful rings, imaged with the Cassini spacecraft.

- Saturn would float on water if you had an ocean large enough.
- Saturn is famous for its rings. The rings are over 200 000 km wide and only a few tens of metres thick.

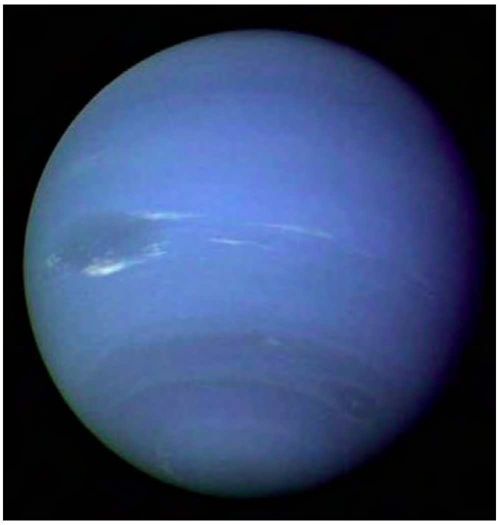
Uranus



Uranus spins on its side. Scientists think Uranus may have been knocked on its side by a collision with a large object early in its history.

- Uranus is believed to have an ocean of liquid water, ammonia, and methane above a rocky core.
- Uranus was the first planet discovered using a telescope.

Neptune



Neptune and its "Great Dark Spot" (middle left). This is a giant storm that was raging on the planet until very recently. The winds reached nearly 1931 km/hour.

- Neptune has the strongest winds in the solar system. With storm winds recorded at over 10 times that of hurricanes on Earth.
- Neptune has the most methane in its atmosphere out of all the gas giants, which gives it its blue colour.

ACTIVITY: Planetary holidays

In this activity you will write a travel brochure for a trip to your favourite planet.

MATERIALS:

- information about the planets
- pictures of the planets
- example travel brochures

INSTRUCTIONS:

1. Research information about your chosen planet.
2. Write a travel brochure for a trip to your chosen planet. Include real facts about the planet and think about what unusual things you could see and do on the planet.

ACTIVITY: Planet fact sheet

In this activity you will make a one page fact sheet about your chosen planet.

MATERIALS:

- information about the planets
- pictures of the planets

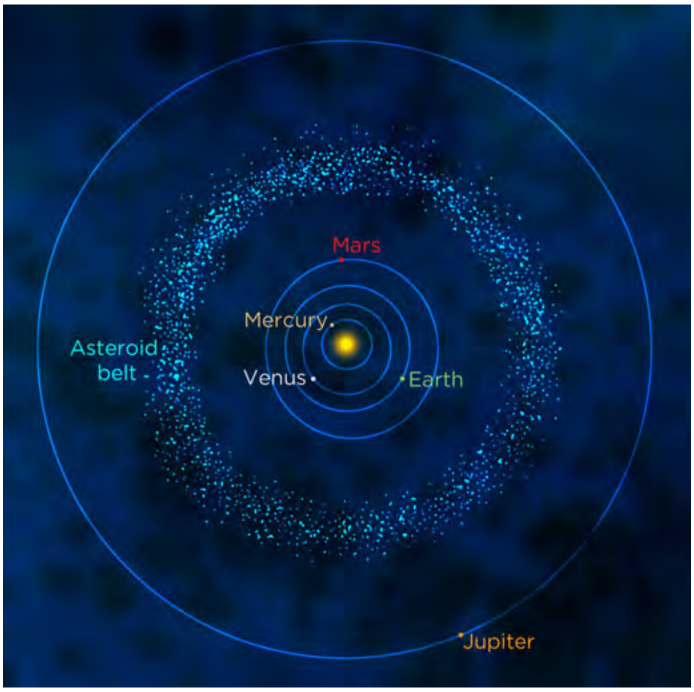
INSTRUCTIONS:

1. Research information about your chosen planet.
2. Write a one page fact sheet about your chosen planet.

Let's now look at some of the other objects that we find in our solar system.

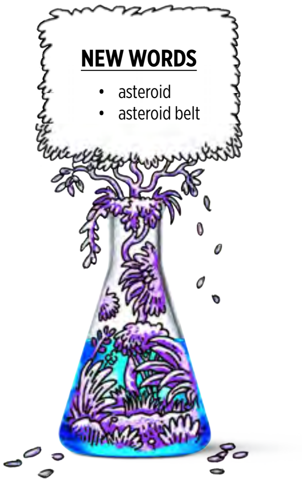
Asteroids

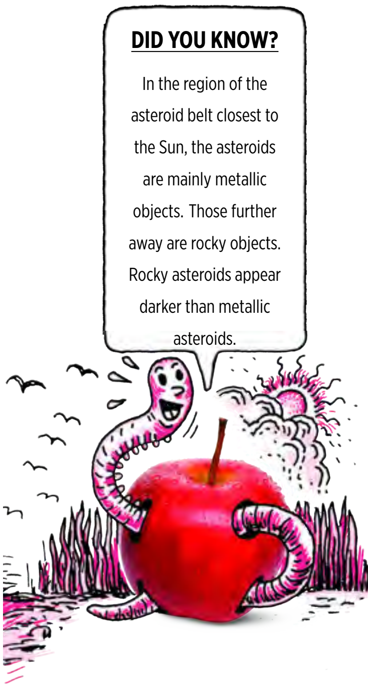
Asteroids are small rocky objects that are believed to be left over from the formation of our solar system 4.6 billion years ago. They range in size from tens of metres across to several hundred kilometres across and come in a variety of shapes. Most asteroids are found in the **asteroid belt**, which lies between the orbits of Mars and Jupiter. More than 100,000 asteroids lie in the asteroid belt and several thousand of the largest ones have been named.

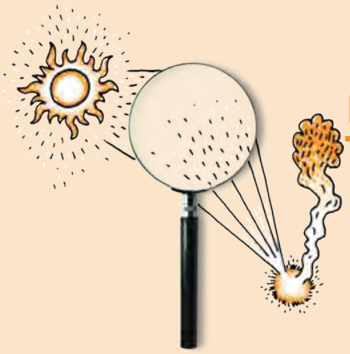


An image of asteroid 951 Gaspra taken with the Galileo spacecraft 5300 kilometres away. Gaspra is 19 x 12 x 11 km. Notice how the asteroid's surface has many craters.

Although science fiction movies give the impression that the asteroid belt is a tightly packed region of dangerous rocks, in reality the asteroids are separated from each other by millions of kilometres. However, very rarely, collisions between asteroids do occur which is why asteroids are covered with impact craters. We will look at impact craters more closely in the following activity.







INVESTIGATION: Impact craters

INVESTIGATIVE QUESTIONS: How does the mass of an object affect the size of the crater it leaves? How does the height at which an object is dropped affect the size of the crater it leaves?

HYPOTHESIS:

What do you think will happen?

___

___

IDENTIFY VARIABLES:

1. What are you keeping constant in this experiment?

___

___

___

2. What are you changing in this experiment?

___

___

MATERIALS:

- deep tray or large plastic container
- measuring scales
- ruler
- sand
- a marble
- a ball bearing
- chair or step ladder
- measuring tape (at least 2 m long)

METHOD:

1. Fill the tray or plastic container with sand to a depth of 10 cm.
2. Smooth the surface of the sand using the long edge of a ruler.
3. Measure the mass of the marble and record it in the table below.
4. Drop the marble from a height of 1 m into the tray of sand and observe the crater that forms.
5. Carefully remove the marble, without disturbing the shape of the crater and measure the diameter of the crater using the ruler.
6. Record the diameter of the crater in the table below.
7. Smooth the sand.
8. Repeat steps 3-7

9. Measure the mass of the ball bearing and record it in the table below.
10. Drop the ball bearing from a height of 1 m into the tray of sand and observe the crater that forms.
11. Carefully remove the ball bearing and measure the diameter of the crater using the ruler.
12. Record the diameter of the crater in the table below.
13. Smooth the sand.
14. Repeat steps 9 -13.
15. Drop the ball bearing into the sand from a height of 2 m. You may need to stand on a chair or step ladder to do this.
16. Record the size of the crater formed in the table below.
17. Smooth the sand.
18. Repeat steps 15-17, dropping the ball bearing from heights of 1.5m, 0.5m and 0.25m. Record all your measurements in the table below.
19. If you have time you can make repeated measurements.

RESULTS AND OBSERVATIONS:

Record your results and observations in the following table.

Object	**Mass (kg)**	**Drop Height (m)**	**Crater diameter - reading 1 (cm)**	**Crater diameter - reading 2 (cm)**	**Average crater diameter (cm)**
marble		1			
ball bearing		1			
ball bearing		2			
ball bearing		1.5			
ball bearing		0.5			
ball bearing		0.25			

EVALUATION:

How reliable was your experiment? How could it be improved?

CONCLUSIONS:

Write a conclusion for this investigation based on your results.

QUESTIONS:

1. How did the mass of the object affect the size of the crater?

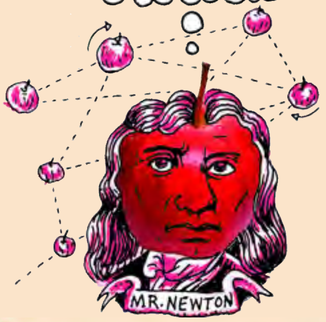
2. How did the height at which the object was dropped affect the size of the crater?

3. Why do you think the drop height affected the size of the crater?

4. What does this investigation tell us about craters on the surfaces of planets?

DID YOU KNOW?

As Jupiter is more massive than all the other planets in the solar system, its large gravity attracts many asteroids and comets travelling in towards the inner solar system that would otherwise potentially crash into the Earth.

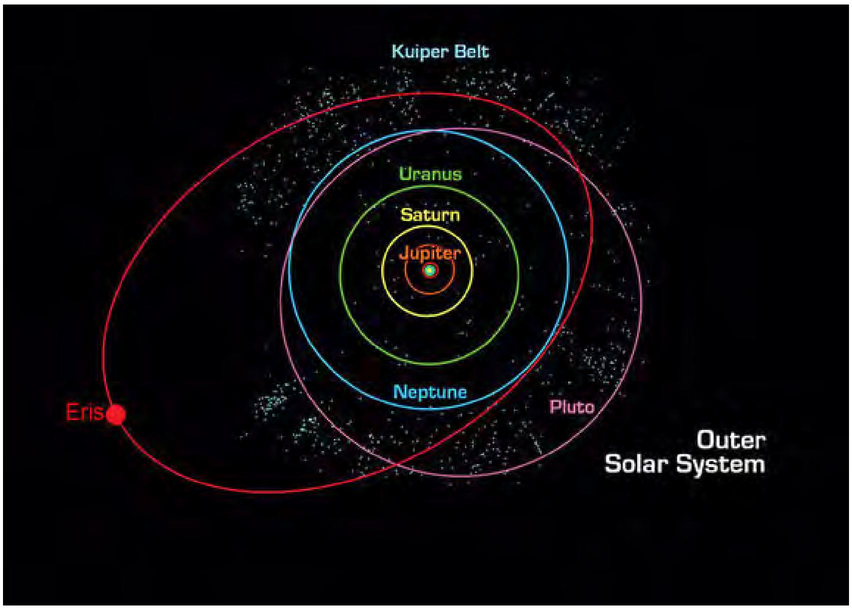


NEW WORDS

- Kuiper Belt
- Kuiper Belt object
- dwarf planet
- comet
- Oort Cloud

Kuiper Belt objects

The Kuiper belt is a region of space filled with trillions of small objects that lies in the outer reaches of the solar system, past the orbit of Neptune. The Kuiper belt is a region between 30 and 50 times the Earth's distance from the Sun. This belt is similar to the closer asteroid belt, except that the objects are not made of rock, but rather of frozen ices. These icy objects can range in size from a fraction of a kilometre to more than a 1000 km across and are called Kuiper belt objects. The two largest known members of the Kuiper Belt are Eris and Pluto, both dwarf planets.



The Kuiper belt (the pale blue dot dots) is shown beyond the orbit of Neptune. Its members include the dwarf planets Pluto and Eris.

What keeps the objects in the Kuiper Belt in orbit around the Sun?

Dwarf planets

Dwarf planets are objects that orbit the Sun, just like the planets. However, they are smaller than planets. Due to their small size, they are unable to meet the official definition of a planet. Can you remember what the three criteria are to be classed as a planet? List them below.

To be classed as a planet an object must:

Asteroids are clearly not planets as they have irregular shapes and they are not spherical. Some dwarf planets are spherical, but they do not meet the third criterion. With their weak gravities they are unable to clear out other objects from their orbits. Which famous ex-planet is now considered a dwarf planet because it failed to meet the third criterion?

For many years the object Pluto was considered to be a planet. However, since the 1990s many more objects very similar to Pluto have been discovered orbiting the Sun out past Neptune's orbit. This resulted in new criteria to be drawn up to be considered a planet and Pluto was demoted to dwarf planet status

VISIT

Gerard Kuiper (1905 - 1973) is regarded by many as the father of modern planetary science. He is well known for his many discoveries. Read more about them here.

bit.ly/16mZX7C

DID YOU KNOW?

NASA launched a space probe called New Horizons to study Pluto and other Kuiper Belt objects in more detail in 2006. It will arrive at Pluto in 2015.

VISIT

Why Pluto is not a planet anymore (video).

bit.ly/1fQiWLd

VISIT

Read more about dwarf planets.

bit.ly/H6nJtd

This image shows the five dwarf planets that have been discovered to date, Pluto, Haumea, Makemake, Eris and Ceres in relation to the size of the Earth. Some even have their own moons, which are shown. Ceres is in the asteroid belt and the other four are in the Kuiper Belt.

DID YOU KNOW?

Scientists estimate that there might be 200 or more dwarf planets in the Kuiper Belt, and thousands more beyond the Kuiper Belt.

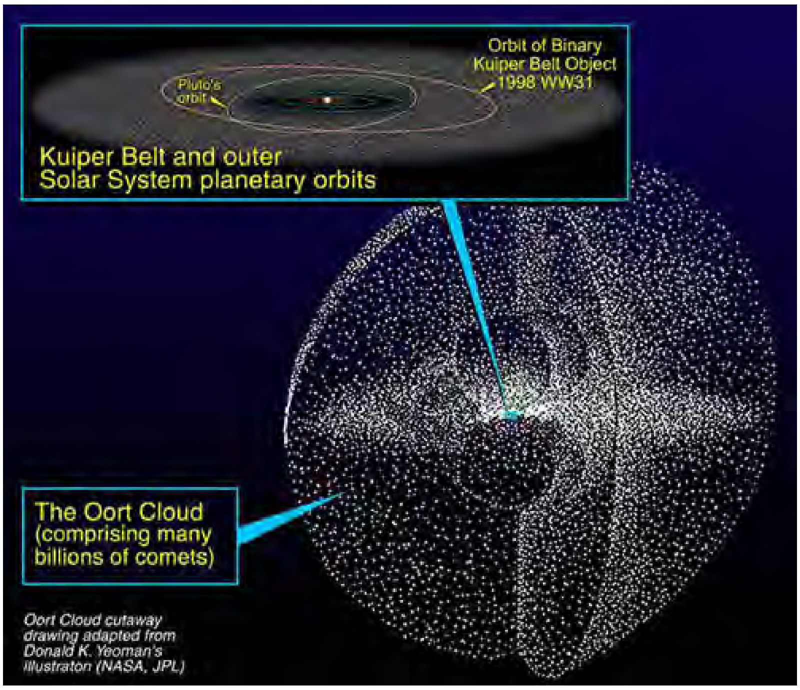
Comets and the Oort Cloud

Comets are icy, dusty objects, orbiting around the Sun at great distances. Comets are found in the Kuiper Belt and in the predicted **Oort Cloud**. The Oort Cloud is thought to be a huge cloud of icy objects surrounding the Sun at the very edge of our solar system at a distance between 5,000 and 100,000 times the Earth's distance from the Sun!

A comet will remain in the Kuiper Belt or Oort Cloud unless it is disturbed by another comet. If this happens, then the comet's orbit changes and occasionally the comet will come into the inner solar system for us to see.

DID YOU KNOW?

Pluto was named by Venetia Burney, an 11 year old from Oxford, England, in 1930. She suggested that the planet be named after the Roman god of the underworld, Pluto.



The hypothetical Oort Cloud is a huge cloud of icy objects or comets surrounding the outer reaches of our solar system.

We can only see comets directly when they come into the inner solar system because they are small and only visible by reflected sunlight. As a comet approaches the Sun, the Sun's heat evaporates the dust and ices it consists of, forming a bright dust tail which is visible from Earth. Some comet dust tails can be millions of kilometres long. The dust tail usually points back along the path of the comet.

Comets often have a second tail called an ion tail. The ion tail is made of ions that are pushed away from the comet's head by particles emitted from the Sun's atmosphere, called the solar wind. Let's find out more about this type of tail.

ACTIVITY: A comet's ion tail

In this activity you will make your own comet and explore how a comet's ion tail moves.

MATERIALS:

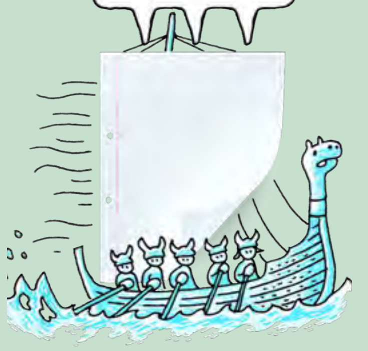
- table tennis ball
- sellotape
- tissue paper or crepe paper
- scissors

INSTRUCTIONS:

1. Cut the tissue paper or crepe papers into several strips (at least 4) about 1 cm wide by about 15 cm long.
2. Attach the paper strips to the ping pong ball, evenly spread around the equator of the ball using the sellotape. Wrap the sellotape around the ball a few times if needed to secure the paper in place. You have now made your comet and ion tail.
3. Hold out your comet in front of you and blow on the ball hard so that the ion tail is blown away from you. You are representing the Sun and your breath represents the solar wind, blowing on the comet's ion tail.
4. Continuing to blow fairly hard on the ball, move the ball from left to right and observe which way the paper moves.

TAKE NOTE

An **ion** is an atom with an electrical charge due to the gain or loss of electrons.



QUESTIONS:

1. Which direction did the ion tail move when you held up the comet in front of you and blew on the comet?

____

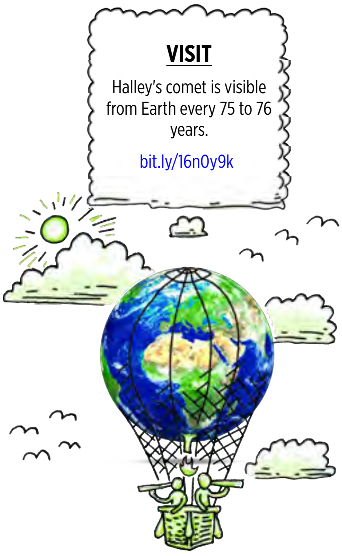
2. Which direction did the ion tail move when you moved the ball left and right while still blowing?

____

____

In a similar way, a comet's ion tail always points away from the Sun.

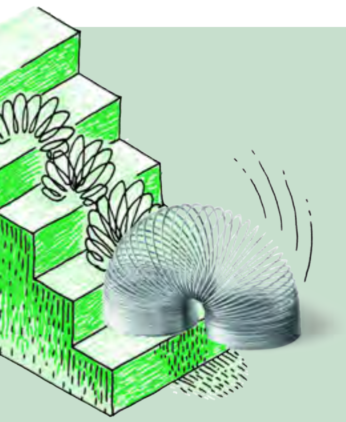
Comet West, photographed in 1995. Here you can see that the comet actually has two tails. The white tail is the dust tail and the blue tail is the ion tail made of charged particles evaporated from the comet's surface.



Comets that come into the inner solar system do not live forever. The Sun's heat melts comets, just like a snowman melts out in the Sun. After several thousand years the remains are so small that they no longer form a tail. Some comets completely melt away.

1.3 Earth's position in the solar system

As you discovered in the last section, the Earth, along with the other planets, orbits around the Sun. The Earth is the third most distant planet from the Sun, lying in between Venus and Mars. Let's compare the Earth and its two neighbours in more detail.



ACTIVITY: The Sun's Habitable Zone

Property	Venus	Earth	Mars
Distance from Sun (AU)	0.7	1.0	1.5
Average Temperature (°C)	464	15	-63

MATERIALS:

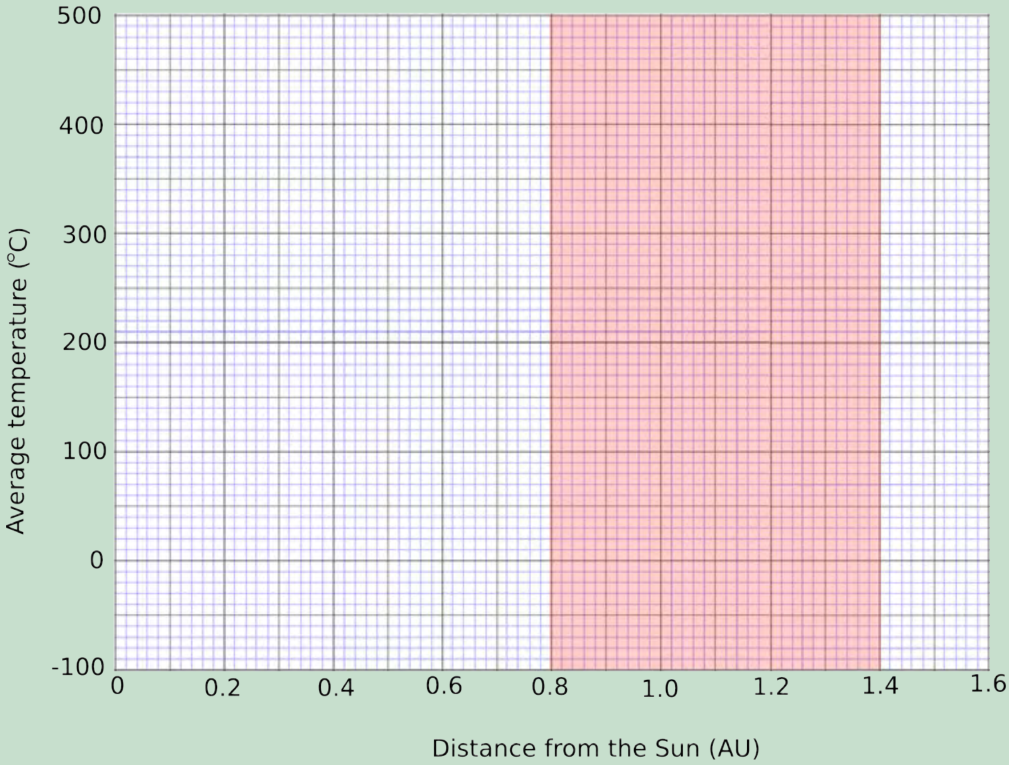
- pencil
- ruler

INSTRUCTIONS:

1. Look at the data provided in the table. It shows the distance from the Sun for three planets (in units of one Earth-Sun distance or Astronomical Unit). It also shows the average temperature on each planet in degrees Celsius.
2. Plot a graph to show the data in the table. Mark each point with an X.

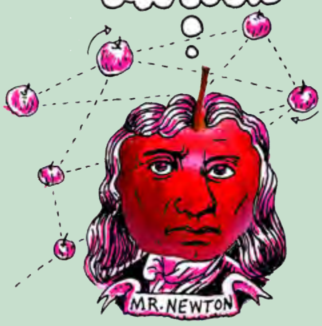
3. The Sun's habitable zone extends from 0.8 to 1.4 AU and is shaded in red in the graph paper. This is the region where scientists think a planet has to lie in order for there to be life on the planet.

Graph showing the average temperature and the distance from the Sun of Venus, Earth and Mars.



DID YOU KNOW?

It is completely safe to fly through a comet's tail. The only thing that would hit your space ship would be microscopic pieces of dust.



QUESTIONS:

1. What is the average temperature on Venus?

2. Can liquid water exist on Venus? Why?

3. What is the average temperature on Mars?

4. Is liquid water likely to be found on Mars? Why?

5. What is the average temperature on Earth?

VISIT

Our solar system's habitable zone (video)

bit.ly/15XwDT1

6. Can liquid water exist on Earth? Why?

7. Which planet/s lie within the Sun's habitable zone (the red shaded region in the graph)?

TAKE NOTE

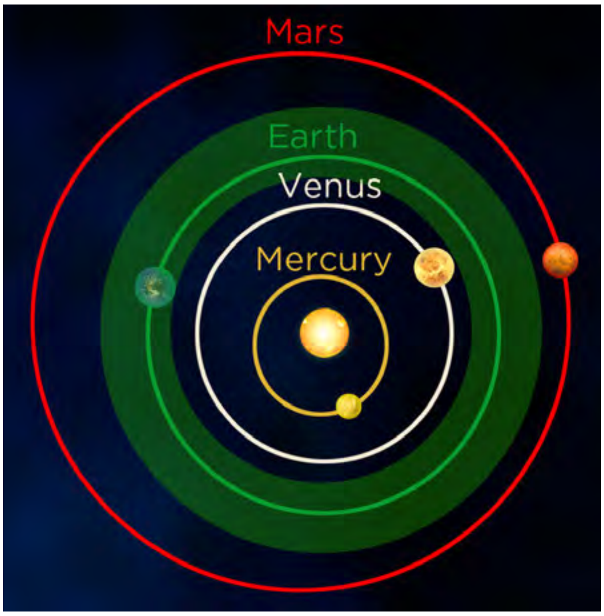
An astronomical unit corresponds to the average distance between the Earth and the Sun.

The average temperature on Earth is a moderate 15 °C. Because of this, water can exist in liquid form on Earth. *This is important because scientists think that liquid water is one of the key things needed for life.* Venus has an average temperature of 464 °C and no liquid water exists on Venus because it is too hot. On Mars, the opposite is true. The average temperature on Mars is -63 °C and any water on Mars would be frozen. Earth is unique in our solar system as it is the only planet known to have liquid water on its surface and to harbour life.

If the Earth were too close to the Sun it would be too hot and all the water would evaporate from the oceans, like it has on Venus. If the Earth were too far from the Sun it would be too cold, and all the water would be frozen, like on Mars. Earth is at just the right distance from the Sun to have liquid water on its surface. The other planets in the solar system are either too close or too far from the Sun. The range of distances that a planet can lie from the Sun and still have liquid water on the planet's surface is called the **habitable zone**. Estimates for the habitable zone in our solar system range from 0.8 - 1.4 astronomical units (AU).

DID YOU KNOW?

The habitable zone is sometimes called the Goldilocks zone after the famous children's story where Goldilocks eats the porridge that is neither too hot nor too cold.



Our Sun's habitable zone (light green). The Earth is the only planet in our solar system which lies within our Sun's habitable zone. It is just the right distance from the Sun for liquid water to remain on the planet, something which scientists think is essential for life.

What other conditions do you think are necessary for life on Earth or other planets? List your answers in the space below.

TAKE NOTE

Other stars also have habitable zones. Scientists believe that planets orbiting other stars within the habitable zone could support life forms.

Scientists think that in order for life to arise and survive on a planet:

- there must be sunlight for plants to grow.
- the planet must be located in the habitable zone of a star so that there are moderate temperatures and liquid water.
- there must be oxygen for respiration.

Which of the planets in the solar system receive light from the Sun?

VISIT

Kepler Mission: A search for habitable planets.

bit.ly/HdBXYI

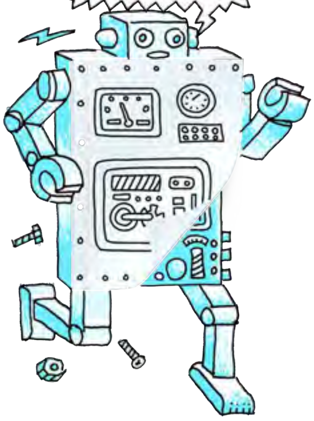
Which of the planets in the solar system have moderate temperatures and liquid water on their surface?

Which of the planets in the solar system have significant amounts of oxygen in their atmosphere or oceans?

As you can see the Earth is very fortunate, because it lies at just the right distance from the Sun to have moderate temperatures and abundant liquid water. The Sun provides the energy for plants to grow. There is plenty of oxygen in Earth's present day atmosphere and oceans, which means that life can survive on land and in the Earth's oceans. The Earth is unique in that it is the only planet we know of that has life.

TAKE NOTE

You may have heard a lot about global warming and the greenhouse effect in the news and in our studies in Energy and Change.



The greenhouse effect

During the day, the Sun shines through the atmosphere heating the Earth's surface. At night, the Earth's surface cools, releasing the heat back into space. Some of the heat is trapped by greenhouse gases in the air like carbon dioxide, which causes the Earth to remain warmer than it would have otherwise. This is called the **greenhouse effect**.

Scientists think that due to human activities, like cutting down forests and burning fossil fuels, the greenhouse effect is now too strong. Scientists are more than 90 % certain that the increase in greenhouse gases has caused the average temperature on Earth to rise. This is known as global warming.

Venus provides us with a clue as to what might happen to the Earth if global warming continues. Venus' thick atmosphere has led to a runaway greenhouse effect on the planet, heating it to 462 °C. Venus's oceans have boiled away leaving behind a hot, inhospitable planet. We should therefore try our best to look after our precious planet!

The beginnings of life

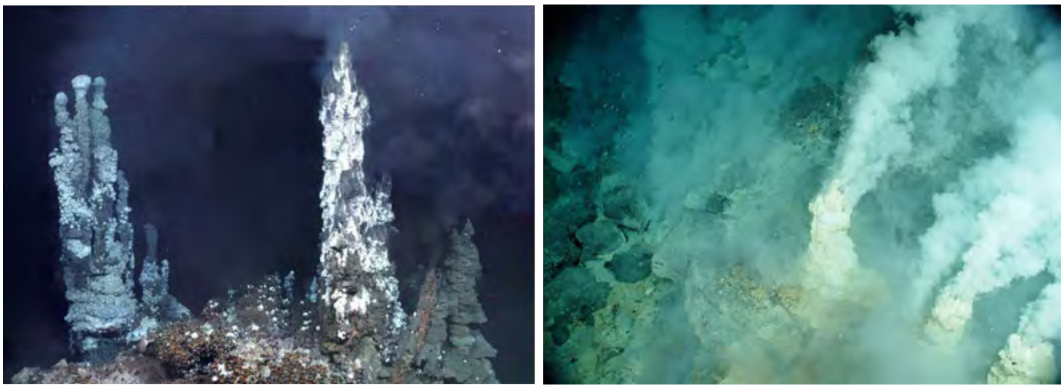
DID YOU KNOW?

The moons Europa (orbiting Jupiter) and Titan (orbiting Saturn) are considered to be places where life may exist. Europa's surface is covered in smooth water ice and scientists think that there might be a water ocean beneath the icy surface. Titan has liquid lakes and seas on its surface, although they are not made of water, but rather liquid methane and ethane. Some scientists think that life may be able to survive in these lakes.

Scientists do not know how life began on Earth, but they estimate that the early ancestor of modern bacteria was alive on Earth 3.5 billion years ago. The early Earth's atmosphere had almost no oxygen. Instead, it was composed mainly of carbon dioxide, nitrogen and water vapour with some methane and ammonia. Carbon dioxide and water vapour were pumped into the atmosphere during volcanic eruptions, which caused the atmosphere to change over time. Eventually the water vapour in the atmosphere condensed to form rain, forming the first oceans. Eventually living organisms (bacteria) appeared in the oceans. These simple organisms used sunlight, water and carbon dioxide in the oceans to produce sugars and oxygen. What is this process called?

____

This is where the first oxygen in the ocean and atmosphere came from. That oxygen made it possible for other organisms to develop and flourish and is the reason that you are here today.



Scientists are busy exploring the possible locations for the origin of life, including hot springs and tidal pools. Recently, some scientists have started to support the hypothesis that life originated in deep sea hydrothermal vents, as shown in the image. These vents are like underwater volcanoes. The investigation continues to try to understand how life originated on Earth.

VISIT

Do you enjoy English and Science? Read more about a career as a science writer.

bit.ly/18CxYiZ

SUMMARY:

Key Concepts

- The Sun produces its energy at its centre via nuclear fusion reactions, where hydrogen nuclei are squeezed together to form helium nuclei.
- The Sun's energy is transported to the surface and radiates equally in all directions.
- Our solar system consists of the Sun and all the objects that are held in orbit around the Sun by gravity.
- Objects such as planets, dwarf planets, asteroids, comets and Kuiper Belt objects orbit around the Sun.
- The 8 planets in our solar system have their own properties and characteristics.
- The planets can be split into two groups, the inner small rocky terrestrial planets and the outer large gas giants.
- The asteroid belt is the area where most asteroids are found in our solar system, lying between the orbits of Mars and Jupiter
- The Oort Cloud is a hypothetical huge cloud of icy objects (comets) surrounding the Sun at the very edge of our solar system.
- Sometimes, comets from the Oort Cloud come close to the sun. We can only see them when they come into the inner solar system because they are small and only visible by reflected sunlight.
- Scientists think that some of the conditions necessary for sustained life include moderate temperatures, liquid water, sunlight (energy) and oxygen.
- The Earth is the third planet from the Sun and the only planet in the solar system known to harbour life.
- The Earth lies within the Sun's habitable zone; the range of distances that a planet can lie from a star and still have liquid water on the planet's surface.

Concept Map

Complete the concept map which summarises the key concepts from this chapter about our solar system.

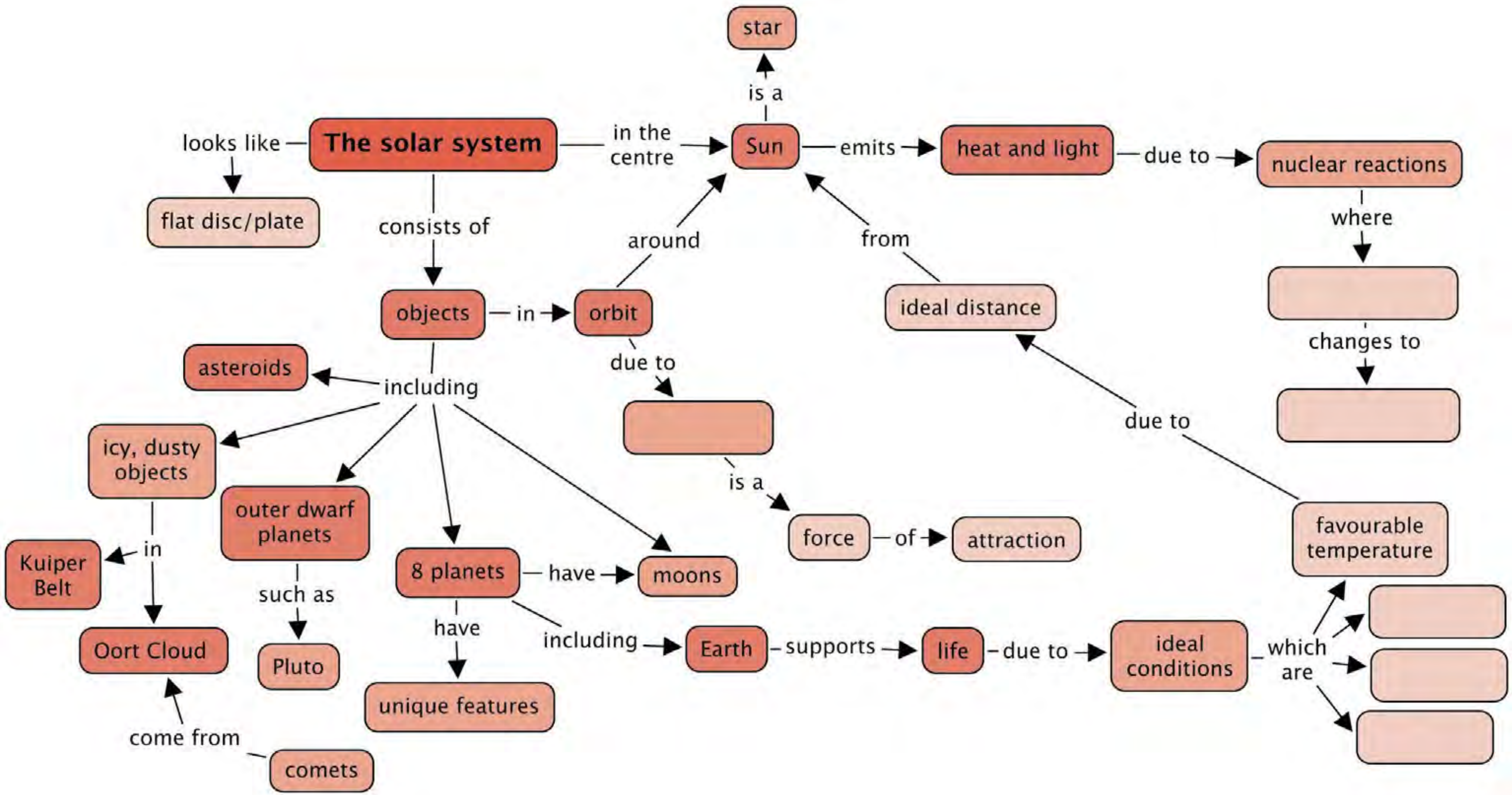
VISIT

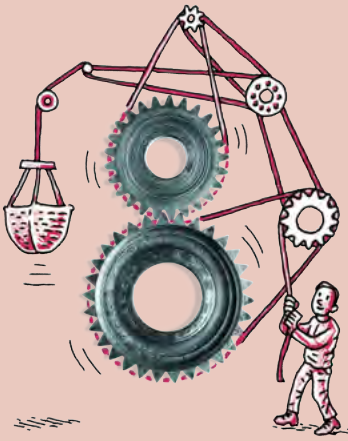
Global warming: How humans are affecting our planet.

bit.ly/1c9toNa

DID YOU KNOW?

Average temperatures on Earth have increased by 0.8°C around the world since 1880, with the biggest increase in the last few decades. The rate of warming is also increasing.


The solar system
looks like
flat disc/plate
in the centre
Sun
is a
star
emits
heat and light
due to
nuclear reactions
where
changes to
consists of
objects
in
orbit
around
due to
is a
force
of
attraction
from
ideal distance
due to
including
asteroids
icy, dusty objects
in
Kuiper Belt
Oort Cloud
come from
comets
outer dwarf planets
such as
Pluto
8 planets
have
moons
have
unique features
including
Earth
supports
life
due to
ideal conditions
which are
favourable temperature



REVISION:

1. How does the Sun produce its energy? [2 marks]

2. Why do sunspots look darker than the rest of the surface of the sun? [2 marks]

3. What keeps the planets and other bodies in our solar system in orbit? [1 mark]

4. Name the terrestrial planets. [4 marks]

5. Name the gas giants. [4 marks]

6. Where is the asteroid belt located? [1 mark]

7. Where is the Kuiper belt located? [1 mark]

8. Why are the gas giants so much larger than the terrestrial planets? [2 marks]

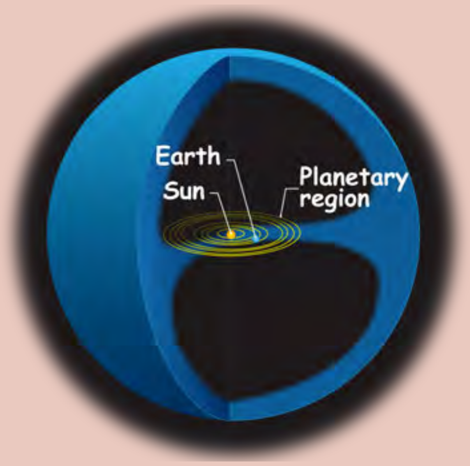
9. List the planets in increasing distance from the Sun. [4 marks]

10. Which planets have rings? [4 marks]

11. Why is Venus so hot? [2 marks]

12. On which planet have landers found frozen water in the rocks under the planet's surface? [1 mark]

13. The following diagram shows the solar system at the centre.



a) What does the blue space represent? [1 mark]

b) What is mostly found in this space? [1 mark]

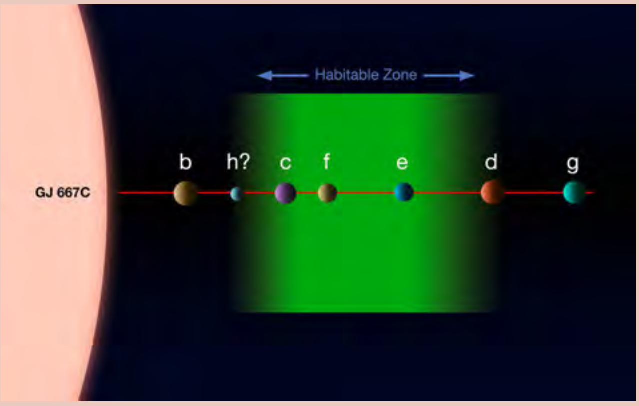
14. Why can we only see comets as they come close to the Sun? [3 marks]

15. What is the official definition of a planet and why was Pluto downgraded to a dwarf planet? [4 marks]

16. Why can the Earth support life? [4 marks]

17. What would happen to the Earth if it warmed significantly, like Venus has in the past? [2 marks]

18. The following diagram shows the system of planets around the star Gliese 667C.



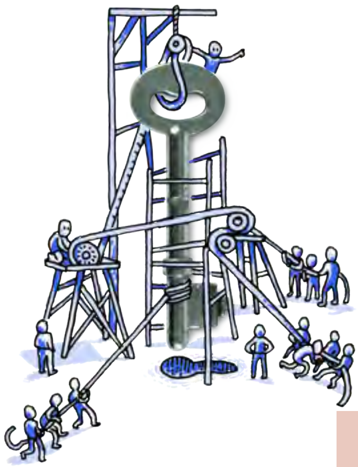
The planets around another star.

a) Which of these planets are possible candidates for life? [1 mark]

b) Explain your answer above. [2 marks]

Total [46 marks]

2 Beyond the solar system



KEY QUESTIONS:

- How far is our second closest star, Proxima Centauri?
- What is a galaxy and how many different types of galaxy are there?
- Where is our Sun located within our own Milky Way Galaxy?
- How do galaxies arrange themselves on the largest scales in the Universe?
- How large is the observable Universe and how many galaxies does it contain?

2.1 The Milky Way Galaxy

NEW WORDS

- galaxy
- galaxy disk
- galaxy bulge
- spiral arm

At the darkest places on Earth, far away from city lights, you can see thousands of stars at night using nothing but your eyes. In fact there are many more stars in the sky which are too faint for us to see.

All of the individual stars that you can see are members of our **Milky Way Galaxy**. A galaxy is a massive collection of stars, gas and dust all held together by gravity. The Milky Way has about 200 billion stars and our Sun is just one of those stars in the Milky Way Galaxy.

From the Earth, the Milky Way looks like a bright hazy band of light across the sky, mixed in with dark dusty patches. This was called *Galaxies Kuklos* by the Greeks which means the *Milky Circle* because they thought it looked like milk spilled across the sky. The Romans changed the name to *Via Lactea* which means the *Milky Road* or the *Milky Way.*

The Milky Way stretching across the sky viewed from Sutherland. The dark shape of the SALT telescope can be seen in the foreground with the night sky in the background (SAAO)

VISIT

Time lapse video of the Milky Way.

bit.ly/19Ylkal

If you could travel outside the Milky Way and look down on it from above, the galaxy would look like a giant spiral in space as shown in the following image.

This is what the Milky Way would look like if you could see it from far away in space. Scientists only know this from many observations made from Earth. No one has actually been that far away from our galaxy to look at it. The structure is what we have inferred from other observations.

The image shows what scientists think our galaxy looks like. You can see the spiral arms of our Milky Way. These are bluish in colour and are filled with dust and gas and hot young stars. The thin dark wisps in the image are dust lanes, regions where the gas is very dusty. The central part of the galaxy is more orangey in colour than the spiral arms. This is because the stars found at the centre of the galaxy tend to be older and cooler than the young hot blue stars.

Scientists think that there are five major spiral arms in our galaxy. These are the Norma Arm, the Scutum-Crux Arm, the Sagittarius Arm, the Perseus Arm and the Cygnus Arm.

Our Sun is located in a small spiral arm called the Orion (or Local) Arm which lies between the Sagittarius Arm and the Perseus Arm. Our Sun is about halfway out from the centre of the galaxy.

All the stars in this galaxy are revolving around the centre of the galaxy. Just as the Earth travels around the Sun, the Sun and our entire solar system is travelling around the centre of the Milky Way Galaxy at a speed of 250 km/s. Even though we are travelling incredibly fast, it takes the Sun about 225 million years to complete one orbit around the galaxy centre. The Milky Way is truly massive, measuring a staggering 950 000 000 000 000 000 km across!

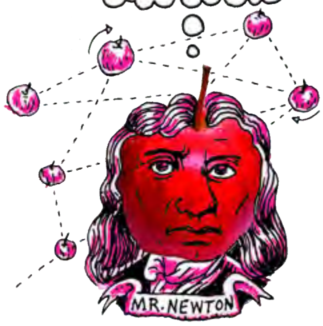
VISIT

Discover more online and read about missions beyond our solar system.

bit.ly/1iaQrog

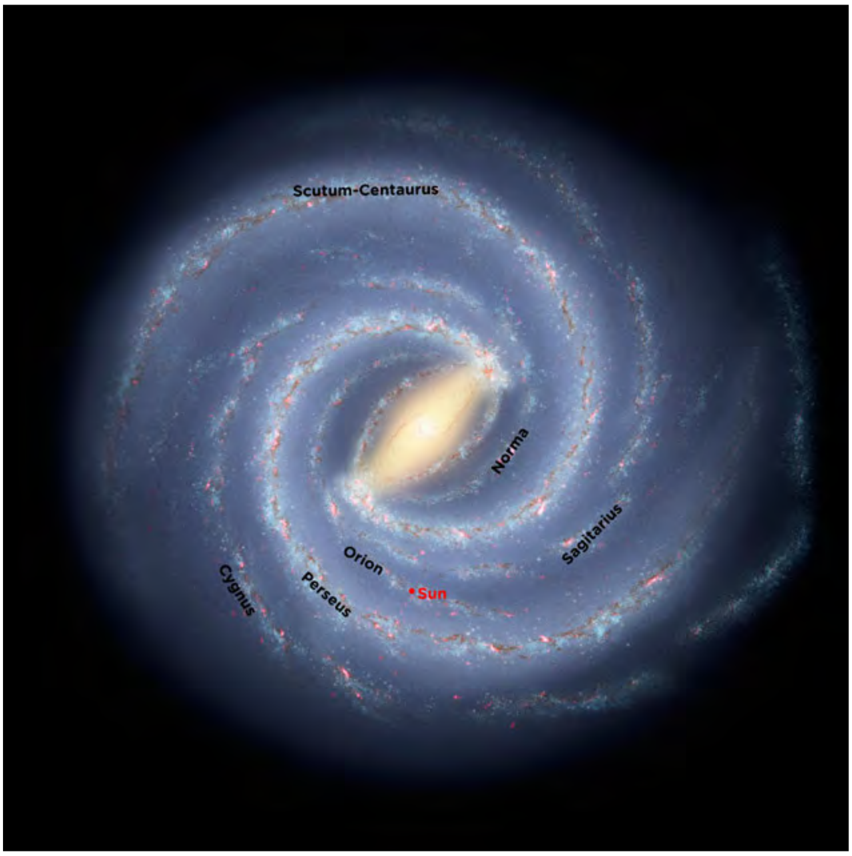
DID YOU KNOW?

Our Solar System is orbiting around the centre of the Milky Way at thousands of kilometres per hour. But even at that speed, it still takes over 200 million years for us to make one complete orbit around the Milky Way Galaxy.



DID YOU KNOW?

If you could shrink the solar system so that the distance from the Sun to Pluto is 2.5 cm, the Milky Way would have a diameter of 2000 km (about the distance from Durban to Windhoek!)

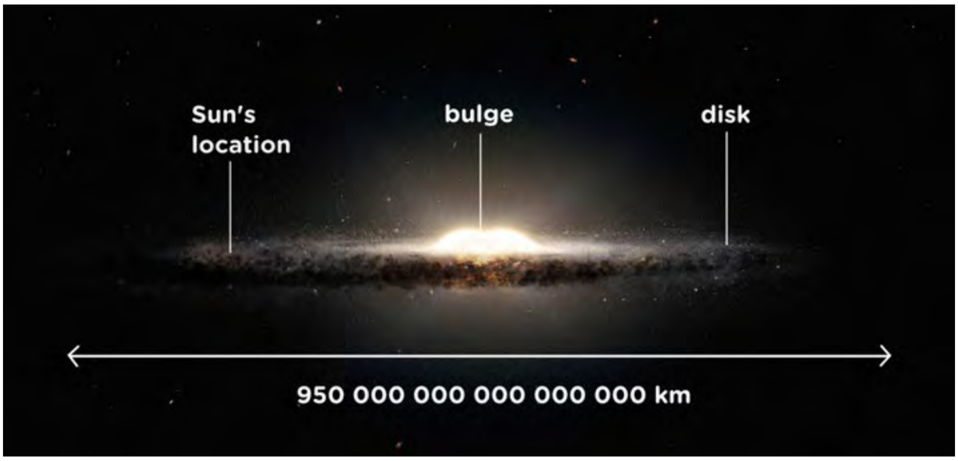


The Sun's position in the Milky Way.

If, instead of looking down on the Milky Way Galaxy, you looked at it from one side you would see that the Galaxy looks like this:

TAKE NOTE

To us the Earth seems big, but the Earth is only a very small part of the Solar System. And our Solar System is a very small part of the Milky Way Galaxy. And our galaxy is only a very small part of the whole Universe.



Looking at the Milky Way from the side.

The Milky Way is shaped like a giant fried egg. It is about a hundred times wider than it is thick, and it bulges in the middle. The central lump is called the **bulge** and the rest of the galaxy outside the bulge is called the **disk**.

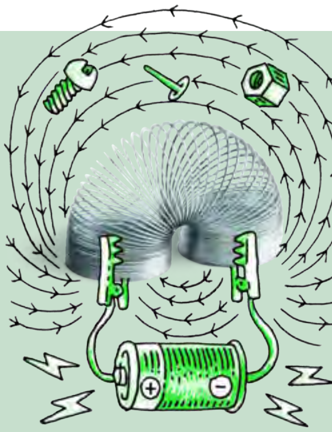
As you know, we are inside the Milky Way Galaxy. So when you look at the thin milky-looking band stretching across the sky at night, what do you think you are actually looking at?

The thin band of light that you see is actually the stars in the Sagittarius arm as you look inwards towards the centre of the galaxy. There are so many stars densely packed together that you cannot make out individual stars with your eyes. Therefore you just see a haze of light. Above and below the plane of the disk there are very few stars.

If you look closely at the image of the Milky Way above, you can see several round fuzzy blobs dotted about above and below the disk. These are called **globular clusters** and are vast collections of hundreds of thousands of ancient stars tightly packed together by gravity. The Milky Way has an estimated 160 globular clusters. The oldest stars in the galaxy are found in these globular clusters, some are almost as old as the Universe itself.

A globular cluster called M80. The stars in this globular cluster are around 12.5 billion years old. Our Sun is a mere 4.5 billion years old.

ACTIVITY: Draw the Milky Way



MATERIALS:

- black paper
- white crayon, pencil or paint
- glue - optional
- glitter or sand - optional
- newspaper for working on
- white or silver pencil/pen for labelling
- sticker - optional

INSTRUCTIONS:

1. Draw or paint a picture of the Milky Way. You can use the picture in the text above as a guide. The galaxy has five major spiral arms, and some smaller ones including our Orion Arm. The galaxy also has a bulge in the middle.
2. If you are going to use glitter or sand, glue along your spiral arms and in the central bulge.
3. Scatter glitter or sand over the picture, each grain represents a star in our Milky Way.
4. Tilt the picture onto the newspaper to remove any excess glitter.
5. Label each of the major arms of the Milky Way Galaxy.
6. On the Orion Arm place a sticker or mark a point halfway out from the galaxy centre. This marks the position of the Sun.

VISIT
The sound of interstellar space.
bit.ly/1cbfjil

How do you think astronomers know what the Milky Way looks like from the outside when they have never been outside the Milky Way? The task is similar to trying to figure out the shape of a forest from outside when you are in the middle of the forest. How would you go about this?

___

___

___

Astronomers look at the sky in all directions and count the number of stars that they see, they also measure the distance to each of the stars so that they can build up a three dimensional map of the galaxy. One of the difficulties that astronomers have in doing this is seeing through all the dust in the galaxy which dims the optical light coming from the stars.

ACTIVITY: Make the Milky Way

MATERIALS:

- thick piece of black cardboard at least 30 cm across
- other materials for your model, either collected by you or supplied by your teacher

INSTRUCTIONS:

1. You need to build a 3 dimensional model of the Milky Way Galaxy. You will either need to collect the most appropriate materials for your model beforehand, or else your teacher will supply you with a selection of materials to use in class.
2. Cut out a circle of radius 15 cm from the black card and use this to build your 3D model.
3. You must show the central bulge, the spiral arms and the different coloured stars.
4. Mark the position of our Sun on your model.
5. Using your model, view it from different angles and compare the view you have with the images of the Milky Way in this chapter.

QUESTIONS:

1. What are the two main parts that make up our Milky Way Galaxy?

___

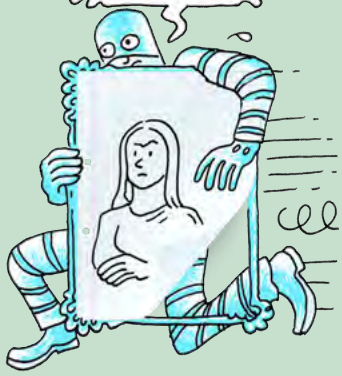
2. Where are the spiral arms located; in the disk or the bulge of our galaxy?

___

3. Is our Sun found in the central bulge or in a spiral arm in the disk?

___

TAKE NOTE
We will learn more about the life cycle of stars in Gr 9. Younger stars are hotter and bright white or blue in colour, while older stars are cooler and more yellow and red in colour.



4. How far from the centre of the galaxy is our Sun located?

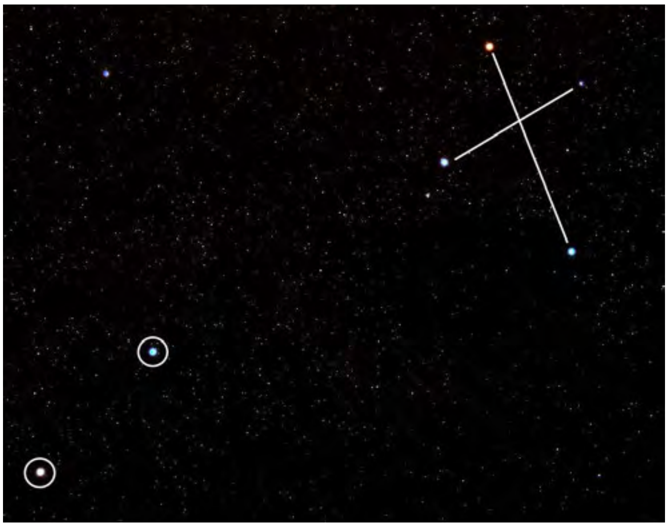
2.2 Our nearest star

NEW WORDS

- Proxima Centauri
- Alpha Centauri
- constellation

The Sun is our closest star, and is *only* 150 million kilometres from Earth. When you look up at the sky at night, if you are lucky enough to be far from the glare of city lights, you can see thousands of stars. For those of you in a city, perhaps you can see hundreds of stars, depending on the amount of light pollution from street lights and other light sources. As you know, there are actually billions of stars in our galaxy but most of them are too faint to see from Earth.

A **constellation** is a group of stars that, when viewed from Earth, form a pattern in the sky. One famous **constellation** that is visible, even from big cities in South Africa, is the Southern Cross or Crux. The two bright stars at the bottom left pointing towards the cross are called the pointers.



The Pointers (circled) and the Southern Cross.

DID YOU KNOW?

You can find south using the Southern Cross Constellation. Just extend the long axis of the cross 4 times and then go straight down to the horizon to find south.

The brightest of the Pointers looks slightly orange if you look closely. This star is called **Alpha Centauri** and is our closest easily visible star after the Sun. Alpha Centauri is actually part of a triple star system which is where three stars are in orbit around each other. The two main stars of the system are called Alpha Centauri A and Alpha Centauri B. They orbit close together, on average about eleven times the Earth-Sun distance from each other.

A smaller, fainter star, called **Proxima Centauri**, orbits much farther out. If you were to look at Alpha Centauri through a small telescope, instead of one star you would be able to make out the two separate stars Alpha Centauri A and B next to each other. Proxima Centauri is much fainter and further away from the other two so you would not see this one with the other two.

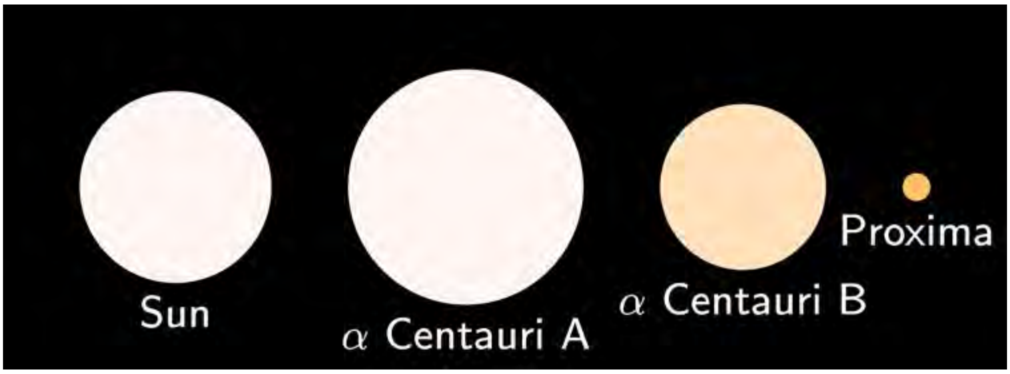
VISIT

Scale of the Universe.

bit.ly/185Yjlc

DID YOU KNOW?

Proxima Centauri was discovered in 1915 by the Scottish astronomer Robert Innes. He was the director of what was then the Union Observatory in South Africa.



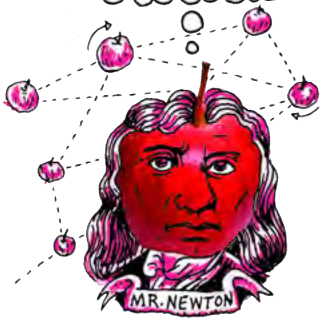
A comparison of the sizes of the Alpha Centauri star system and the Sun.

Proxima Centauri, the closest star to our own Sun, is about 40 trillion km away from the Earth. Alpha Centauri A and B are slightly farther away, at 42 trillion km away from us. Our closest star is 694 times farther away than Pluto is. These numbers are astronomically large! As the numbers are so large, astronomers do not use kilometres to measure the distances to stars, but use larger units based on the speed of light, which you will discover in the next section of this chapter.

Do you know how much a trillion or a billion is? Have a look at the following table:

DID YOU KNOW?

Astronomers have recently discovered a planet similar in size to the Earth orbiting around Alpha Centauri B, but we think it is too close to the star to have life on it.



In words	In number format
one thousand	1 000
one million	1 000 000
one billion	1 000 000 000
one trillion	1 000 000 000

2.3 Light years, light hours and light minutes

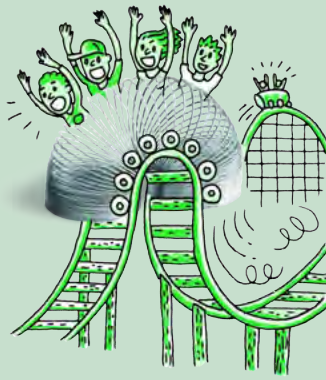
NEW WORDS

- light minute
- light hour
- light year

Our solar system is a pretty big place. Our nearest neighbour, the Moon, is on average 384 400 kilometres away, and the closest to us that our nearest planet Venus gets is about 42 million kilometres. The Sun is about 150 million kilometres away and the closest that Pluto can ever get to us is 4.3 billion kilometres. These large numbers are impractical to use and so we rather use much larger distance units based on the speed of light. This makes the numbers smaller and easier to deal with.

This is just like using metres instead of centimetres to make the numbers smaller when you measure a distance. For example, if you are telling a friend how far it is from your house to school, you would say it is 7.5 km, and not 7 500 000 cm. Let's begin by comparing the speed of light with the speed of some other things that move very fast.

ACTIVITY: Travelling fast



A cheetah, the fastest land mammal, can reach speeds of 120 km/h, as fast as cars on the highway.

A Peregrine Falcon, the fastest animal, can fly as fast as 389 km/h.

Japan's high speed train the JR-Maglev MLX01 has reached 581 km/h.

NASA's scramjet the X-43 flies at 7000 km/h.

VISIT
How to break the speed of light.
bit.ly/H7MoO0

The international space station (ISS) orbits the Earth at a speed of 27 744 km/h.

What about light? Light travels at about 1080 million km/h, or 299 792 458 m/s.

INSTRUCTIONS:

1. Imagine you are going on a trip from Cape Town to Durban, which is a distance of 1753 km.
2. Calculate how long it would take you to complete the trip travelling at the speeds of the animals and modes of transport in the examples above.
3. Fill in your answers in the table below.

Remember the formula: $\text{time} = \dfrac{\text{distance}}{\text{speed}}$

Mode of transport	Speed (km/h)	Distance between Cape Town and Durban (km)	Time taken for the journey
cheetah	120	1753	14.6 hours
peregrine falcon		1753	______ hours
high speed train		1753	______ hours
NASA's scramjet		1753	______ minutes
International space station		1753	______ seconds
light		1753	______ seconds

Light is amazingly fast. Look at the examples below.

In one second light can travel...	Light takes...
between Cape Town and Johannesburg 214 times.	0.0000003 seconds to travel 100 m.
between Cape Town and London, England, 31 times.	1.3 seconds to travel from the Earth to the Moon.
around the Earth 7.5 times.	8 minutes to travel from the Sun to the Earth.

For distances within the solar system, astronomers use units called **light hours** and **light minutes**.

A light hour is the **distance** that light travels in one hour. Despite its name, a light hour is not a unit of time, it is a **unit of distance**.

What do you think a light minute corresponds to?

____

____

Which do you think is a smaller distance, a light hour or a light minute, and why?

____

____

____

VISIT
How far is a light year?
bit.ly/GZCzBy

Astronomers use units called **light years** to measure the distances between stars and galaxies. One light year is almost 10 trillion kilometres. As you can see, a light year is very, very far.

Light years, light hours and light minutes measure distances. They also tell us something else very interesting. If you measure the distance to a light source in light travel time, you can work out how long light emitted from the distant source takes to reach you. Light that is emitted from an object one light year away from you, takes one year to reach your eyes. Similarly, light that is emitted from an object one light hour away, takes one hour to reach your eyes.

How long do you think light emitted from one light minute away takes to reach your eyes?

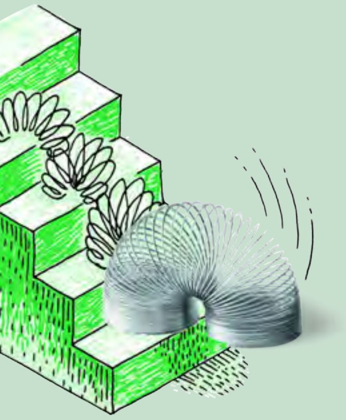
____

VISIT
Scale of the Universe interactive animation.
bit.ly/1bavNSv

This may sound very strange to you because when you switch on a lamp in your home you see the light straight away. You do not have to wait for the light from the lamp to reach you. You do not notice that it actually takes some time for the light from the lamp to reach your eyes because light travels extremely fast.

Light travels so fast, that if you were standing a metre away from the lamp it would only take only three billionths of a second for the light from the lamp to reach your eyes. It is therefore no surprise that you don't notice the delay.

VISIT
How big is the Universe?
bit.ly/1ddYnaj



ACTIVITY: Scale of the solar system

INSTRUCTIONS:

1. The table below shows the distance that each planet lies from the Sun in kilometres (km) and then in light hours or light minutes.
2. Study the table and answer the questions that follow.

Distances of each planet from the Sun.

Planet	Distance from the Sun (million km)	Distance from the Sun in light hours or minutes
Mercury	57.9	3.2 light minutes
Venus	108.2	6.0 light minutes
Earth	149.6	8.3 light minutes
Mars	227.9	12.7 light minutes
Jupiter	778.6	43.3 light minutes
Saturn	1433.5	1.3 light hours
Uranus	2872.5	2.7 light hours
Neptune	4495.1	4.2 light hours

QUESTIONS:

1. How far away from the Sun is Earth?

2. How long does light take to travel from the Sun to the Earth?

3. What does the answer to (2) imply about our view of the Sun?

4. How many times further away from the Sun than the Earth is Neptune?

5. How far away from the Sun is Neptune in light hours?

DID YOU KNOW?

The speed of light is special, nothing can move faster than the speed of light, it is like a cosmic speed limit.

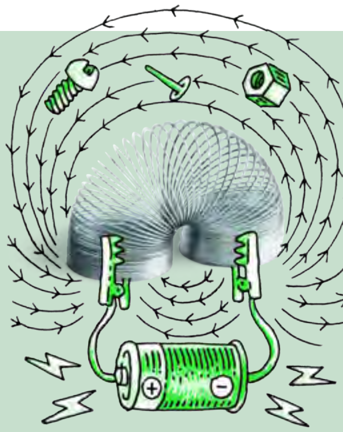
6. How long does light from the Sun take to reach Neptune?

7. Imagine you have a cousin living on Neptune. You and your cousin both decide to look at the Sun, each of you using a telescope with a special solar filter so as not to damage your eyes. As you are watching the Sun you suddenly notice a big blob of gas thrown off in a massive solar flare. You cousin says she cannot see it. Why is that?

As you can see, the solar system is very large. The orbit of Neptune is over 4 light hours from the Sun and the Kuiper Belt and Oort Cloud extend out even further than this.

The distance to the next closest star, Proxima Centauri, is 40 trillion km. This corresponds to 4.24 light years. This means that light from the star takes just over four years to reach Earth. Let's investigate the distances to some of our closest stars.

ACTIVITY: Our closest stars



INSTRUCTIONS:

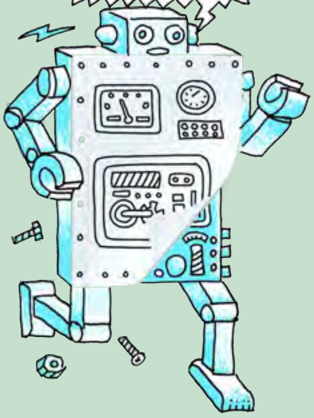
1. Look at the table showing our closest stars and the star map.
2. Answer the questions below.

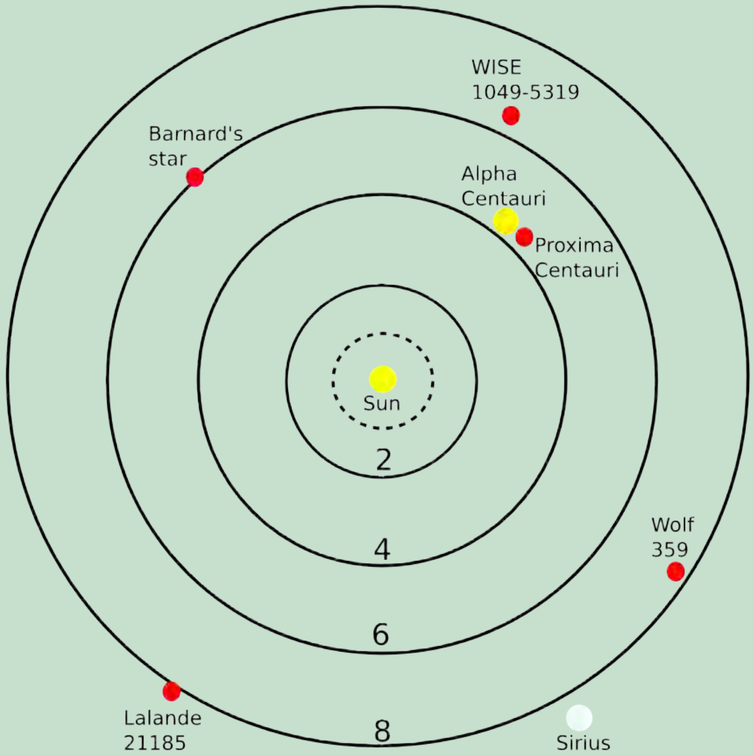
Star	Distance (light years)
Proxima Centauri	4.24
Alpha Centauri	4.37
Barnard's Star	5.96
WISE 1049-5319	6.52
Wolf 359	7.78
Lalande 21185	8.29
Sirius	8.58

The following map shows the Sun in the centre with the locations of our closest stars. Each solid ring represents a distance of 2, 4, 6 and 8 light years from the Sun respectively. The dotted circle represents the Oort Cloud.

TAKE NOTE

The star map is shown in two dimensions, on a flat plane. Remember that the stars are located in 3 dimensions in space.





QUESTIONS:

1. Which star is our closest neighbour, excluding the Sun?

2. How far is Sirius?

3. How long does light from Barnard's Star take to reach us?

4. Explain in your own words what the statement "Sirius is 8.58 light years away from Earth" means.

DID YOU KNOW?

The Milky Way is so large that light takes 100 000 years to cross from one side to the other side.

Our closest stars are less than ten light years away, however most stars in our galaxy are much farther away. The distances to stars are generally measured in tens, hundreds or even thousands of light years and the distances between galaxies are truly enormous as you will discover in the next section.

2.4 What is beyond the Milky Way Galaxy?

Our galaxy, the Milky Way, is only one out of a total of about 100 to 200 billion galaxies that astronomers estimate to be in the **Universe**. That's more than 10 times the total number of people on Earth.

As well as stars, galaxies contain vast amounts of gas and dust. Galaxies come in a variety of shapes and sizes. The Milky Way is an average-sized spiral galaxy: it is 100 000 light years across and contains around 200 billion stars.

Small galaxies may contain only a few million stars, while large galaxies can have several trillion stars.

Our closest galaxy neighbour is called the Andromeda Galaxy. Andromeda is 2.5 *million* light years away from the Milky Way. If you wanted to travel to Andromeda and could travel as fast as light, it would still take you 2.5 million years to get there.

Our closest neighbouring galaxy, Andromeda. Light from the galaxy takes 2.5 million years to reach Earth and so the light that hits your eyes now from that galaxy was emitted before there were humans on Earth.

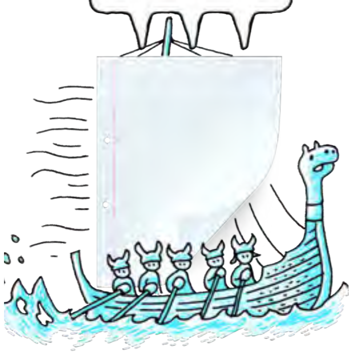
This illustration shows a stage in the predicted collision between our Milky Way Galaxy and the neighboring Andromeda Galaxy, as it will unfold over the next several billion years. This image shows how we think Earth's night sky will look like in 3.75 billion years time.

NEW WORDS

- galaxy
- galaxy group
- galaxy cluster
- filament
- void
- Universe

TAKE NOTE

The distances between galaxies are even larger than the sizes of galaxies and are measured in millions or even billions of light years.



DID YOU KNOW?

The Milky Way and Andromeda galaxies are on a collision course. Astronomers estimate that the two galaxies will collide in around 4 billion years time. No need to worry just yet!

VISIT
The largest known galaxy in the Universe.
bit.ly/1ddXJcR

VISIT
What is the Universe?
bit.ly/1eFW3XI
How big is the Universe?
bit.ly/16sqdlB

VISIT
Something fun - have a look at this picture of a cheetah created using thousands of images of galaxies from Galaxy Zoo.
bit.ly/177itzq

There are five main types of galaxies. You do not need to know these names. This is included for your interest.

- spiral
- barred spiral
- elliptical
- lenticular
- irregular

Spiral galaxy named NGC 4414.

Barred spiral galaxy named NGC 1300.

Elliptical galaxy NGC 1132.

A lenticular galaxy, called NGC 5866.

Irregular galaxy named NGC 1427A.

Let's do an activity to explore the different types of galaxies we see.

ACTIVITY: Comparing galaxies

MATERIALS:

- images of the galaxies to be compared

INSTRUCTIONS:

1. Look at the images of the of six galaxies used in this activity.
2. Using the information in this chapter, write down in the table what type of galaxy our Milky Way Galaxy is.
3. Write down in the table below what type of galaxy (spiral, barred spiral, elliptical or irregular) you think each galaxy is.

Galaxy Name	Galaxy type
 The Milky Way Galaxy.	
 Galaxy M 89. The galaxy is 60 million light years away.	
 Galaxy NGC 4622. The galaxy is 111 million light years away.	

Galaxy Name	Galaxy type
 The Large Magellanic Cloud galaxy. This satellite galaxy of our own Milky Way is only 163 000 light years away.	
 The Spindle Galaxy, 44 million light years away.	

VISIT
What is dark matter?
bit.ly/1ab5oFO

QUESTION:

List the galaxies in the table above in increasing order of distance from our Milky Way Galaxy.

____

____

____

____

____

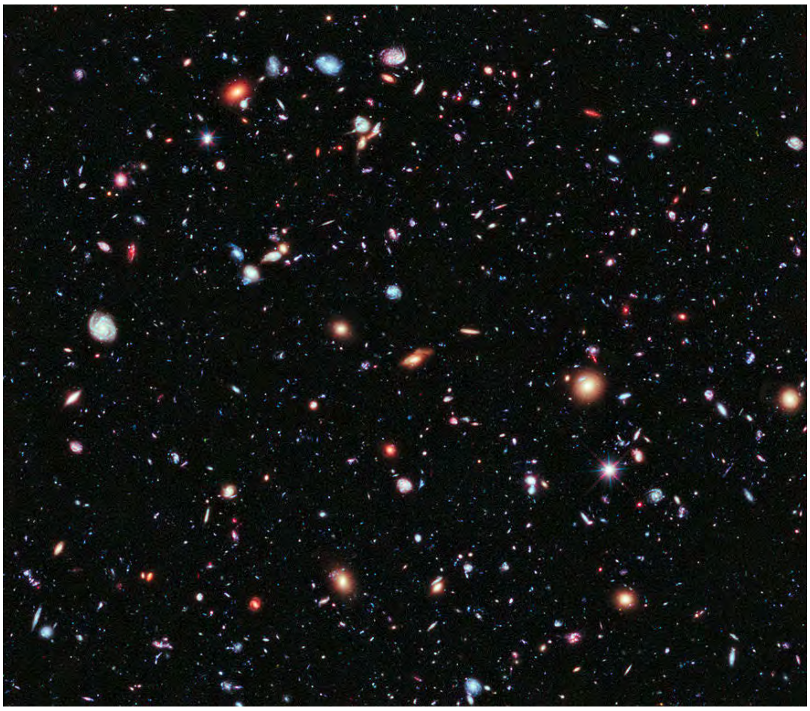
Have a look at the following diagram which shows the location of Earth in the Universe. **You do not need to know this classification**; this is included for your interest.

VISIT

How do we know how many galaxies there are in the Universe?

bit.ly/HfeA0O

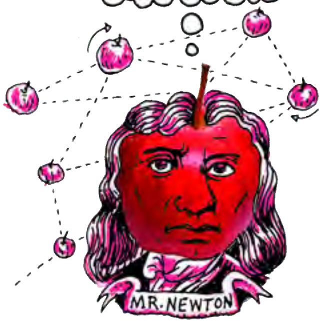
- Most galaxies are found gathered together in gigantic galaxy neighbourhoods, called **galaxy groups**. Our Milky Way is found in a group of galaxies called **The Local Group.**
- **Galaxy clusters** are even larger, spanning tens of millions of light years, and can contain hundreds or even thousands of galaxies.
- Many clusters of galaxies come together to form **superclusters** of galaxies. Our own local group is part of the Virgo supercluster.
- Gravity holds the galaxies in groups, clusters and superclusters together.



Galaxies in the Hubble Extreme Deep Field. Every smudge in the image is a distant galaxy.

DID YOU KNOW?

The Hubble Extreme Deep Field is the most distant picture of the Universe ever taken. Astronomers used the Hubble Telescope to take an image of a small patch of sky. Around 5500 galaxies of all shapes, sizes and colours were discovered in the image.



DID YOU KNOW?

On the largest scales the Universe resembles a giant bath sponge. The galaxy clusters are arranged in thin walls called **filaments**. Between the filaments are huge gaps which contain very few galaxies and so are called **voids**.

VISIT

Do we expand with the Universe?

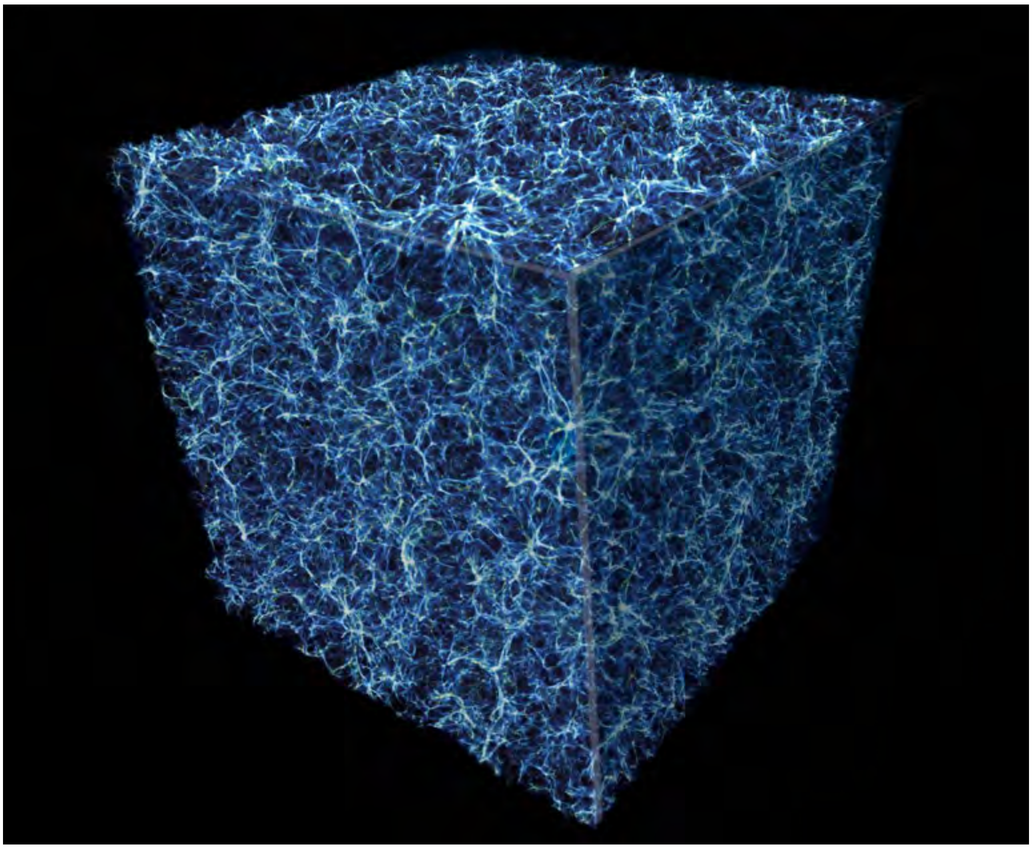
bit.ly/1ddYyTd

VISIT

Read interesting articles on the latest developments in astronomical research on **Space Scoop**, an astronomy news service.

bit.ly/1fSxJ84

The Observable Universe



This computer generated graphic represents a slice of the sponge-like structure of the Universe. All the galaxies lie along thin walls called filaments. The darker areas show the voids where there are no galaxies.

Astronomers estimate that the age of the Universe is 13.7 billion years old. This might make you imagine that you can see objects from as far as 13.7 billion light years away in all directions. If you were to draw a sphere around the Earth, with a radius of 13.7 billion light years, with the Earth placed at the centre, the surface of the sphere would represent the limit of how far light could travel to Earth in 13.7 billion years. The surface would represent the edge of the *observable Universe as seen from Earth*. You might therefore assume that the diameter of the **observable Universe** is 27.4 billion light years (2 times 13.7).

However, you would actually be wrong. Astronomers estimate the size of the observable Universe to be 93 billion light years in diameter, which is much, much larger. The reason that the size is much larger than expected is because the Universe is expanding and galaxies are moving further and further away from the Earth as the space between them expands. So we are able to see galaxies that are now very far away because when they emitted their light they were closer to Earth. The size of the whole Universe, which includes regions too far from Earth for us to see at this time, is unknown.

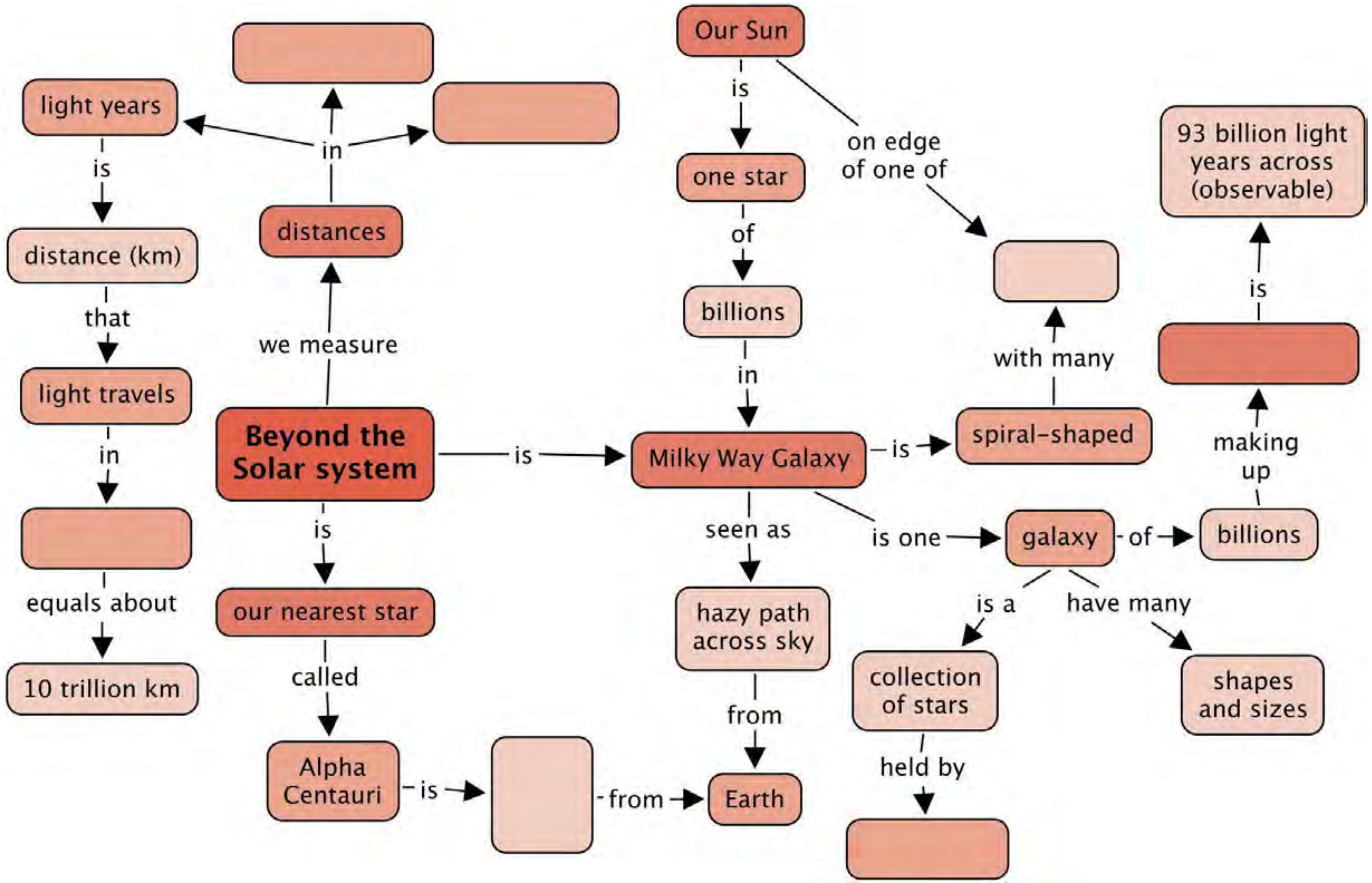
SUMMARY:

Key Concepts

- A galaxy is a collection of millions or billions of stars, together with gas and dust, held together by gravity.
- Galaxies come in all shapes and sizes.
- Our home galaxy, the Milky Way Galaxy, is a spiral galaxy containing around 200 billion stars. Our Sun is just one of those stars.
- After the Sun, our nearest star is Alpha Centauri, the brighter of the two pointer stars in the Southern Cross Constellation
- Light minutes, light hours and light years are used to measure distances in space because the distances are so immense.
    - A light minute is the distance that light can travel in one minute.
    - A light hour is the distance that light can travel in one hour.
    - A light year is the distance that light can travel in one year.
- Beyond the Milky Way Galaxy, are many more galaxies.
- Astronomers estimate the size of the observable Universe to be 93 billion light years in diameter.

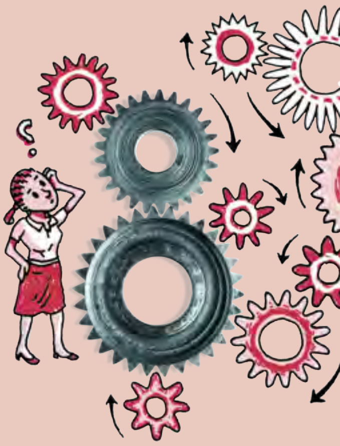
Concept Map

Remember that you can also add your own notes to the concept maps to expand and personalise them.


Beyond the Solar system
we measure
distances
in
light years
is
distance (km)
that
light travels
in
equals about
10 trillion km
is
our nearest star
called
Alpha Centauri
is
from
Earth
is
Milky Way Galaxy
Our Sun
is
one star
of
billions
in
on edge of one of
with many
spiral-shaped
is
seen as
hazy path across sky
from
is one
galaxy
of
billions
making up
is
93 billion light years across (observable)
is a
collection of stars
held by
have many
shapes and sizes

REVISION:

1. What is the name of our second closest star? How far away is it? [2 marks]



2. What is the name of our second closest easily visible star? Is it really a single star? [2 marks]

3. What is the definition of a light year? [2 marks]

4. What is a galaxy? [3 marks]

5. Where is the Sun located within the Milky Way? [2 marks]

6. How many stars are in our Milky Way Galaxy? [1 mark]

7. Name the 4 main types of galaxies. [4 marks]

8. What kind of galaxy is the Milky Way? [2 marks]

9. Draw an image of the Milky Way Galaxy as viewed from the top and as viewed from the side. Note the position of the Sun in both images. Include the labels: spiral arm, bulge, disk. [8 marks]

10. Why does it look as though the Milky Way is a splash of milk or a starry road across the sky? [2 marks]

11. What is a group of galaxies? [2 marks]

12. What is the name of the group of galaxies that the Milky Way is a member of? [1 mark]

13. What are clusters of galaxies and superclusters of galaxies? [2 marks]

14. What is the size of the observable Universe? [1 mark]

15. **Bonus question:** On the largest scales what does the Universe look like? Name the two types of structure which make up the Universe on the largest scales? [2 marks]

Total [34 marks]

Total with extension [36 marks]

3 Looking into space

KEY QUESTIONS:

- How did early cultures observe and interpret the night sky?
- How does a telescope help us to see more objects in the sky and in greater detail?
- What kind of telescopes are there?
- Why is South Africa a good place for locating telescopes?

3.1 Early viewing of space

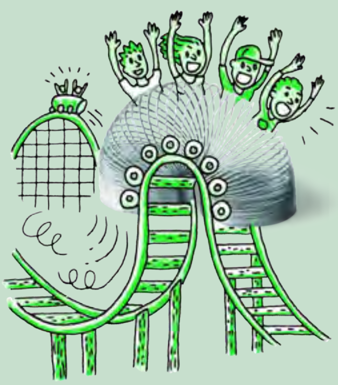
NEW WORDS

- constellation
- starlore

In dark conditions away from city lights, thousands of stars are visible in the night sky. Early cultures around the world gazed at the stars in wonder. They noted the movement of the stars and planets across the sky and used this to mark the passage of time. People often grouped the stars they saw into patterns called **constellations**. Early cultures tended to associate the stars and planets they saw in the night sky with animals or gods and told stories, which were passed on from generation to generation, about the patterns in the sky which were passed down from generation to generation.

The stars that are visible depend upon your location on Earth and also the time of year. The southern sky, which we see from South Africa, is full of beautiful stars and several prominent constellations are visible in the sky including the Southern Cross or Crux, Orion and Pavo the Peacock.

In the following activities you will have the opportunity to observe the night sky and familiarise yourself with some of the most famous southern constellations.

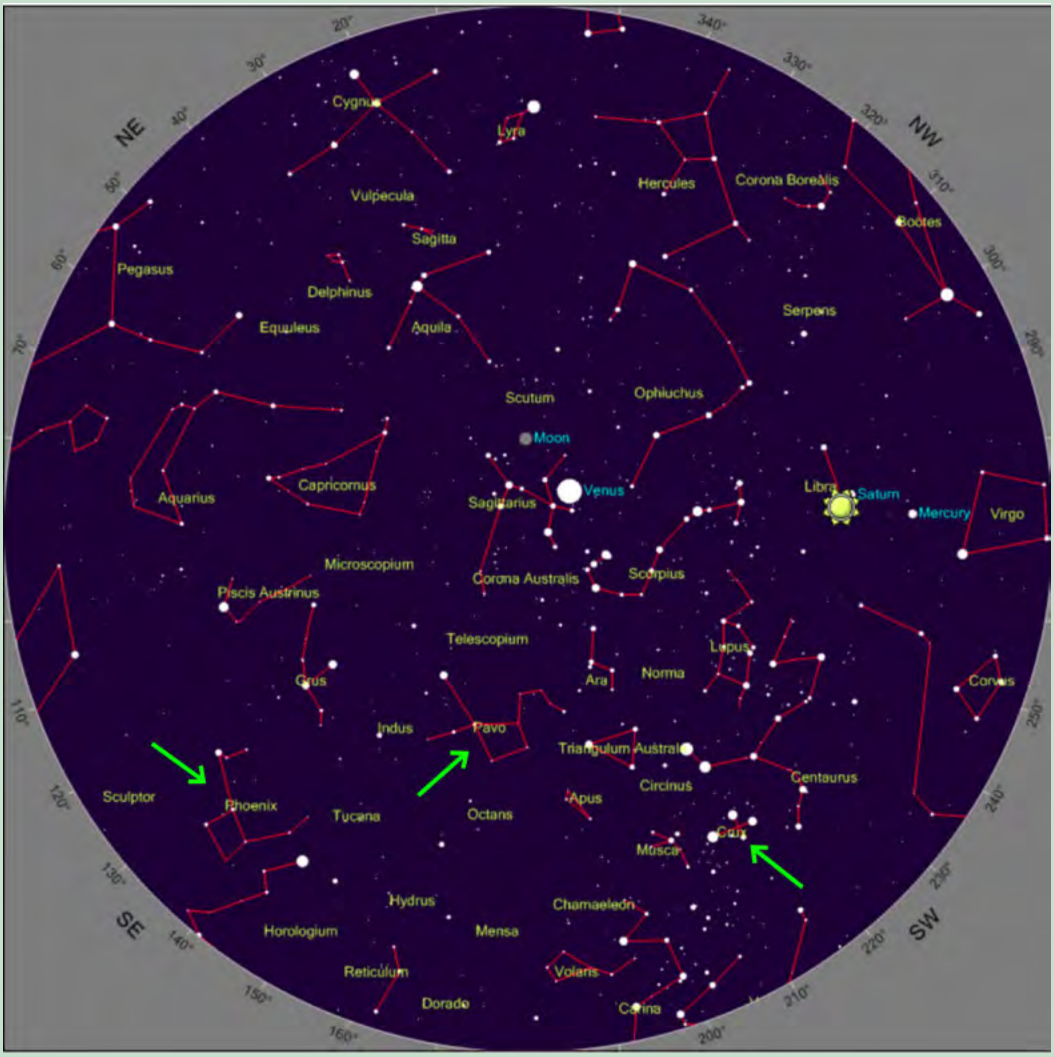


ACTIVITY: Using star maps to observe the night sky

MATERIALS:

- star map
- clear skies
- pencil
- paper or this workbook
- torch - optional

Below is an example star map of the Southern Hemisphere. Ignore the positions of the Moon and the planets. You can generate your own, customised star map for your exact location using the link in the **Visit** margin box.



VISIT

Create your own star map for your area.

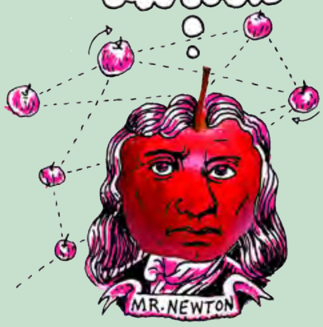
bit.ly/1a4N1nU

INSTRUCTIONS:

1. Go outside at night with your star map.
2. Wait a few minutes to let your eyes adjust to the dark.
3. Try to identify the following constellations in the sky: Pavo, Phoenix and Crux (indicated with green arrows on the star map).
4. Draw a picture of each of the constellations as you see them.
5. See if you can spot any of the planets, these will not twinkle like the stars do.

DID YOU KNOW?

Early telescopes were used by merchants to spot approaching trade ships or pirates. Telescopes also gave rise to the first high-speed telecommunications networks, as spyglasses were used to observe signals from kilometres away.



TAKE NOTE

Today professional astronomers formally recognise 88 constellations, 23 of which are in the southern hemisphere.

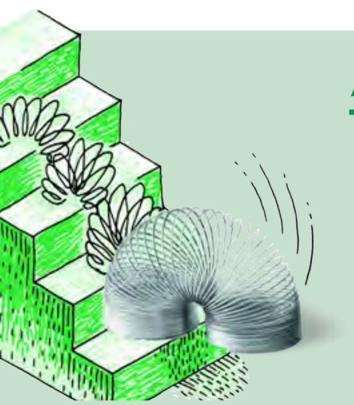
VISIT

Learn how to view the night sky with Google Earth.

bit.ly/16pYL3u

DRAWINGS:

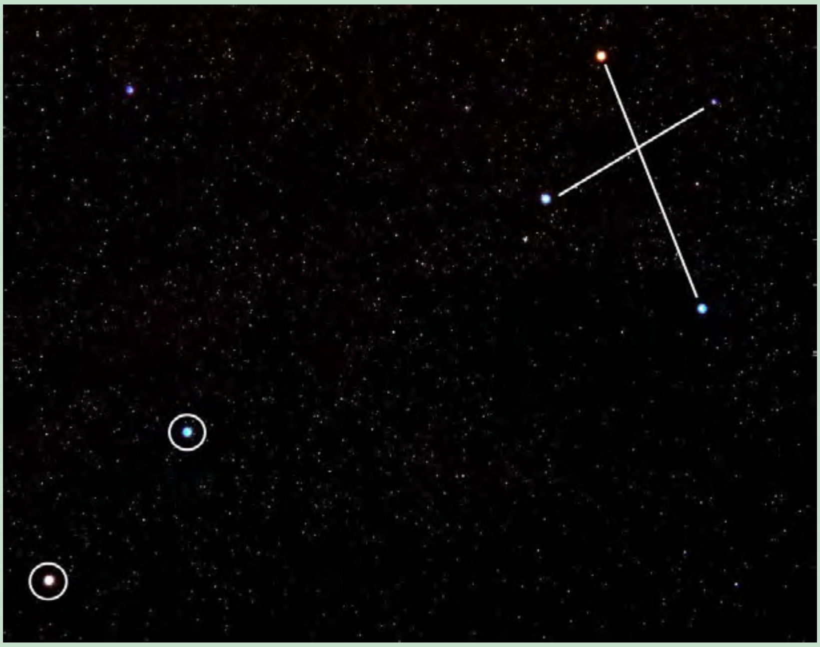
Draw your pictures in the space below. If you have used separate paper you can stick your pictures in here.



ACTIVITY: Observing the Southern Cross (Crux)

MATERIALS:

- picture of the Southern Cross constellation and star map
- clear skies
- pencil
- paper or this workbook



The Southern Cross or Crux (top right) and the Pointers (bottom left).

INSTRUCTIONS:

1. Go outside around 8 pm with your star map (if in the Western half of the country, closer to Cape Town), or if you live in the Eastern half of the country, (closer to Johannesburg or Durban) go out an hour earlier around 7pm.
2. Wait a few minutes to let your eyes adjust to the dark.
3. Try to identify the Southern Cross constellation using the star map.
4. Draw a picture of the Southern Cross and the Pointers as you see them. Make a note of the date and time of your picture and in roughly which direction you are facing (north, south, east or west).
5. Draw or paste your image (if you have used separate paper) into the space below.
6. Repeat the observations at least twice so that you have a minimum of three observations on different nights, over the course of a few weeks, and try as best as possible to make your observations at the same time each night.

DRAWINGS:

DID YOU KNOW?

These workbooks were created by Siyavula with the help of contributors and volunteers. Read more about Siyavula here.

www.siyavula.com

DID YOU KNOW?

Some people believe that the builders of the ancient pyramids of Giza in Egypt placed them specifically to look the same from above as the three "belt stars" of the constellation Orion look from Earth.

QUESTION:

What did you notice about the orientation of the Southern Cross as you made your observations?

____

____

____

____

VISIT

Stellarium - a free, open source programme for your computer to to generate a realistic, real-time 3D simulation of the night sky.

bit.ly/1aE2Imj

Although the stars appear to lie in patterns when viewed from the surface of the Earth, in reality the stars within a constellation are unrelated, and they can lie at vastly different distances from Earth. When we look at the stars at night, we only see a two dimensional projection on the sky of three dimensional space, as you can see in this photograph showing the constellation, Orion.

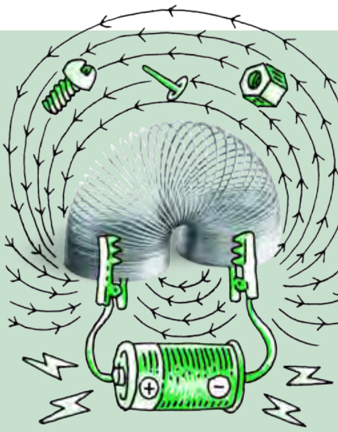
The Orion Constellation, seen here as the three bright stars in the middle making up Orion's belt and the 4 stars in each corner.

You might imagine that all the stars lie at the same distance from Earth. This isn't true, the stars lie at different distances. The closest star in Orion is called Bellatrix and is around 250 light years away. The furthest star Meissa is around 1100 light years away, roughly the same distance as the Orion nebula (1300 light years). But, when viewed from Earth, we see them making up a pattern in relation to each other.

Now that you are familiar with some of the constellations in the Southern sky, including the Southern Cross you can learn what some of the early cultures in Southern Africa thought about them.

As you can imagine there are many stories associated with the constellations in the sky. In the following activity you will carry out research to find an example story to tell to your class.

ACTIVITY: Constellation starlore



The /Xam Bushman imaged that the two pointer stars of the Southern Cross were two male lions who had once been men before they were thrown up into the sky to be stars by a magical girl. The three brightest stars in the Southern Cross were seen as female lions, perhaps women also changed into stars by the magical girl.

The Khoikhoi thought that the Pointers were the eyes of some great beast and they were called *Mura* which means *the eyes*.

In Sotho, Tswana and Venda cultures, these stars are called *Dithutlwa* which means *the Giraffes*. The bright stars of the Southern Cross are male giraffes, and the two Pointer stars are female giraffes. The Venda named the fainter stars of the Southern Cross *Thudana*, which means *the Little Giraffe*. The Sotho used these stars to indicate the beginning of the cultivating season which began when the giraffe stars were seen close to the south-western horizon just after sunset.

VISIT

Read more about some South African starlore:
bit.ly/1cbF7uu and bit.ly/1abUL5z

INSTRUCTIONS:

1. Search for a story about a constellation found in the South African sky.
2. Use a South African starmap as a guide to the constellations found in South Africa.
3. Research information on the origin of the story and any beliefs associated with it.
4. Tell your classmates about the constellation and story you have found out about.
5. Your teacher will decide on the format of this presentation which might be a poster or oral presentation.

In their quest to find out more about planets, stars and galaxies, people invented instruments to observe them in more detail. In the next section we will learn about the telescope: an invention used to study the stars.

NEW WORDS

- celestial
- telescope
- chromatic aberration
- primary mirror

3.2 Telescopes

Unfortunately, we cannot visit distant stars or galaxies to study them directly as they are so far away. Instead astronomers study stars and galaxies by analysing the visible light, radio waves and electromagnetic radiation that they receive from them.

The Andromeda galaxy, viewed with the Hubble Space Telescope. Humans can only see it as a tiny faint smudge in the sky with the naked eye.

Human eyes can see very far. Andromeda Galaxy which is 2.5 million light-years away is visible to the naked eye. However, we cannot make out any detail as it appears as only a tiny smudge on the sky to our eyes even though in reality it is 220 000 light years across.

VISIT

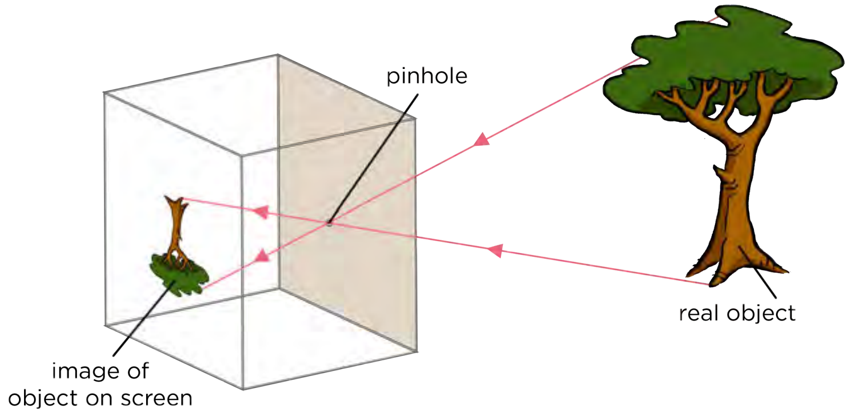
History of the telescope.
bit.ly/1ibkZ9o

Light is emitted from stars and galaxies and travels in a straight line in all directions. When you look at a star, you only see the light rays that hit your eye. In Energy and Change, we learnt about visible light. How is the energy of light transferred through space?

___

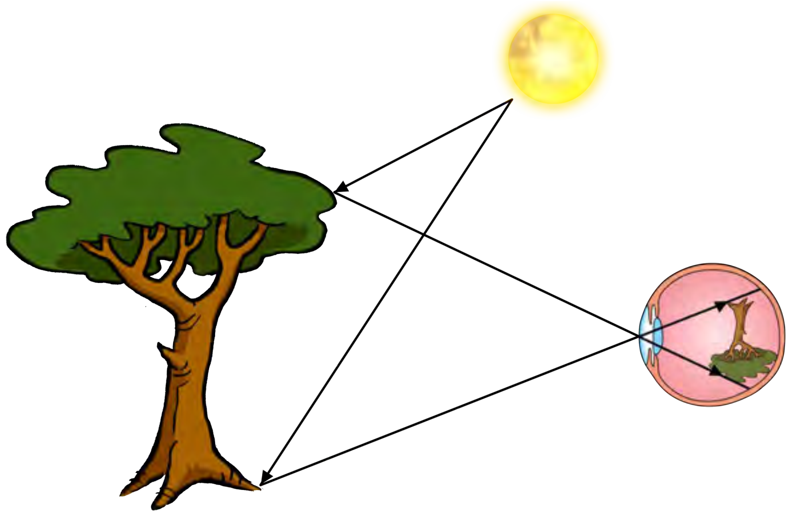
The further away a star is, the more the starlight is spread out and so less of the total light from the star reaches your eye. This makes distant objects faint and difficult to see clearly. If we had huge eyes we would be able to see distant objects more clearly because our eyes would gather more of their light.

Do you remember making a pinhole camera in Energy and Change? Have a look at the following diagram which illustrates this again.



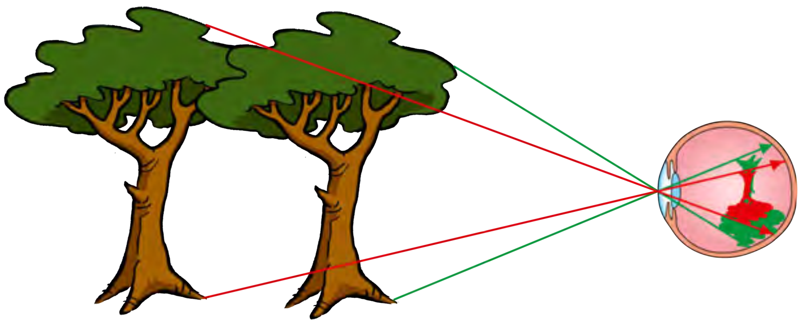
Which way is the image projected onto the screen?

This is the same way in which images are formed on your retina when you view an object, as shown in the following image.



Images formed on the light-detecting retina at the back of your eye are upside down.

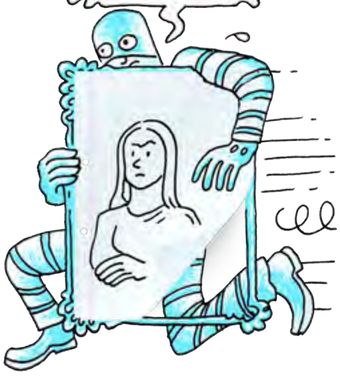
An object that is far away projects a small image of the object onto the retina at the back of your eye making it difficult to see fine details in the image.



More distant objects appear smaller on our retina.

TAKE NOTE

Luminous objects, such as the Sun and other stars, emit light. The tree is NOT a luminous object as it does not emit its own light light. It reflects the light from the Sun.



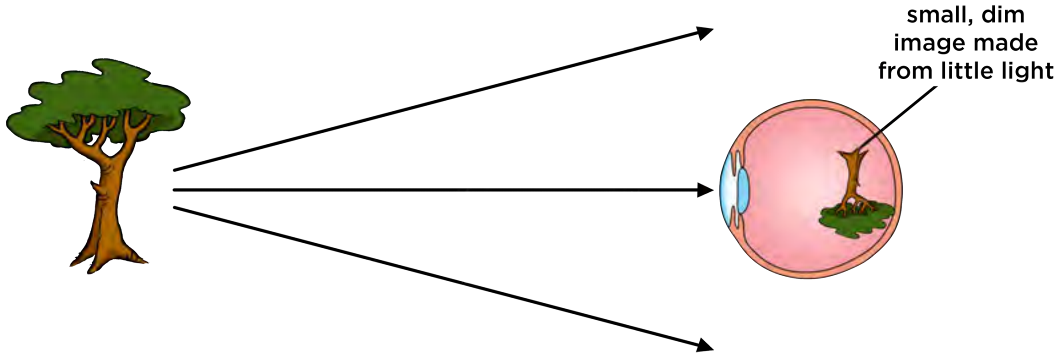
VISIT

The beauty of the night sky.

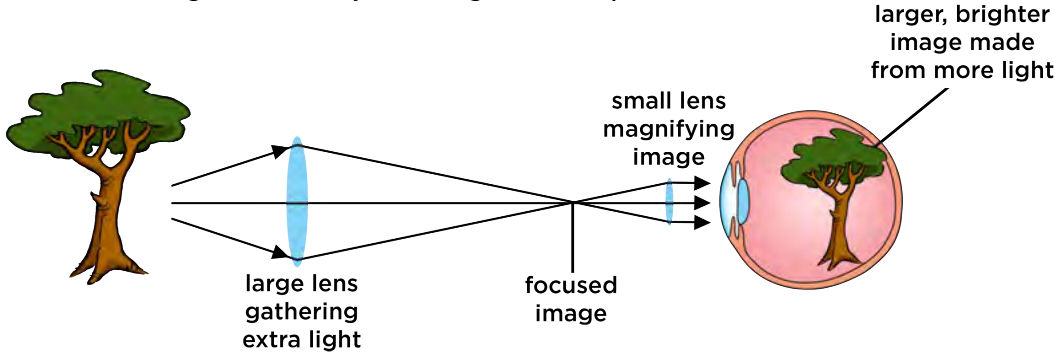
bit.ly/1h5dy5M

Telescopes help us see faint, distant objects more clearly because they collect more light from the objects than our eyes do. They also magnify the image.

Viewing a distant object



Viewing the same object through a telescope



DID YOU KNOW?

Images formed on your retina are actually upside down. Your brain "corrects" the image, so that you do not notice.

As revision of what we learned in Energy and Change last term, answer the following questions.

What type of lens is shown in the above image?

___

What happens to the light when it passes through the lens?

___

___

Let's take a closer look at how a telescope works.

ACTIVITY: Telescopes as light buckets

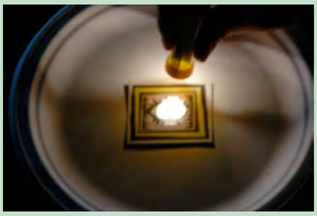
There is only so much light emitted from an object each second. Little packets of light are called **photons**. Our eye needs at least 500 photons, or packets of light, coming into it every second for our brains to sense that something is there. In this activity you are going to represent photons from a distant galaxy using pepper grains or hundreds and thousands.

MATERIALS:

- paper plate
- piece of paper 3cm by 3cm
- pencil or pen
- torch
- pepper grains or hundreds and thousands
- wooden skewers
- foam (bath sponge will do, ideally as wide as the paper plate in one direction)
- tape - optional
- scissors

INSTRUCTIONS:

1. On the piece of paper draw an image of your eye including the pupil and iris.
2. Tape or place the image of your eye onto the middle of a paper plate. The paper plate represents a telescope mirror or lens.
3. Take the foam and cut it into a thin strip about 3 cm wide and as long as the paper plate across.
4. Stick six skewers into the foam equally spaced along the strip. Trim the pointed edges off that are sticking out for safety. You will use this foam strip later in this activity.
5. Shine a torch light just above the picture of the eye on the plate.

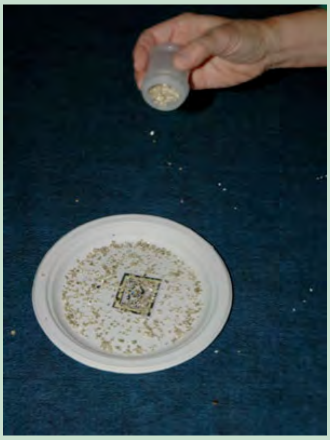


When an object is closer, more light reaches your eye.

6. Slowly move the torch further away from the plate and watch how the light spreads out and dims.
7. Note how much of the torch light the eye's pupil receives compared to the paper plate.
8. Now remove the torch and get ready to use the pepper grains or hundreds and thousands. These will represent *photons* or *packets of light.*

The further away an object, the less light that reaches your eye.

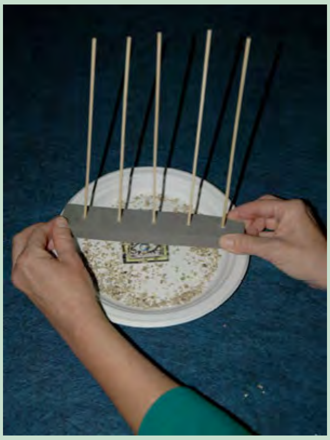
9. Sprinkle these photons for one second over the plate.



Sprinkle the pepper grains or hundreds of thousands.

10. Note roughly how many photons get into the eye compared with how many hit the paper plate representing the telescope mirror or lens.

11. Now place the foam across the centre of the paper plate. The skewers should be pointing straight up. This represents a strip of the telescope mirror with the skewers representing light rays from distant objects.



The skewers represent the light rays hitting the mirror of the telescope.

12. The telescope mirror is actually curved. Bend the foam upwards at either end so that the skewers begin to come together in the middle.

13. Turn the foam over and direct the skewers into the picture of the eye. The light rays from a large strip of the mirror are now entering the small pupil of the eye.

Now you can see how a telescope's mirror can collect lots of light and direct it into a small detector, like your eye.

QUESTIONS:

1. Which collects more of the torch light as the torch moves further away: the eye's pupil or the paper plate?

---

2. Did the eye collect enough photons in one second to detect the light?

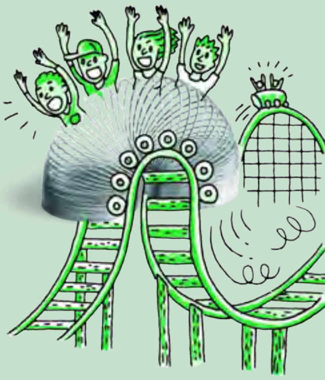
---

3. Did the telescope mirror (paper plate) collect enough photons for the eye to detect the light?

4. How do you think all the light that hits the telescope mirror is concentrated so that it can enter our eyes or a small telescope detector?

Telescopes have big lenses or mirrors to collect as much light as possible. This is how they are able to see faint objects. Telescopes also concentrate or focus the light and redirect it into our small eye so that we can see the dim object. Alternatively, telescopes can redirect the light into special detectors that record images, similar to a cell phone camera.

ACTIVITY: Compare your eye with SALT



The Southern African Large Telescope (SALT) takes pictures of some of the most distant and faintest objects in the Universe. SALT's camera takes images with exposure times typically of twenty minutes, after which the camera shutter closes and the resulting image is displayed on a computer. The longer the exposure, the more light that the telescope can gather to make the image. The human eye does not have a shutter. We seem to see continuously, rather than as a succession of still images. However, the eye does have a kind of exposure time. In this activity you will estimate the exposure time of your eye by estimating your reaction time and then compare it with SALT's typical exposure time.

MATERIALS:

- ruler
- calculator
- pencil or pen

INSTRUCTIONS:

1. Work in pairs for this activity.
2. Look at your partner's eyes. Estimate the diameter of their pupils using a ruler held close to their eye. Be careful not to actually touch your partner's eyes.
3. Write down the diameter of pupil in the table below.
4. Compare the diameter of the pupil with that of the Southern African Large Telescope (SALT) which is roughly 10 m in diameter.
5. Calculate how many times larger than an eye SALT is. (Remember to compare the areas rather than the diameters).
6. One of the pair: hold a pen or pencil directly in front of you, while the other person stands opposite you and prepares to catch it.

7. Drop the pen or pencil and see if you partner can catch it.
8. Estimate the reaction time of your partner. Is it a second? Is it a tenth of a second? Is it a thousandth of a second?
9. Repeat steps 6 - 8 swapping places.
10. Fill in your reaction times in the table below, these represent the exposure time of your eye.
11. Complete the questions.

Table to record your results:

	Eye	**SALT**	**SALT / Eye**
Diameter of collecting lens / mirror	________ cm	________ cm	
Area of collecting lens / mirror	________ cm^2	________ cm^2	
Exposure time	_____ seconds	_____ seconds	

Hint: Convert the diameter of SALT to cm. Convert the exposure time of SALT to seconds. To simplify the calculation of the area of the SALT mirror assume it is a circle with a radius of 5 m. The area of a circle is given by the formula $A = \pi r^2$.

QUESTIONS:

1. Why should you compare the area of the telescope and eye's pupil rather than their diameters?

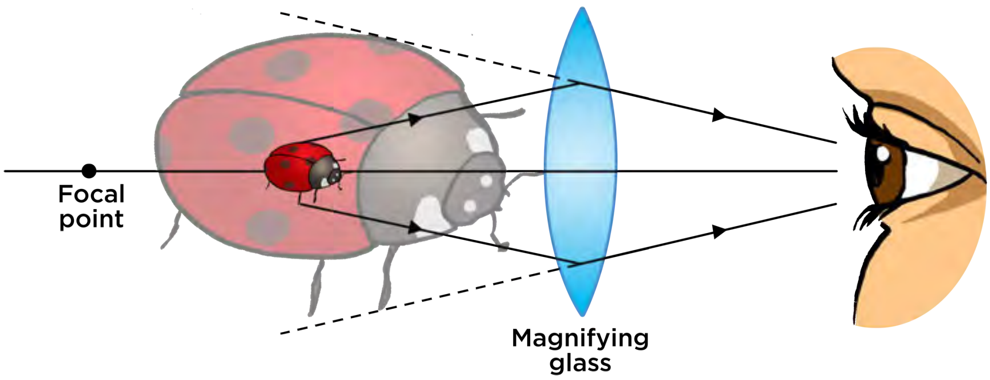
2. How many times more light does the SALT telescope collect compared with your eye?

3. What would happen to your reaction time if your eye had to accumulate light over a longer interval before sending an image to the brain?

4. How many times longer can SALT expose for than your eye?

Telescopes can collect more light from faint and distant objects because they have larger collecting areas *and* because they can accumulate light over longer periods of time to make an image. This means that you can see fainter objects with telescopes that you would be able to see using just your eye.

Telescopes also **magnify** (enlarge) the image that you see, so it takes up more room on your retina allowing you to see the object more clearly.



A convex (converging) lens used as a magnifying glass. The resulting image is larger than the object. Telescopes magnify images from distant stars and galaxies.

Magnification comes at a price however. A fixed amount of light is received from any object, so if you make the image larger, its gets fainter as the light is spread out within the image. This is why it is so important to collect as much light as possible.

The larger a telescope's mirror or lens, the better it is at seeing narrowly separated objects as individual objects and the sharper the images look.

The most important feature of a telescope is how much light it can collect, which depends upon the area of the lens or mirror. The larger the light collecting area, the more light a telescope gathers and the higher resolution (ability to see fine detail) it has. So the size of a telescope is far more important than its magnification.

Now that we have briefly looked at how telescopes work, we are going to look at the different types of telescopes, namely:

- optical telescopes
- radio telescopes
- space telescopes

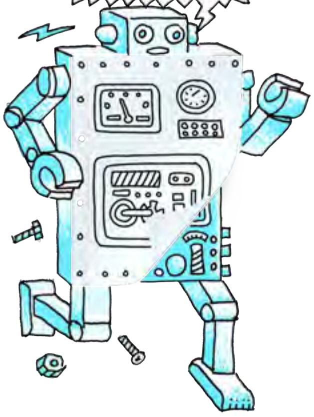
Optical telescopes

Optical telescopes collect visible light from celestial objects. There are two types of optical telescopes.

1. Refracting telescopes use **lenses** to collect and focus the light from distant objects.
2. Reflecting telescopes use **mirrors** to collect and focus the light from distant objects.

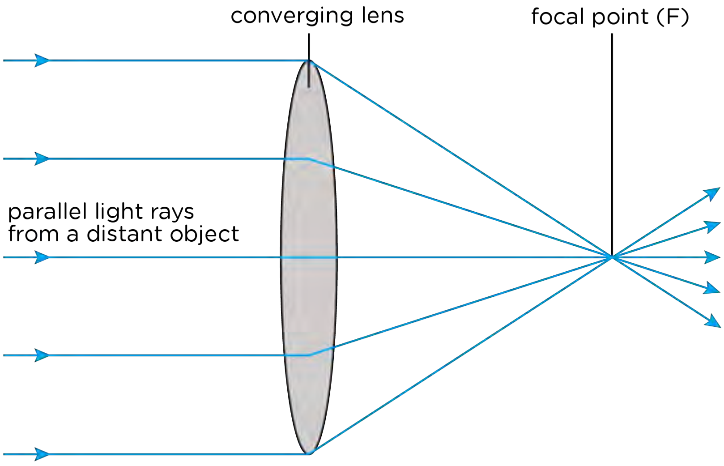
TAKE NOTE

As astronomical objects are so far away, their light rays are considered to be parallel to each other.



1. Refracting telescopes

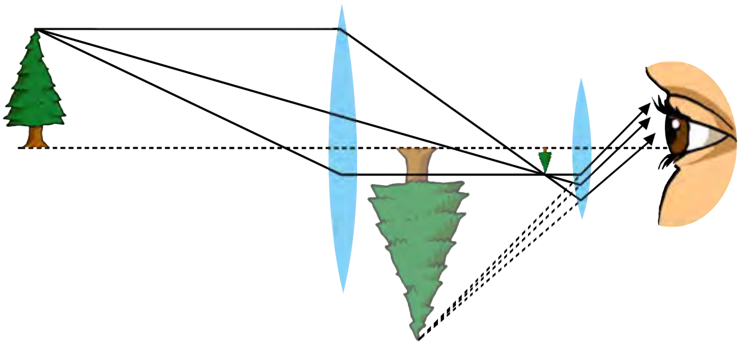
Refracting telescopes use a converging (convex) lens to collect and bend the light rays inwards to the focal point (also called the focus) of the telescope. The light collecting lens is called the objective lens.

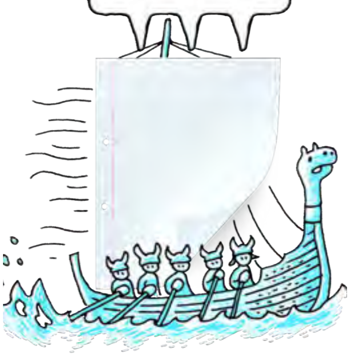


Once light is brought to a focus, it is then magnified by another lens called the eyepiece lens. Look at the optical ray diagram below showing a simple refracting telescope.

TAKE NOTE

A **real image** is called real because light rays actually pass through the point where the image is formed. A **virtual image** is called a virtual image because the light rays do not actually come from the image, they just appear to have come from the image.





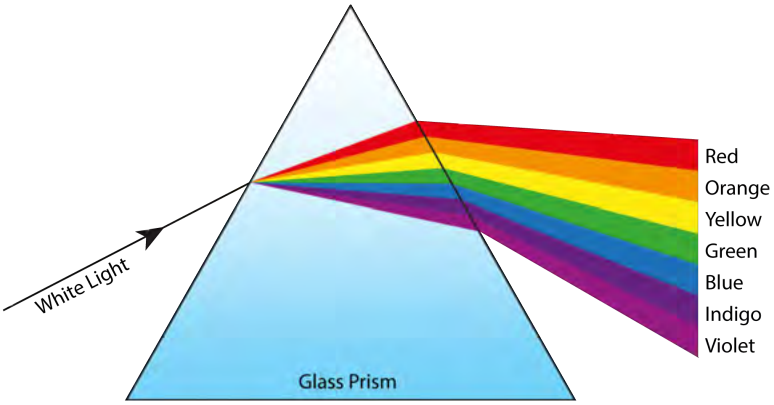
The telescope objective lens collects and focuses the light from a distant tree forming a real inverted image of the tree. The eyepiece lens, like a magnifying glass, then enlarges the image collected by the objective lens, producing a larger, virtual image. This images is what we see when we look through the telescope.

What kind of lenses are the objective lens and the eyepiece lens?

____

____

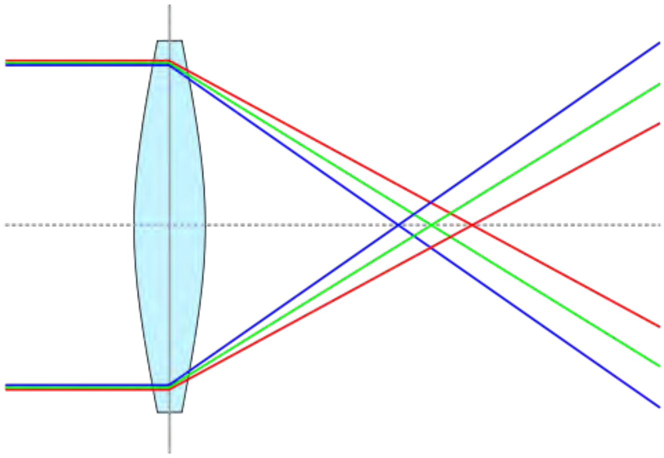
Look at the following picture which shows how white light is refracted (bent) as it travels through a prism. As we learnt in Energy and Change, when light travels through glass it slows down and so it bends or refracts.



Do all the colours undergo the same amount of refraction? Which colour is bent the most?

White light is a mixture of all the colours of the rainbow. Different colours are refracted by different amounts as they travel through the prism so the white light is split into its different colours. How do you think this affects the images produced by refracting telescopes?

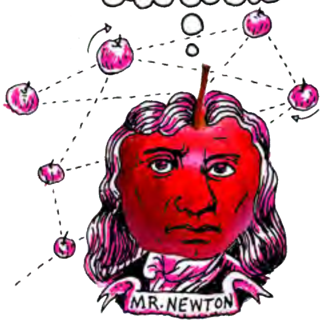
Lenses are shaped to bend light by a certain desired amount. However, the different colours that make up white light bend by slightly different amounts. This means that different colours come to a focus at slightly different distances from the objective lens. Each colour will produce its own image and they will be slightly misaligned with each other resulting in a slightly blurry image. This effect is called **chromatic aberration** and all lenses suffer from this effect.



Blue light is bent more than red light and so different colours are focused at different distances from the lens. The different coloured images are overlaid upon each other and, because they are misaligned, the resulting image is blurry.

DID YOU KNOW?

The first successful reflecting telescope built was the Newtonian Telescope, by Isaac Newton, which is where the design gets its name.



TAKE NOTE

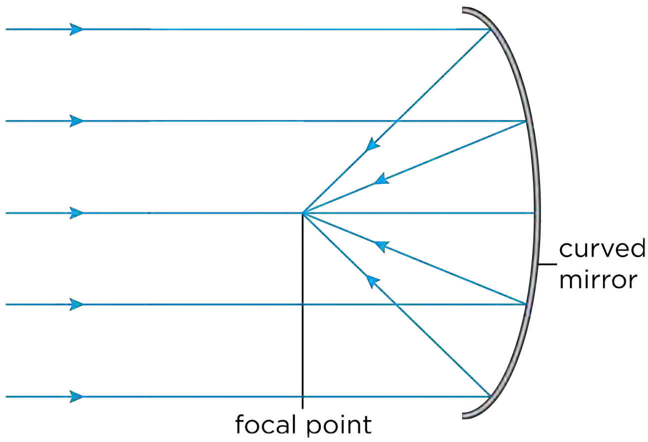
Remember that for each ray, the angle of incidence is equal to the angle of reflection, as you learned in Energy and Change.

The main disadvantages of refracting telescopes are:

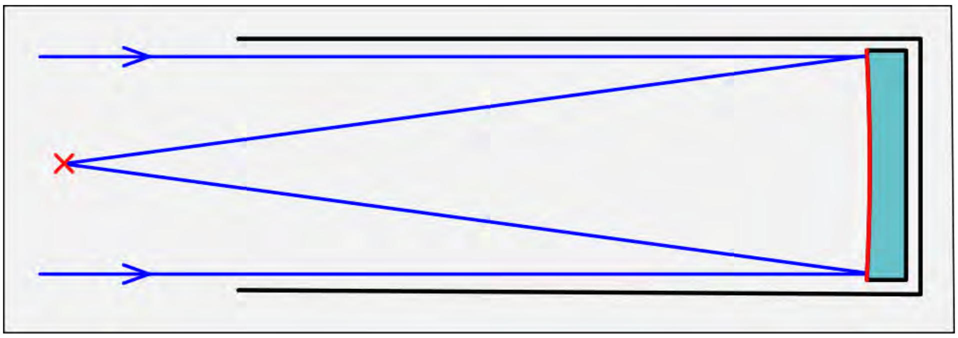
1. Light travels through the lenses in the telescope and so the lenses have to be perfect. There must be no bubbles of air in the glass which would distort the image. It is difficult to and expensive to make large perfect lenses.
2. The light travels through the lenses and so they can only be supported around their edges, where they are thinnest and weakest. This limits the size of refracting telescopes because if a lens is too large it will sag under its own weight and distort the image.
3. Lenses suffer from chromatic aberration which blurs the image.

2. Reflecting telescopes

In the 1680s, Isaac Newton invented the reflecting telescope. Reflecting telescopes use a curved primary mirror to collect light from distant objects and reflect it to a focus.



There are many different types of reflecting telescopes. A **prime focus reflector** is the simplest type of reflector telescope. In this design, a recording structure is placed at the focal point to obtain the focused image. In the old days, in very large telescopes, a person would actually sit in an "observing cage" to view the image directly or operate a camera. However, now a detector is used and is operated from outside of the telescope. The position of the detector is shown in the following diagram with a red cross.



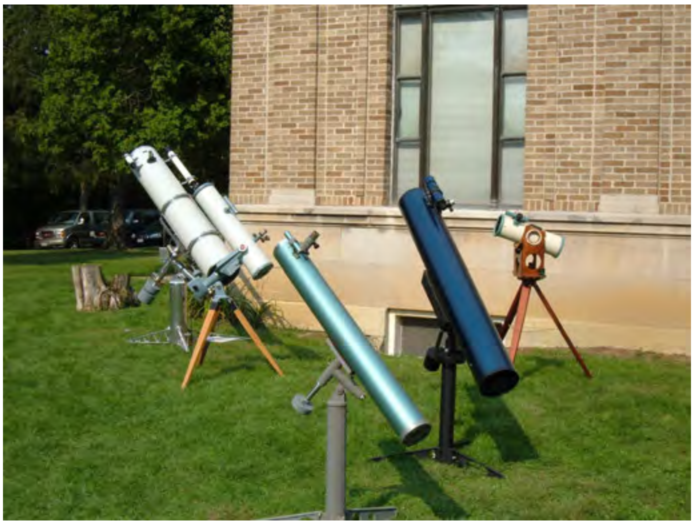
A prime focus reflector with a detector at the focal point, marked with an X.

More complex designs of reflecting telescopes use a secondary small mirror to reflect the light towards the eyepiece lens.

- A **Newtonian reflector** reflects the light to an eyepiece on the side of the telescope tube. This design is often used for amateur telescopes because having the eyepiece on the side of the tube makes the telescope easy to use.
- A **Cassegrain reflector** reflects light through a small hole in the primary mirror. This kind of telescope is often used for large professional telescopes as it allows heavy detectors to be placed at the bottom of the telescope. This makes them easy to reach for repairs and also means that the weight of the detectors does not affect the telescope tube.

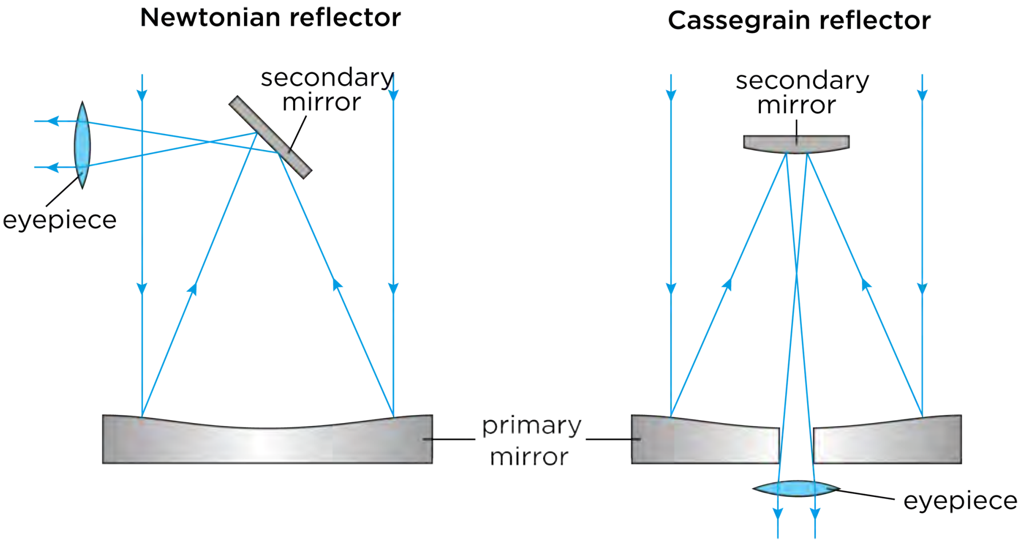
DID YOU KNOW?

The Cassegrain reflector is named after a reflecting telescope design that was published in 1672 and has been attributed to Laurent Cassegrain.



A group of Newtonian telescopes.

The following ray diagrams show the difference between a Newtonian and Cassegrain reflector.

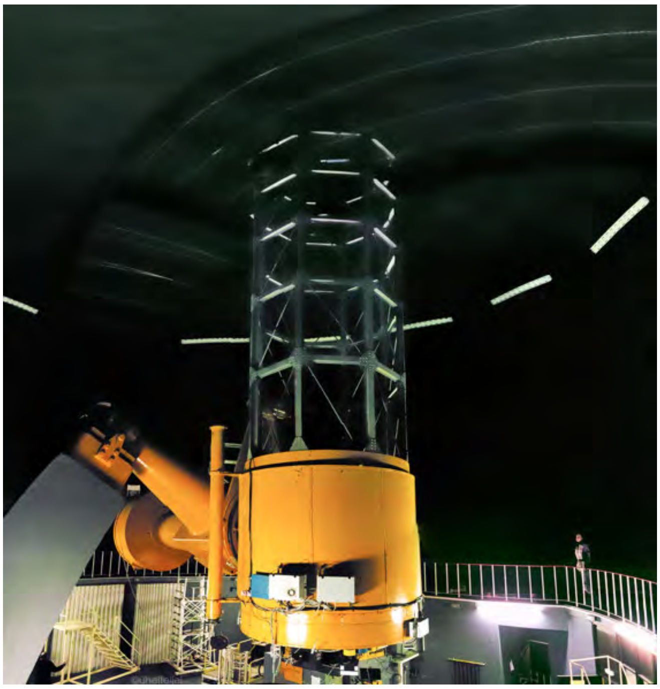


Ray diagrams for some example reflecting telescopes. The Newtonian reflector is often used in amateur telescopes. The Cassegrain telescope is often used at large observatories.

VISIT

Curious about the Universe, but don't know where to start? Have a look at this step-by-step guide to becoming an awesome amateur astronomer.

bit.ly/1gBwrQ8



The SAAO 1.9 m reflecting telescope. Detectors are bolted onto the Cassegrain focus at the bottom of the telescope (metal boxes under the orange tubing). (Credit: SAAO).

The secondary mirror in a reflecting telescope must be very small. Why do you think this is so?

DID YOU KNOW?

NASA is currently planning the successor for the Hubble Space Telescope, called the James Webb Space Telescope (JWST). It will be launched into space in 2018.

Do you think that reflecting telescopes suffer from chromatic aberration? Why?

The advantages of a reflecting telescope include:

1. The glass of the mirror does not have to be perfect throughout, only the surface has to be perfect.
2. The mirror can be supported across the whole of its back so it won't sag.
3. Making large mirrors is easier and cheaper than making big lenses.
4. They do not suffer from chromatic aberration.

Optical telescopes on the ground do however have some disadvantages:

1. They can only be used at night.
2. They cannot be used in bad weather (rain, cloud, snow etc).

Optical telescopes are best placed on the tops of remote mountains. Discuss within your class why you think this is. Take some notes in the space below.

The largest telescopes in the world today are reflecting telescopes. In the next section you will learn about one of the largest reflecting telescopes in the world which is located right here in South Africa.

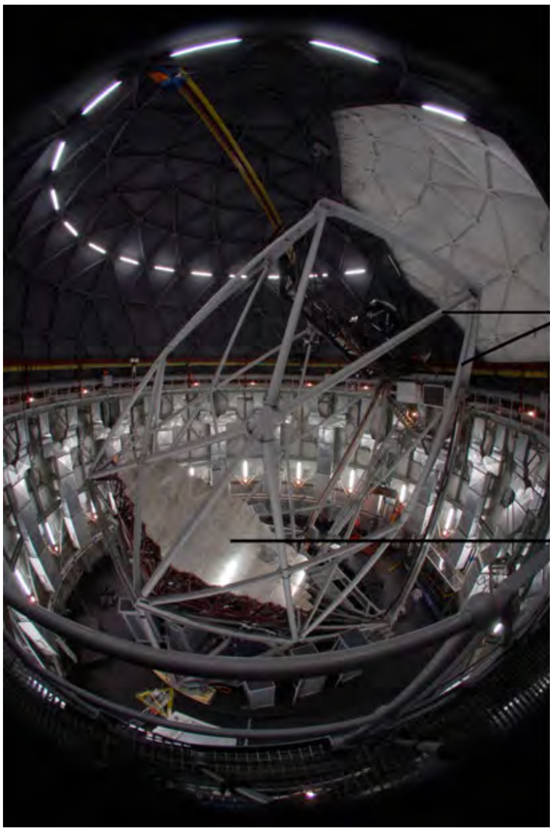
SALT

The Southern African Large Telescope (SALT) is the largest optical telescope in the southern hemisphere and among the largest in the world. SALT was completed in 2005 and is located in the Karoo in the Northern Cape, near the town, Sutherland. Astronomers use telescopes like SALT to study planets, stars and galaxies. SALT can detect the light from faint or distant objects in the Universe a billion times too faint to be seen with the naked eye.

The SALT telescope just outside Sutherland.

The SALT telescope has a large mirror which collects light. SALT's primary mirror is a hexagonal shape measuring 11.1 m by 9.8 m across and is made up of 91 individual 1.2 m hexagonal mirrors. SALT is a prime focus reflector. What does this mean?

SALT does not have a telescope tube. Instead there is a network of metal struts which support the tracker and payload at the top of the telescope. The whole telescope structure weighs 85 tons. The payload contains detectors which take pictures of the night sky.



SALT's giant mirror, made up of 91 individual mirrors.

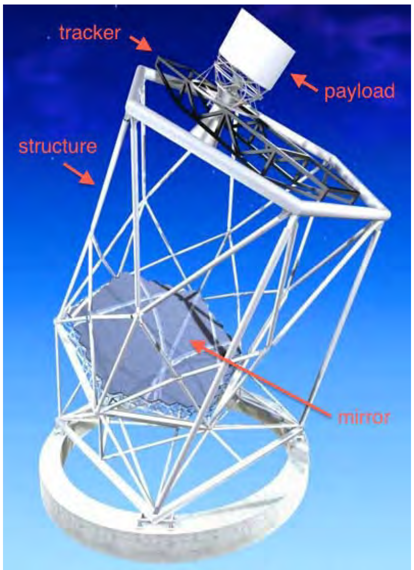
VISIT

What is a radio telescope?

bit.ly/1a4LbTW

Stars move during the night just as the Sun moves across the sky in the day. The telescope must follow the stars as they move. The tracker at the top of SALT is used to follow the drifting stars carrying the detectors along with it as it tracks the stars.

SALT is currently being used to study stars, in particular binary star systems where two stars orbit around each other. Astronomers also use the telescope to study galaxies and some of the most violent explosions in the Universe called supernovae and Gamma Ray Bursts which occur when massive stars explode at the end of their lives. SALT is also looking at the Universe on the largest of scales, in order to answer the questions how did the Universe begin, and what will happen to it in the future?



The SALT telescope structure. (Credit: SALT)

NEW WORDS

- antenna
- receiver
- amplifier
- SKA

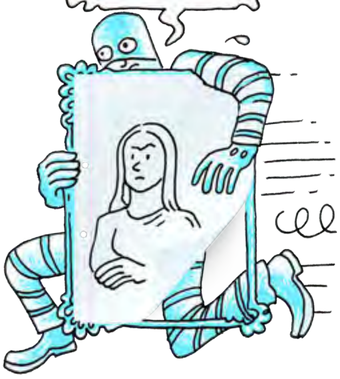
The Karoo is an ideal place to host SALT because it is far away from towns and cities so there is very little light pollution. The area is also at a high elevation, dry and there are no extreme weather conditions, such as flooding or storms. Despite it being so remote at the observatory site there is good infrastructure, including roads and electricity, in the surrounding area of Sutherland.

Radio telescopes

Radio waves are a type of electromagnetic radiation (or light) that humans cannot see with their eyes. They have very long wavelengths compared to optical light. Purple light, for example, has a wavelength of 400 nm whereas red light has a wavelength of 700 nm. Radio wavelengths are much longer; radio waves have wavelengths from approximately one millimeter to hundreds of metres.

TAKE NOTE

Do you remember learning about wavelength in Energy and Change? A wavelength is the distance between two corresponding points on two consecutive waves.



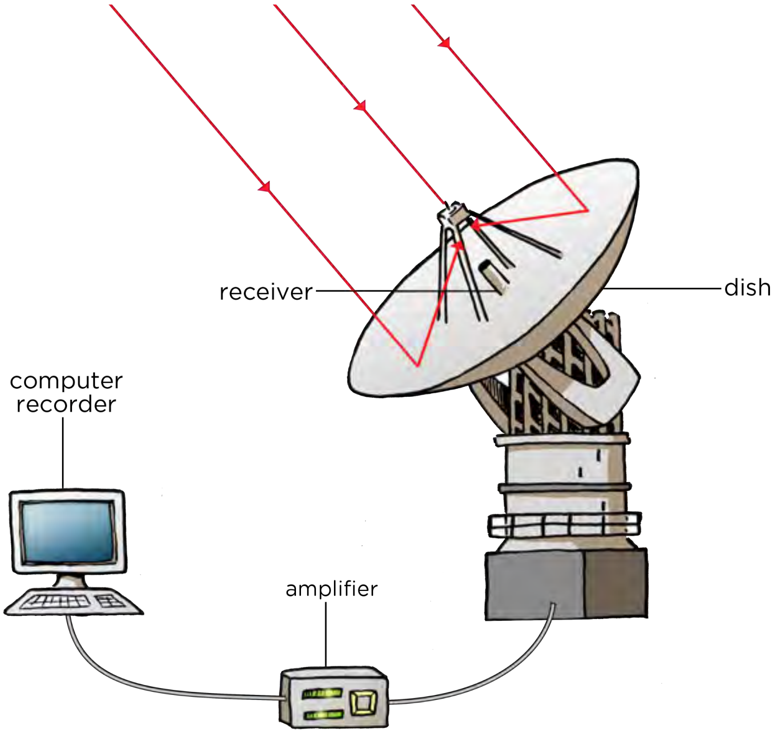
Radio telescopes detect radio waves coming from distant objects. Radio telescopes have several advantages over optical telescopes. They can be used in bad weather, as radio waves are not blocked by clouds. They can also be used during the day and at night.

Many objects in space emit radio waves, for example some galaxies, stars and nebulae which are giant clouds of dust and gas where stars are born. Some objects emit radio waves but do not emit optical light, therefore looking at the sky at radio wavelengths reveals a completely different picture of our Universe.

An optical (white) and radio (orange) image of the galaxy NGC 1316. The radio emission spans over one million light years and engulfs the optical light at the centre.

If your eyes could see radio waves at night, rather than white light, instead of seeing pointlike stars, you would see distant star-forming regions, bright galaxies and beautiful giant clouds around old exploded stars.

Radio telescopes typically look like large dishes. The **dish** or **antenna**, acts like the primary mirror in a reflecting telescope, collecting the radio waves and reflecting them up to a smaller mirror which then reflects the radio waves to a radio wave detector. Radio wave detectors are called **receivers**. An **amplifier** amplifies the signal and sends it to a computer which processes the information from the receiver to create colour images which we can see.



Radio telescopes need to be placed far away from cities and towns as man-made radio interference can interfere with the telescope's observations.

Part of the KAT-7 radio telescope array in the Northern Cape.

VISIT

Atacama Large Millimeter/sub-millimeter Array (ALMA) is a new radio telescope in Chile's Atacama desert. It will open an entirely new 'window' into the Universe, allowing scientists to search for our cosmic origins. Watch a video on some of the latest research and images released from ALMA.

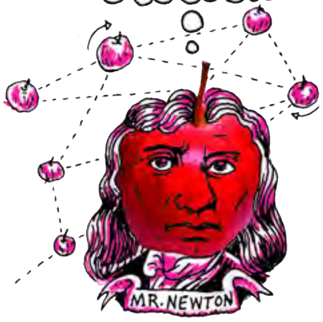
bit.ly/17GzMnB

DID YOU KNOW?

Rapidly rotating star remnants, called pulsars, were first discovered using a radio telescope in 1967. Astronomers initially considered the possibility that the regular pulses of radio waves were signals from an alien civilisation but quickly realised that this was not the case.

DID YOU KNOW?

The SKA central computer will have the processing power of about one hundred million computers. The dishes of the SKA will produce 10 times the global internet traffic.



VISIT

A video on the SKA.
bit.ly/1aE3b2A
Read more about the SKA online.
bit.ly/H02OHY

DID YOU KNOW?

The data collected by the SKA in a single day would take nearly two million years to playback on an ipod.

MeerKAT and the SKA

The MeerKAT radio telescope array is currently under construction in the Northern Cape. MeerKAT is scheduled to be complete in 2016 and when it is finished it will have 64 radio dishes each 13.5 m in diameter. The MeerKAT array will be the largest and most sensitive radio telescope in the southern hemisphere until the Square Kilometre Array (SKA) is completed around 2024.

The KAT-7 test array in the Northern Cape is a test array for the larger MeerKAT array.

The Square Kilometre Array (SKA) will be the most powerful telescope ever. It will have a total collecting area of one square kilometer. It will have 3000 radio dishes each about 15 m wide which will act together as one large telescope. As well as the 3000 radio dishes there will be two other types of radio wave detectors.

The location of SKA in South Africa, and other African countries.

Many different countries are working together to build, and pay for, the SKA. At least 13 countries and close to 100 organisations are already involved, and more are joining the project. Most of the SKA will be located in South Africa. There will also be locations in Australia and some stations in eight African partner countries namely, Botswana, Ghana, Kenya, Madagascar, Mauritius, Mozambique, Namibia, and Zambia.

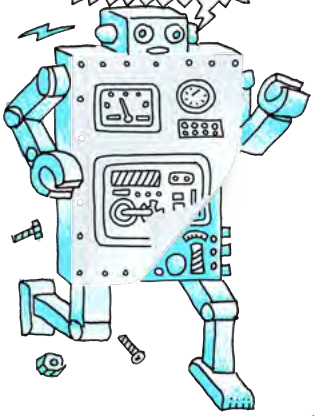
DID YOU KNOW?

Radio astronomy observatories use diesel cars around the telescopes because the ignition of the spark plugs in petrol cars can interfere with radio observations.

One of the SKA dishes.

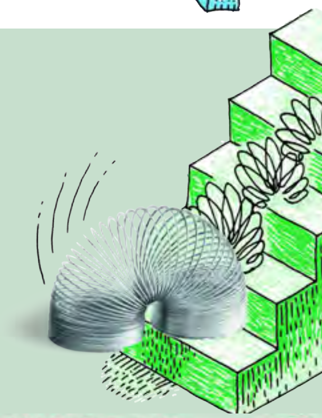
TAKE NOTE

Jobs are not just limited to astronomers: engineers, computer scientists and administrative staff are needed to run the telescopes.



MeerKAT and the SKA will be used to investigate how galaxies change over time, our understanding of gravity, the origin of cosmic magnetism, how the very first stars formed, other planets around other stars, and whether we are alone in the Universe.

ACTIVITY: Careers in Astronomy



INSTRUCTIONS:

Discuss in class with your teacher and classmates what sorts of careers you think are now available in astronomy in South Africa because of the construction of SALT and MeerKAT / SKA. Think about and discuss the skills needed in each of the roles you discuss.

ACTIVITY: Draw a telescope

MATERIALS:

- paper
- pencils or crayons

INSTRUCTIONS:

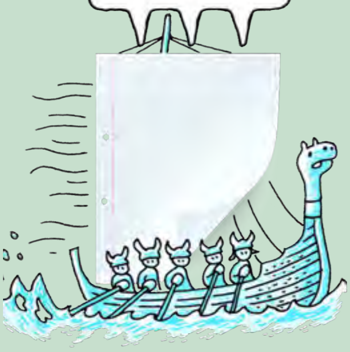
1. Pick either an optical telescope or radio telescope and draw a picture of the telescope.
2. Label the different parts of the telescope and describe what they do.

DID YOU KNOW?

The SKA will be so sensitive it could detect TV signals from planets orbiting other stars.

TAKE NOTE

The sensitivity of a radio telescope depends upon the area of the collecting dish and the sensitivity of the radio receiver. In order to produce sharp radio images comparable to images from optical telescopes a radio telescope must be much larger than an optical telescope.



Space telescopes

Radio waves and visible light form part of what is called the electromagnetic spectrum of light. There are other types of light at different wavelengths that we cannot see with our eyes including X-rays, ultraviolet and infrared light.

The Earth's atmosphere blocks X-rays, ultraviolet and infrared light and stops them from reaching the ground. So if we want to observe this kind of light from stars and galaxies, we need to put telescopes in space. This is why X-ray telescopes and infrared telescopes are placed in space.

A picture of an X-ray telescope called XMM-Newton.

The advantages of space telescopes are that they can observe the whole sky and operate during both night and day. Images taken with space telescopes are far sharper than images taken with telescopes on the ground, because images are not smeared or blurred by turbulence in the Earth's atmosphere, as with images take from ground telescopes. This is why the Hubble Space Telescope images are so detailed even though it is a relatively small reflective telescope. The major disadvantages of space telescopes are their costs and the fact that if something goes wrong they are extremely difficult to fix.

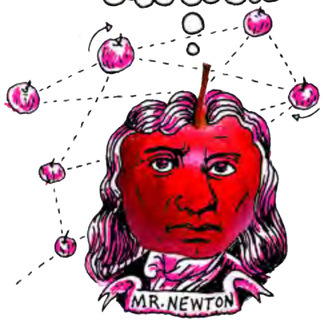
The Hubble Space Telescope has a 2.4m diameter collecting mirror.

VISIT

The Hubble Space Telescope (videos) bit.ly/1873WQd and bit.ly/1abWIiG and some of the best images from Hubble bit.ly/1deIJLS

DID YOU KNOW?

The Hubble Space Telescope is named after Edward Hubble, considered to be one of the most important cosmologists of the 20th century. Hubble discovered there were galaxies beyond our own and helped confirm that the universe is expanding.



ACTIVITY: Telescope information poster

MATERIALS:

- paper
- pencils or crayons
- pictures downloaded from the internet or copied from books - optional

INSTRUCTIONS:

1. Pick a telescope that you want to make a poster about. It can be a ground-based or space-based telescope.
2. Describe the telescope and explain how it works. Include a diagram or picture of the telescope and label its main parts in your poster.
3. List some of the science that the telescope is used for in your poster.
4. List some of the advantages and disadvantages of the type of telescope you have chosen in your poster.

VISIT

How many people are in space right now? Find out here.

bit.ly/18Gzr83

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VISIT

Learn more about the James Webb Space Telescope (video).

bit.ly/1h5hUd9

SUMMARY:

Key Concepts

- Early cultures observed the stars and grouped them together in patterns or constellations.
- Telescopes allow astronomers to see distant, faint objects in more detail.
- The performance of a telescope is measured by how much light it can collect. Larger telescopes can collect more light and see finer details than smaller telescopes.
- Optical telescopes detect optical light from distant objects.
- Most modern day optical telescopes use mirrors to collect and focus the light from distant objects.
- Radio telescopes collect and focus radio waves, emitted from distant objects in space.
- South Africa is host to one of the the most advanced optical telescopes in the world, the Southern African Large Telescope (SALT).
- South Africa will also host a large part of the soon to be constructed SKA radio telescope which will be the largest radio telescope in the world once complete.

Concept Map

The concept maps in this workbook we made using an open source, free programme. If you would like to make your own concept maps for your other subjects, you can download the programme from the link in the visit box.

VISIT

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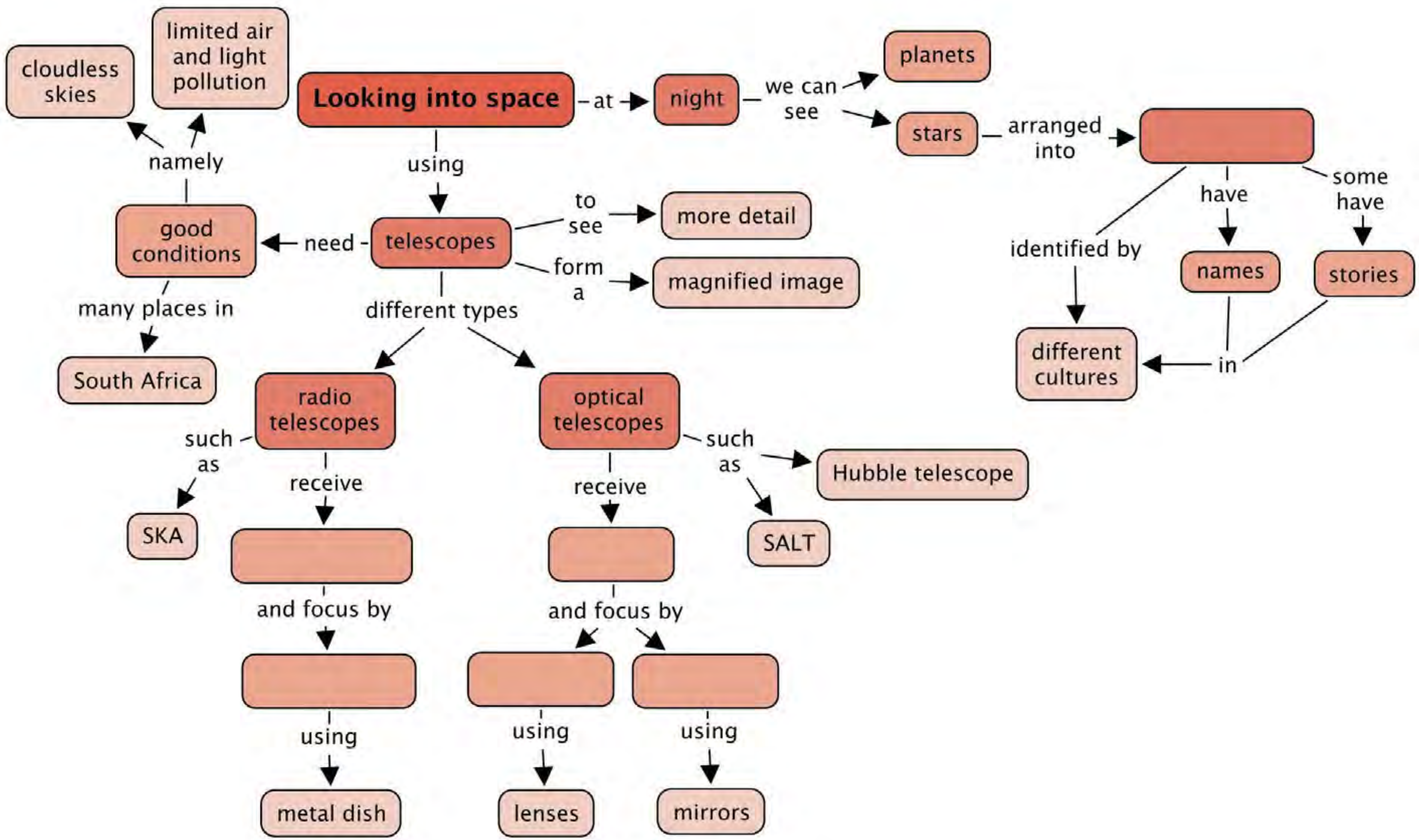
bit.ly/1fSWS2s

VISIT

Science is about curiosity, discovery and innovation!

bit.ly/18GzSyZ



Looking into space
at
night
we can see
planets
stars
arranged into
some have
stories
have
names
identified by
different cultures
in
using
telescopes
need
good conditions
namely
cloudless skies
limited air and light pollution
many places in
South Africa
to see
more detail
form a
magnified image
different types
radio telescopes
such as
SKA
receive
and focus by
using
metal dish
optical telescopes
such as
Hubble telescope
SALT
receive
and focus by
using
lenses
using
mirrors

REVISION:

1. What do astronomers call patterns of stars in the sky? [1 mark]

2. Name three famous southern constellations. [3 marks]

3. What do optical refracting telescopes use to collect and focus light from distant objects? [1 mark]

4. What do optical reflecting telescopes use to collect and focus light from distant objects? [1 mark]

5. List two advantages that reflecting telescopes have over refracting telescopes. [2 marks]

6. What sort of light do radio telescopes detect? [1 mark]

7. List two advantages that radio telescopes have over optical telescopes. [2 marks]

8. Why are X-ray telescopes located in space? [1 mark]

9. Why does the Hubble Space Telescope produce such sharp images even though it is much smaller than most professional ground based telescopes? [1 mark]

10. Why should astronomers look at objects at different wavelengths? [1 mark]

11. What is the name of the largest optical telescope located in the Northern Cape? [1 mark]

12. List three reasons why the SALT telescope is located near Sutherland in the Northern Cape. [3 marks]

13. How many dishes will the MeerKAT array have? [1 mark]

14. How many dishes will the SKA array have? [1 mark]

15. List two areas of astronomy that will be studied using the SKA telescope. [2 marks]

Total [22 marks]

What can you transform our Earth into? Be curious!

GLOSSARY

Alpha Centauri:	our second closest *easily visible* star after the Sun; it is actually two stars orbiting very close together
amplifier:	a device which amplifies (to make something bigger) the radio wave signals
antenna:	the dish or other device used to collect radio waves in a radio telescope
asteroid belt:	the area where most asteroids are found in our solar system, lying between the orbits of Mars and Jupiter
asteroid:	a small rocky object orbiting the Sun
astronomical unit (AU):	the average distance between the Earth and the Sun, equal to around 150 million kilometres
celestial:	positioned in or relating to the sky, or outer space as observed in astronomy
chromatic aberration:	an optical effect where different colours are refracted by different amounts in a lens leading to a distorted image
comet:	a small object made of ice and dust which sometimes enters the inner solar system; when a comet enters the inner solar system, part of it evaporates to form a long tail of ice and dust pointing away from the Sun
constellation:	a group of stars that form a pattern in the sky when viewed from Earth
convection:	one of the three ways to transport heat energy (the other two are conduction and radiation); as a liquid or gas is heated, it becomes less dense and rises; while denser colder material sinks, creating a flow of moving liquid or gas which transports heat energy along with it
dwarf planet:	a large, roughly spherical object orbiting a star which cannot be classed as a planet because it is not large enough to sweep out other objects from its orbit
filament:	a threadlike structure in space containing galaxies and galaxy groups and clusters
galaxy bulge:	a spheroidal (rugby ball shaped) distribution of old stars at the centre of a galaxy
galaxy cluster:	a collection of over 50 or more galaxies, held together by gravity
galaxy disk:	the flat distribution of stars, gas and dust in a galaxy
galaxy group:	a collection of about 50 or less galaxies, held together by gravity
galaxy:	a collection of millions or billions of stars, gas and dust all held together by gravity

gas giant: a large planet made mostly of gas with no solid surface; the four outermost planets in the solar system are gas giants

habitable zone: the region surrounding a star in which water can remain in its liquid state

Kuiper Belt: region of space filled with trillions of small objects that lie in the outer reaches of the solar system, past the orbit of Neptune

Kuiper Belt object: a small icy object orbiting the Sun out beyond the orbit of Neptune

light hour: the distance that light travels in one hour

light minute: the distance that light travels in one minute

light year: the distance that light travels in one year

nuclear fusion: the process by which stars produce their energy; light atomic nuclei come together and merge to form heavier atomic nuclei, releasing energy as they do so; in the Sun, hydrogen nuclei fuse with other hydrogen nuclei to form heavier helium nuclei

Oort Cloud: a hypothetical huge cloud of icy objects (comets) surrounding the Sun at the very edge of our solar system at a distance between 5,000 and 100,000 times the Earth's distance from the Sun

photosynthesis: the process by which green plants and some other organisms use sunlight to synthesise foods from carbon dioxide and water producing oxygen as a byproduct

primary mirror: the light-collecting mirror in an optical telescope

Proxima Centauri: our second closest star after the Sun

receiver: a device that detects radio wave signals

SALT: the Southern African Large Telescope, the largest optical telescope in the southern hemisphere

SKA: the Square Kilometre Array, the largest planned radio telescope array in the world

solar system: the Sun, and the collection of planets and smaller objects that orbit around the Sun

solar wind: the continuous flow of charged particles from the Sun that extends out to the far reaches of the solar system

spiral arm: a region of stars, gas and dust forming a curved shape spiralling out from the centre of a spiral galaxy

star: a huge ball of burning gas which emits energy in the form of light and heat

starlore: mythical stories about the stars, planets and constellations

sunspot: a dark region or spot which appears on the surface of the Sun from time to time; sunspots are cooler than the rest of the Sun's surface

telescope: an instrument used to look at distant objects, which makes distant objects appear brighter, larger and clearer; optical telescopes collect visible light and radio telescopes collect radio waves

terrestrial planet:	a planet with a rocky surface like the Earth's surface; the four innermost planets in the solar system are terrestrial planets
Universe:	all of existence, including all planets, stars, galaxies, the space between objects, and all matter and energy
void:	a vast empty bubble in space found between filaments

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1 http://www.flickr.com/photos/chefranden/3507963245/ 106
2 http://en.wikipedia.org/wiki/File:Prime_focus_telescope.svg 228